Off on a Tangent

Meditations on the derivative

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Introduction

Prerequisites

This text assumes basic familiarity with point-set topology (specifically, metric spaces) and with linear algebra. At a few points, there may be some ideas from point-set topology and linear algebra that you have not encountered before. A sort of "bare minimum" exposition of some of these ideas is included in chapter 0.

How to use this book

You are advised to *not* spend time reading chapter 0 thoroughly before jumping into the main part of the text (ie, chapter 1 onwards). When ideas from chapter 0 are invoked in the main part, a reference to the relevant part of chapter 0 is included. My suggestion is to only look at chapter 0 when you run into a reference to it, chasing references back as needed.

I also suggest reading sections in their entirety to get a grasp of the big picture (including reading the statements of the exercises), and then working on cementing your understanding of the details by working through the exercises.

Suggestions for improvement

This is a first draft, and there are bound to be many errors. Please be on the lookout for them! If you think you've found one, please share it with me. I'll also be very interested to hear about non-errors: sentences that are maybe correct but phrased confusingly, places where adding a picture could be helpful, etc.

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0 Prerequisites

You should consult a dedicated linear algebra textbook to review the definitions of vector spaces, subspaces, linear independence, span, dimension, and linear maps. You should also consult an analysis textbook to review the definitions of metric spaces, equivalence of metrics, open subsets, closed subsets, continuous maps, compactness, connectedness, etc.

0.1 Product metric

Exercise 0.1.1. Suppose X and Y are metric spaces with metrics d_X and d_Y, respectively. Define a function $d : (X \times Y) \times (X \times Y) \to \mathbb{R}$ by

$$d((x_1, y_1), (x_2, y_2)) = \max\{d_X(x_1, x_2), d_Y(y_1, y_2)\}.$$

Show that d is a metric on $X \times Y$.

Exercise 0.1.2. Suppose X and Y are metric spaces with metrics d_X and d_Y, respectively. Define a function $d : (X \times Y) \times (X \times Y) \to \mathbb{R}$ by

$$d((x_1, y_1), (x_2, y_2)) = \sqrt{d_X(x_1, x_2)^2 + d_Y(y_1, y_2)^2}.$$

Show that d is a metric on $X \times Y$, and that it is equivalent to the metric from section 0.1.

0.2 Norms on vector spaces

Definition 0.2.1. Let V be a vector space. A *norm* $\| - \|$ on V is a function $V \to \mathbb{R}$ satisfying the following two axioms.

(N1) $|v| \geqslant 0$.

(N2) $|v| = 0$ if and only if $v = 0$.

(N3) $|\lambda v| = |\lambda||v|$ for all $\lambda \in \mathbb{R}$ and $v \in V$.

(N4) $|v + w| \leqslant |v| + |w|$ for all $v, w \in V$.

0.2.A Norms induce metrics

Exercise 0.2.2. Suppose $\| - \|$ is a norm on a vector space V. Show that the function $d : V \times V \to \mathbb{R}$ given by $d(v, w) = |v - w|$ is a metric.

Exercise 0.2.3. Suppose $|-|$ is a norm on a vector space V. Show that the function $V \times V \to V$ given by $(v, w) \mapsto v + w$ is continuous, where $V \times V$ is regarded as a metric space via one of the metrics defined in section 0.1.

Exercise 0.2.4. Suppose $\| - \|$ is a norm on a vector space V. Show that the function $\mathbb{R} \times V \to V$ given by $(\lambda, v) \mapsto \lambda v$ is continuous, where $\mathbb{R} \times V$ is regarded as a metric space via one of the metrics defined in section 0.1.

0.2.B Equivalence of norms

Definition 0.2.5. Two norms $| - |$ and $| - |'$ on a vector space V are *equivalent* if there exist nonzero constants C_1 and C_2 such that

$$C_1|v| \leqslant |v|' \leqslant C_2|v|$$

for all $v \in V$.

Exercise 0.2.6. If V is a vector space, show that equivalence of norms is an equivalence relation on the set of all norms on V.

Exercise 0.2.7. Suppose V is a vector space and $| - |$ and $| - |'$ are two equivalent norms on V. Show that the metrics d and d' corresponding to $| - |$ and $| - |'$ (cf. exercise 0.2.2) are equivalent.

0.2.C Norms on finite dimensional vector spaces

If V is a finite dimensional vector space, we can construct a number of norms on V by choosing a basis. Of these, two are of key importance.

Example 0.2.8. Suppose V is a finite dimensional vector space, and $v_1, \ldots, v_n$ is a basis for V. The L^2-*norm* on V with respect to this basis is defined by

$$|a_1 v_1 + \cdots + a_n v_n| = \sqrt{a_1^2 + \cdots + a_n^2},$$

where $a_1, \ldots, a_n \in \mathbb{R}$. Axioms (N1) through (N3) are straightforward to verify. For the triangle inequality (N4), suppose $v = a_1v_1 + \cdots + a_nv_n$ and $w = b_1v_1 + \cdots + b_nv_n$. Then

$$\begin{aligned} |v+w|^2 &= (a_1+b_1)^2 + \cdots + (a_n+b_n)^2 \\ &= |v|^2 + 2(a_1b_1 + \cdots + a_nb_n) + |w|^2 \\ &\leqslant |v|^2 + 2|v||w| + |w|^2 \\ &= (|v| + |w|)^2, \end{aligned}$$

where we have used the Cauchy-Schwarz inequality[1] for the third step. Taking square roots yields (N4).

Example 0.2.9. Suppose V is a finite dimensional vector space, and $v_1, \ldots, v_n$ is a basis for V. The L^∞-*norm*, or the *max norm*, on V with respect to this basis is defined by

$$|a_1v_1 + \cdots + a_nv_n|_{\max} = \max\{|a_1|, \ldots, |a_n|\}.$$

Axioms (N1) through (N3) are straightforward to verify. For the triangle inequality (N4), suppose $v = a_1v_1 + \cdots + a_nv_n$ and $w = b_1v_1 + \cdots + b_nv_n$. Then

$$\begin{aligned} |v+w|_{\max} &= \max\{|a_1+b_1| + \cdots + |a_n+b_n|\} \\ &\leqslant \max\{|a_1|+|b_1|, \cdots, |a_n|+|b_n|\} \\ &\leqslant \max\{|a_1|, \ldots, |a_n|\} + \max\{|b_1|, \ldots, |b_n|\} \\ &= |v|_{\max} + |w|_{\max} \end{aligned}$$

where we used the triangle inequality for real numbers for the second step.

The following states that these two norms are actually equivalent.

Exercise 0.2.10. Suppose V is a finite dimensional vector space and $v_1, \ldots, v_n$ is a basis for V. Let $|-|$ denote the L^2-norm from example 0.2.8 and $|-|_{\max}$ the max norm from example 0.2.9. Show that, for any $v \in V$, we have

$$|v|_{\max} \leqslant |v| \leqslant \sqrt{n}|v|_{\max}.$$

In fact, here is vast generalization of exercise 0.2.10.

Theorem 0.2.11. *If V is a finite dimensional vector space, all norms on V are equivalent.*

[1] The *Cauchy-Schwarz inequality* states that

$$(a_1b_1 + \cdots + a_nb_n)^2 \leqslant (a_1^2 + \cdots + a_n^2)(b_1^2 + \cdots + b_n^2)$$

for all $a_1, \ldots, a_n, b_1, \ldots, b_n \in \mathbb{R}$. For a proof, see [PM91, theorem 6.1], for instance.

0.3 Euclidean space

For any non-negative integer n, we write $\mathbb{R}^n$ to denote the set of lists of n real numbers. We will sometimes write its elements as horizontal lists of numbers separated by commas, as in

$$(h_1, \ldots, h_n),$$

and sometimes as vertical column of numbers, as in

$$\begin{bmatrix} h_1 \\ \vdots \\ h_n \end{bmatrix}.$$

0.3.A Linear structure on $\mathbb{R}^n$

Given $v_1, v_2 \in \mathbb{R}^n$, we define their sum $v_1 + v_2$ by adding the entries coordinate-wise. Given $\lambda \in \mathbb{R}$ and $v \in \mathbb{R}$, we define λv by multiplying each entry of v by λ. This endows $\mathbb{R}^n$ with the structure of a vector space.

For each $i = 1, \ldots, n$, we define the i*th standard basis vector*, denoted e_i, to be the list whose ith entry is 1 and all other entries are 0.

$$\begin{aligned} e_1 &= (1, 0, 0, \ldots, 0, 0) \\ e_2 &= (0, 1, 0, \ldots, 0, 0) \\ &\vdots \\ e_n &= (0, 0, 0, \ldots, 0, 1) \end{aligned}$$

Observe that

$$(h_1, \ldots, h_n) = h_1 e_1 + h_2 e_2 + \cdots + h_n e_n.$$

Thus the list $e_1, \ldots, e_n$ is a basis for $\mathbb{R}^n$.

For any $i = 1, \ldots, n$, we let $\pi_i : \mathbb{R}^n \to \mathbb{R}$ denote the ith *projection map*, given by

$$\pi_i(h_1, \ldots, h_n) = h_i.$$

This is a linear map.

0.3.B Norms on $\mathbb{R}^n$

The L^2-norm (cf. example 0.2.8) with respect to the standard basis $e_1, \ldots, e_n$ of $\mathbb{R}^n$ is the usual norm we use on $\mathbb{R}^n$. Explicitly, if $v = (h_1, \ldots, h_n) \in \mathbb{R}^n$, we define

$$|v| := \sqrt{h_1^2 + \cdots + h_n^2}.$$

Unless explicitly specified otherwise, we regard $\mathbb{R}^n$ as a metric space with the metric induced by this norm (cf. exercise 0.2.2).

On the other hand, we could also endow $\mathbb{R}^n$ with the max norm (cf. example 0.2.9) with respect to the standard basis. We denote this norm by $|-|_{\max}$. Explicitly, if $v = (h_1, \ldots, h_n) \in \mathbb{R}^n$, we have

$$|v|_{\max} = \max\{|h_1|, \ldots, |h_n|\}.$$

This also defines a norm in the sense of definition 0.2.1.

0.4 Little-oh notation

Suppose X is a metric space[2] and $x_0 \in X$ is a point.

Definition 0.4.1. A function $g : X \setminus \{x_0\} \to \mathbb{R}$ is *positive* if $g(x) > 0$ for all $x \in X \setminus \{x_0\}$. Similarly, g is *non-negative* if $g(x) \geqslant 0$ for all $x \in X \setminus \{x_0\}$.

Suppose g is a positive function. Definition 0.4.2 below formulates what it means for a non-negative function $f : X \setminus \{x_0\} \to \mathbb{R}$ to be "little-oh of g as x tends to x_0." Intuitively, this condition means that f is *much smaller* than g near x_0.

Definition 0.4.2. A function $f : X \setminus \{x_0\} \to \mathbb{R}$ is *little-oh of* g *as* x *tends to* x_0, written

$$f(x) = o(g(x)) \text{ as } x \to x_0$$

if, for every $\epsilon > 0$, there exists an open neighborhood U of x_0 such that $f(x) \leqslant \epsilon g(x)$ for all $x \in U \setminus \{x_0\}$. Equivalently, this means that

$$\lim_{x \to x_0} \frac{|f(x)|}{g(x)} = 0.$$

When x_0 can be inferred from context, we write simply $f = o(g)$.

[2] Even more generally, X could be a topological space here. See section 0.7.

Remark 0.4.3. It's worth noting that the use of the symbol "=" above is mathematically abusive. The left-hand side of the "=" is a function and the right-hand side is a property of functions; of course, a function cannot literally be equal to a property. Rather, the "=" is being used here to mean that the function on the left-hand side *has* the property described on the right-hand side. As annoying as it is, this abuse of notation is fairly standard, so it's probably best to just get used to it.

Exercise 0.4.4. Prove that the two conditions in definition 0.4.2 are in fact equivalent.

Exercise 0.4.5. Let $X = \mathbb{R}$ and $x_0 = 0$. For each of the functions f and g described below, sketch graphs of f and g and then determine whether or not $f = o(g)$ as $x \to 0$.

(a) $f(x) = x$ and $g(x) = x^2$.

(b) $f(x) = x$ and $g(x) = |x|$.

(c) $f(x) = x^2$ and $g(x) = |x|$.

Exercise 0.4.6. Let $X = \mathbb{R}^2$ and $x_0 = 0$. For each of the following functions f, determine whether or not $f(v) = o(|v|)$ as $x \to 0$.

(a) $f(h_1, h_2) = h_1 - h_2$.

(b) $f(h_1, h_2) = h_1^2 + h_2^2$.

Exercise 0.4.7. For a fixed positive function $g : X \setminus \{x_0\} \to \mathbb{R}$, prove that the set of functions $f : X \setminus \{x_0\} \to \mathbb{R}$ which are $o(g)$ as $x \to x_0$ is a vector space under the natural operations. In other words, verify the following three facts.

(1) ("Zero is small") The zero function is $o(g(x))$.

(2) ("Scalar multiples of small are still small") If $c \in \mathbb{R}$ is a scalar and $f(x) = o(g(x))$, then

$$cf(x) = o(g(x)).$$

(3) ("Sum of smalls is still small") If $f_1(x) = o(g(x))$ and $f_2(x) = o(g(x))$, then

$$(f_1 + f_2)(x) = o(g(x)).$$

Exercise 0.4.8 ("Smaller than small is still small")**.** Suppose $f_1, f_2 : X \setminus \{x_0\} \to \mathbb{R}$ are functions such that $|f_1(x)| \leqslant |f_2(x)|$ for all x in a punctured neighborhood of x_0, and $f_2(x) = o(g(x))$ as $x \to x_0$. Then $f_1(x) = o(g(x))$ also.

0.5 Matrices

The following is a whirlwind review of some facts about matrices, mostly intended to establish notation. The reader is expected to remember definitions of matrix multiplication, determinants, and minors. Few proofs are included in this section; for more details, you are encouraged to reference a dedicated linear algebra text.

Definition 0.5.1. We write $\mathcal{M}_{n,m}$ for the set of all $n \times m$ matrices (ie, the matrices with n rows and m columns). This is naturally a vector space, where addition and scalar multiplication are defined entrywise.

Definition 0.5.2. Let GL_n denote the subset of $\mathcal{M}_{n\times n}$ consisting of invertible $n \times n$ matrices.

Lemma 0.5.3. *Let A be a matrix. Then all of the following numbers are equal.*

(1) The dimension of the span of the columns of A.

(2) The dimension of the span of the rows of A.

(3) The size of largest nonzero minor of A.

This integer is called the rank *of A, and is denoted* $\mathrm{rank}(A)$.

0.5.A Matrix representations

Definition 0.5.4. If V and W are both vector spaces, we write $\mathcal{L}(V, W)$ for the set of all linear maps $V \to W$. This is naturally a vector space.

Definition 0.5.5. If V is a vector space, let $\mathrm{GL}(V)$ denote the set of invertible linear maps $V \to V$, regarded as a subset of $\mathcal{L}(V, V)$.

Throughout the rest of this section, we assume that U, V, and W are finite dimensional vector spaces.

Definition 0.5.6. Suppose B denotes a basis $v_1, \ldots, v_n$ for V. Then any $v \in V$ can be written uniquely as $a_1 v_1 + \cdots + a_n v_n$ for some scalars $a_1, \ldots, a_n \in \mathbb{R}$, and we define the *representation of v with respect to* B, denoted $[v]_B$, to be the column vector with n entries that records the scalars $a_1, \ldots, a_n$. In other words,

$$[v]_B = \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix}.$$

Lemma 0.5.7. *If* B *denotes a basis* $v_1, \ldots, v_n$ *is a basis for* V*, then the function* $V \to \mathbb{R}^n$ *given by* $v \mapsto [v]_B$ *is an isomorphism, with inverse given by*

$$\begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix} \mapsto a_1 v_1 + \cdots + a_n v_n.$$

Definition 0.5.8. Suppose $\ell : V \to W$ is a linear map. Let B and C denote bases $v_1, \ldots, v_m$ and $w_1, \ldots, w_n$ for V and W, respectively. The *matrix representation of* ℓ *with respect to* B *and* C, denoted $[\ell]_{B,C}$, is the $n \times m$ matrix whose ith column of $[\ell]_{B,C}$ is $[\ell(v_i)]_C$. In other words,

$$[\ell]_{B,C} = \begin{bmatrix} [\ell(v_1)]_C & \cdots & [\ell(v_m)]_C \end{bmatrix}.$$

Lemma 0.5.9. *Suppose that* V *and* W *are* m*- and* n*-dimensional vector spaces, respectively, and that and* B *and* C *are bases for* V *and* W*, respectively. Then the function* $\ell \mapsto [\ell]_{B,C}$ *is an isomorphism*

$$\mathcal{L}(V, W) \longrightarrow \mathcal{M}_{n,m}.$$

Lemma 0.5.10. *If* $\ell : V \to W$ *is a linear map between finite dimensional vector spaces and* B *and* C *are bases for* V *and* W*, respectively, then*

$$[\ell]_{B,C}[v]_B = [\ell(v)]_C$$

for any $v \in V$.

Lemma 0.5.11. *If* $\ell : U \to V$ *and* $\ell' : V \to W$ *are linear maps, and* A, B *and* C *are bases for* U, V*, and* W*, respectively, then*

$$[\ell' \circ \ell]_{A,C} = [\ell']_{B,C}[\ell]_{A,B}.$$

Definition 0.5.12. If $\ell : V \to W$ is a linear map, then the *rank* of ℓ, denoted $\operatorname{rank}(\ell)$ is defined to be the dimension of the range of ℓ.

Lemma 0.5.13. *If* $\ell : V \to W$ *is a linear map and* B *and* C *are bases for* V *and* W*, respectively, then*

$$\operatorname{rank}(\ell) = \operatorname{rank}[\ell]_{B,C}.$$

Definition 0.5.14. If $\ell : \mathbb{R}^m \to \mathbb{R}^n$ is a linear map, the *standard matrix representation* of ℓ, denoted $[\ell]$, is the matrix of ℓ with respect to the standard bases on $\mathbb{R}^m$ and $\mathbb{R}^n$.

Lemma 0.5.15. *If* A *is a* $n \times m$ *matrix and* $\ell_A : \mathbb{R}^m \to \mathbb{R}^n$ *is the linear map* $\ell_A(v) = Av$*, then*

$$[\ell_A] = A.$$

0.5.B Matrix norm

Observe that $\mathcal{M}_{n\times m}$ is a finite dimensional vector space. In fact, $\dim \mathcal{M}_{n\times m} = mn$. If $E_{j,i}$ denotes the matrix whose (j,i)-entry is 1 and all other entries are 0 for $i = 1,\dots,m$ and $j = 1,\dots,n$, then the list of all of these mn matrices forms a basis for $\mathcal{M}_{n\times m}$.

We can use this basis to define a L^2 and max norm on $\mathcal{M}_{n\times m}$, as in examples 0.2.8 and 0.2.9. Explicitly, if A is a $n \times m$ matrix whose (j,i)-entry is $a_{j,i}$, we have the following.

$$\|A\| = \sqrt{\sum_{j,i} a_{j,i}^2}$$
$$\|A\|_{\max} = \max_{j,i} |a_{j,i}|$$

We know from exercise 0.2.10 that these two norms are equivalent. All of the following statements are independent of which norm on $\mathcal{M}_{n\times m}$ we have in mind.

Lemma 0.5.16. *The function* $\det : \mathcal{M}_{n\times n} \to \mathbb{R}$ *is continuous.*

Proof. Determinants can be computed using cofactor expansions, so $\det A$ is a polynomial in the entries of A. □

Corollary 0.5.17. GL_n *is an open subset of* $\mathcal{M}_{n\times n}$.

Proof. Since $\det : \mathcal{M}_{n\times n} \to \mathbb{R}$ is continuous, and $\mathbb{R} \setminus \{0\}$ is an open subset of the codomain, we see that

$$GL_n = \det^{-1}(\mathbb{R} \setminus \{0\})$$

must be open in $\mathcal{M}_{n\times n}$. □

Lemma 0.5.18. *The function* $GL_n \to GL_n$ *given by* $A \mapsto A^{-1}$ *is a homeomorphism.*

Proof. This function is its own inverse, so it is sufficient to show that $A \mapsto A^{-1}$ is continuous. But

$$A^{-1} = \det(A)^{-1} \operatorname{adj}(A),$$

where $\operatorname{adj}(A)$ denotes the adjugate matrix of A. We know that $\det A$ is a polynomial in the entries of A. Each entry of $\operatorname{adj}(A)$ is a minor of A and is therefore also a polynomial in the entries of A. The lemma follows. □

0.6 Operator norm

0.6.A Basics

Definition 0.6.1. Suppose $\ell : \mathbb{R}^m \to \mathbb{R}^n$ is a linear map. The *operator norm* of ℓ, denoted $\|\ell\|$, is defined to be

$$\|\ell\| = \sup\{|\ell(v)| : |v| \leqslant 1\}.$$

The operator norm $\|\ell\|$ is *not* the same as the norm of the standard matrix representation $\|[\ell]\|$ as defined in section 0.5.B. The following gives an important reformulation of definition 0.6.1.

Exercise 0.6.2. Suppose $\ell : \mathbb{R}^m \to \mathbb{R}^n$ is a linear map. Show that

$$\|\ell\| = \inf\{\lambda \in \mathbb{R} : |\ell(v)| \leqslant \lambda|v| \text{ for all } v \in \mathbb{R}^m\}.$$

In particular, this means that $|\ell(v)| \leqslant \|\ell\| \cdot |v|$ for all $v \in \mathbb{R}^m$.

Here are some fundamental properties of the operator norm.

Exercise 0.6.3. Show that the operator norm is a norm on $\mathcal{L}(\mathbb{R}^m, \mathbb{R}^n)$ in the sense of definition 0.2.1.

Lemma 0.6.4. *Suppose $\ell : \mathbb{R}^m \to \mathbb{R}^n$ is a nonzero linear map. Then there exists $v_0 \in \mathbb{R}^m$ such that $|v_0| = 1$ and $|\ell(v_0)| = \|\ell\|$.*

Proof. The "unit disk"

$$D = \{x \in \mathbb{R}^m : |x| \leqslant 1\}$$

is a nonempty, connected, and compact subset of $\mathbb{R}^m$. The function $v \mapsto |\ell(v)|$ is also a continuous function $D \to \mathbb{R}$, so its image $|\ell(D)|$ must also be a nonempty, connected, and compact subset of $\mathbb{R}$. In other words, it must be a closed interval in $\mathbb{R}$. This means that there exists $v_0 \in D$ such that

$$|\ell(v_0)| = \sup|\ell(D)| = \|\ell\|.$$

Since $\ell \neq 0$, we know $\|\ell\| \neq 0$ from exercise 0.6.3, which in turn means that $v_0 \neq 0$. To prove that $|v_0| = 1$, assume for a contradiction that $|v_0| < 1$, and let $c = 1/|v_0| > 1$. Then $|cv_0| = |c||v_0| = 1$, so $cv_0 \in D$. But then

$$|\ell(cv_0)| = |c\ell(v_0)| = c|\ell(v_0)| > |\ell(v_0)| = \sup|\ell(D)|,$$

which is absurd. □

Lemma 0.6.5. *Suppose $\ell : \mathbb{R}^m \to \mathbb{R}^n$ is a linear map and $|\ell(v)| = o(|v|)$ as $v \to 0$. Then $\ell = 0$.*

Proof. The fact that $|\ell(v)| = o(|v|)$ tells us that

$$\lim_{v \to 0} \frac{|\ell(v)|}{|v|} = 0. \tag{0.6.6}$$

Suppose for a contradiction that $\ell \neq 0$. Using lemma 0.6.4, choose $v_0 \in \mathbb{R}^m$ such that $|v_0| = 1$ and $|\ell(v_0)| = \|\ell\|$. For scalars c, observe that $cv_0 \to 0$ as $c \to 0$, which means that equation (0.6.6) implies that

$$\lim_{c \to 0} \frac{|\ell(cv_0)|}{|cv_0|} = 0.$$

But observe that, for nonzero c, we have

$$\frac{|\ell(cv_0)|}{|cv_0|} = \frac{|c\ell(v_0)|}{|cv_0|} = \|\ell\|$$

so

$$0 = \lim_{c \to 0} \frac{|\ell(cv_0)|}{|cv_0|} = \lim_{c \to 0} \|\ell\| = \|\ell\|.$$

This contradicts exercise 0.6.3. □

0.6.B Operator norm as a metric ⋆

Since the operator norm is a norm by exercise 0.6.3, we can regard $\mathcal{L}(\mathbb{R}^m, \mathbb{R}^n)$ as a metric space by exercise 0.2.2. It turns out that many natural maps are continuous when we do this. For example, we have already seen that addition and scalar multiplication are continuous in exercises 0.2.3 and 0.2.4.

Exercise 0.6.7 ("Evaluation is continuous")**.** Suppose $v_0 \in \mathbb{R}^m$ is a fixed vector. Then the "evaluate at v_0" function $\mathcal{L}(\mathbb{R}^m, \mathbb{R}^n) \to \mathbb{R}^n$ given by $\ell \mapsto \ell(v_0)$ is continuous.

Possible hint. Since the "evaluate at v_0" function is linear, it is sufficient to check continuity at 0.

Exercise 0.6.8 (Submultiplicativity)**.** Suppose $\ell : \mathbb{R}^m \to \mathbb{R}^n$ and $\ell' : \mathbb{R}^n \to \mathbb{R}^p$ are both linear maps. Show that

$$\|\ell' \circ \ell\| \leqslant \|\ell'\| \cdot \|\ell\|.$$

Give an example to show that this inequality can be strict.

Exercise 0.6.9 ("Composition is continuous")**.** Show that the map

$$\mathcal{L}(\mathbb{R}^n, \mathbb{R}^p) \times \mathcal{L}(\mathbb{R}^m, \mathbb{R}^n) \longrightarrow \mathcal{L}(\mathbb{R}^m, \mathbb{R}^p)$$

given by $(\ell', \ell) \mapsto \ell' \circ \ell$ is continuous.

Exercise 0.6.10. Suppose $\ell : \mathbb{R}^m \to \mathbb{R}^n$ is linear and $A = [\ell]$ is its standard matrix representation. Prove the following.

(a) $\|\ell\| \leqslant \|A\| \leqslant \sqrt{n}\|\ell\|$.

(b) $\|A\|_{\max} \leqslant \|\ell\| \leqslant \sqrt{mn}\|A\|_{\max}$.

Exercise 0.6.11. Show that the isomorphism $\mathcal{L}(\mathbb{R}^m, \mathbb{R}^n) \to \mathcal{M}_{n\times m}$ of lemma 0.5.9, given by $\ell \mapsto [\ell]$, is a homeomorphism.

Under the homeomorphism $\mathcal{L}(\mathbb{R}^n, \mathbb{R}^n) \to \mathcal{M}_{n\times n}$ of exercise 0.6.11, the set $\mathrm{GL}(\mathbb{R}^n)$ of invertible linear maps $\mathbb{R}^n \to \mathbb{R}^n$ corresponds to the set GL_n of invertible $n \times n$ matrices; since GL_n is an open subset of $\mathcal{M}_{n\times n}$ by corollary 0.5.17, we conclude that $\mathrm{GL}(\mathbb{R}^n)$ is an open subset of $\mathcal{L}(\mathbb{R}^n, \mathbb{R}^n)$. Similarly, the following is a consequence of combining lemma 0.5.18 and exercise 0.6.11.

Exercise 0.6.12. Show that the map $\mathrm{GL}(\mathbb{R}^n) \to \mathrm{GL}(\mathbb{R}^n)$ given by $\ell \mapsto \ell^{-1}$ is a homeomorphism.

0.7 Topological spaces

This is a bare-bones introduction to topological spaces, intended for readers who have seen metric spaces but not topological spaces before.

0.7.A Basics

Definition 0.7.1. Let X be a set. A *topology* on X is a set τ_X of subsets of X, called *open subsets*, which satisfy the following axioms.

(T1) $\emptyset$ and X are both open subsets.

(T2) The union of an arbitrary collection of open subsets is open.

(T3) The intersection of a finite collection of open subsets is open.

A pair (X, τ_X) consisting of a set X and a topology τ on X is called a *topological space*. Often, we write simply "X" in place of the pair (X, τ_X).

Topological spaces come up throughout mathematics, and in general can be very very bizarre. We will not delve into the general theory of topological spaces, and won't see any of these bizarre examples. Instead, you are encouraged to content yourself with the following examples.

Example 0.7.2 (Metric spaces)**.** If X is a metric space, the collection of subsets of X that are open with respect to the metric defines a topology on X. Thus, every metric space can be regarded as a topological space in a natural way. It's worth noticing that this process of regarding a metric space as a topological space "forgets information," since equivalent metrics on a set will define the same topology.

Example 0.7.3 (Quotient spaces)**.** Suppose Y is any topological space (for instance, one that comes from a metric space), and let $\sim$ be an equivalence relation on Y. For an element $y \in Y$, let $[y]$ denote its equivalence class, and let $X = Y/\sim$ denote the set of all equivalence classes. There is a natural function $\pi : Y \to X$ given by sending a point $y \in Y$ to its equivalence class $[y]$. We define a topology on X, called the *quotient topology*, by declaring a subset $U \subseteq X$ to be open if and only if its preimage

$$\pi^{-1}(U) = \{y \in Y : [y] \in U\}$$

is open in Y.

Example 0.7.4 (Product spaces)**.** Suppose X and Y are topological spaces. We define a topology τ on the cartesian product $X \times Y$ by declaring a subset U to be open if it is a union of sets of the form $V \times W$ where U and V are open subsets of X and Y, respectively. This is called the *product topology* on $X \times Y$.

Example 0.7.5. Suppose V is a finite dimensional vector space. Then all norms on V are equivalent by theorem 0.2.11, so they all define the same topology on V by example 0.7.2 and exercise 0.2.7. This topology is called the *canonical topology* on V. Unless explicitly specified otherwise, we will always regard finite dimensional vector spaces as topological spaces using the canonical topology.

We now make a few further definitions that readers are invited to compare against the definitions they are familiar with from their experience with metric spaces.

Definition 0.7.6 (Closed subsets)**.** If X is a topological space, a subset F is *closed* if its complement U is *open*.

Definition 0.7.7 (Continuous functions)**.** If X and Y are topological spaces, a function $f : X \to Y$ is *continuous* if $f^{-1}(U)$ is an open subset of X whenever U is an open subset of Y.

Example 0.7.8. If Y is a topological space, $\sim$ an equivalence relation on Y, and $X = Y/\sim$ the quotient space as in example 0.7.3 above, then the map $\pi : Y \to X$ which carries a point $y \in Y$ to its equivalence class is continuous. This follows immediately from the definition of the quotient topology.

Exercise 0.7.9. If $f : X \to Y$ and $g : Y \to Z$ are continuous functions between topological spaces, then $g \circ f$ is also continuous.

Example 0.7.10. If V is a finite dimensional vector space, then the addition map $V \times V \to V$ and the scalar multiplication map $\mathbb{R} \times V \to V$ are continuous functions (using the product topology of example 0.7.4 on the domains). This statement is precisely the same as exercises 0.2.3 and 0.2.4.

Example 0.7.11. Suppose $\ell : V \to W$ is a linear map between two finite dimensional vector spaces. Then ℓ is automatically continuous.

Definition 0.7.12 (Homeomorphisms)**.** Suppose X and Y are topological spaces and $f : X \to Y$ is a function. Then f is a *homeomorphism* if it is bijective, continuous, and $f^{-1} : Y \to X$ is continuous. More generally, f is a *homeomorphism onto its image* if is injective, continuous, and $f^{-1} : f(X) \to X$ is also continuous.

Definition 0.7.13 (Subspace)**.** If X is a topological space and S is a subset, then the *subspace topology* on S is

$$\tau_S := \{S \cap U : U \in \tau_X\}.$$

Further, S equipped with the subspace topology is called a *subspace* of X.

Exercise 0.7.14. If X is a topological space and S is a subspace, prove that the inclusion map $i : S \to X$ is continuous.

0.7.B Hausdorff spaces $\star$

Definition 0.7.15 (Hausdorff)**.** A topological space X is *Hausdorff* if, for every pair of distinct points $x, y \in X$, there exist disjoint open subsets U and V containing x and y, respectively.

Example 0.7.16. If X is a metric space, the corresponding topological space as in example 0.7.2 is automatically Hausdorff.

However, quotient spaces need not be Hausdorff.

Exercise 0.7.17. Let $Y = \mathbb{R} \times \{\pm 1\}$ inside $\mathbb{R}^2$, and define an equivalence relation $\sim$ by declaring that $(a, 1) \sim (a, -1)$ for all $a \neq 0$. Show that $X = Y/\sim$ is not Hausdorff.

0.7.C Compact and σ-compact spaces $\star$

Definition 0.7.18 (Open covers)**.** An *open cover* of a topological space X is a collection $\mathcal{U}$ of open subsets of X which cover X, in the sense that

$$\bigcup_{U \in \mathcal{U}} U = X.$$

A *subcover* of $\mathcal{U}$ is a subset $\mathcal{U}' \subseteq \mathcal{U}$ which is itself an open cover.

Definition 0.7.19 (Compact)**.** A topological space X is *compact* if every open subcover has a finite subcover.

Definition 0.7.20 (σ-compact)**.** A topological space X is σ-*compact* if it is a countable union of compact subspaces.

Example 0.7.21. $\mathbb{R}^n$ is σ-compact, since

$$\mathbb{R}^n = \bigcup_{k=1}^{\infty} [-k, k]^n$$

and each hypercube $[-k, k]^n$ is compact.

1 Single variable derivatives

The goal in this chapter is to formalize derivatives of functions whose input is a single real number and whose output is also a single real number.

1.1 Two definitions of the derivative

1.1.A Slope of the tangent line

Let S be a subset of $\mathbb{R}$ and $a \in S$ an interior point. We would like to define the tangent line to the graph of a function $f : S \to \mathbb{R}$ at a. We know intuitively what "tangent line" should mean: a line that "just barely touches" the graph of f at the point $(a, f(a))$. But it's not immediately clear how to formalize this intuition.

One approach to formalization is to notice that our intuitive idea of "tangent line" can be approximated by secant lines, and secant lines can be formalized without ambiguity. More precisely, for small values of h, the point $a + h$ is close to a and still in S since a is an interior point of S. The slope of the secant line passing through the two points $(a, f(a))$ and $(a + h, f(a + h))$ is computed by the quantity

$$\frac{f(a+h) - f(a)}{h}.$$

This quantity is often called a *difference quotient*. See figure 1.1.1 for a geometric explanation of this difference quotient. By letting $h \to 0$, we get closer and closer to what we intuitively understand to be the slope of the tangent line.

Definition 1.1.2. A function $f : S \to \mathbb{R}$ is *differentiable* at an interior point $a \in S$ if

$$\lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$$

exists. If this limit does exist, its value is called the *derivative of* f *at* a. We will usually denote this derivative as $f'(a)$, but sometimes also as one of the following, where x is the variable that is being used to denote the argument to f.

$$\left.\frac{df}{dx}\right|_{x=a} \qquad \left.\frac{d}{dx}f(x)\right|_{x=a} \qquad \frac{df}{dx}(a)$$

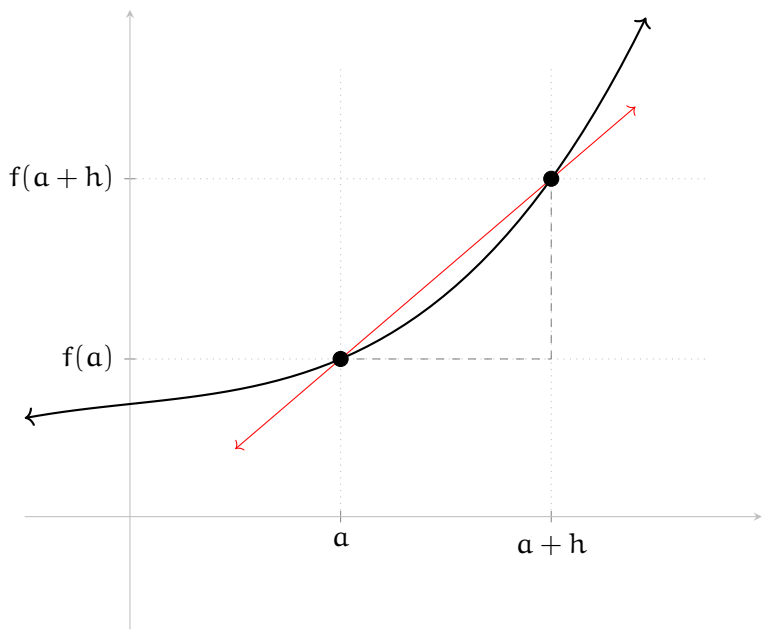

Figure 1.1.1: The graph of a function f is depicted in black. The secant line passing through the two points $(a, f(a))$ and $(a+h, f(a+h))$ is depicted in red. The "run" between these points, is h. The "rise" between these points is $f(a+h) - f(a)$. Thus the slope of the secant line, computed as "rise over run," is precisely the difference quotient $(f(a+h) - f(a))/h$.

Exercise 1.1.3. Use the definition of the derivative to determine whether or not each of the following functions f is differentiable at the given point a. If it is differentiable, calculate $f'(a)$. Check your answers by making sure they agree with geometric intuition and/or rules you learned during your introductory calculus course.

(a) $f(x) = |x|$ at $a = 0$.

(b) $f(x) = x$ at $a = 0$.

(c) $f(x) = x^2$ at $a = 0$.

(d) $f(x) = x^2$ at $a = 2$.

(e) $f(x) = \begin{cases} 1 & x \geqslant 0 \\ 0 & x < 0 \end{cases}$ at $a = 0$.

Exercise 1.1.4 (Power rule for positive integer exponents)**.** Prove that, if $f(x) = x^n$ for a positive integer n, then $f'(a) = na^{n-1}$.

Possible hint. Recall the binomial theorem, which says that

$$(a+h)^n = \sum_{k=0}^{n} \binom{n}{k} a^k h^{n-k}.$$

Exercise 1.1.5 (Differentiability implies continuity)**.** Prove that if $f : S \to \mathbb{R}$ is differentiable at an interior point $a \in U$, then f is also continuous at a.

We now make the following definition.

Definition 1.1.6. We say that a function $f : S \to \mathbb{R}$ is *differentiable* if f is continuous on S and differentiable at every interior point $a \in S$. Letting S° denote the interior of S, the function $S^\circ \to \mathbb{R}$ defined by $a \mapsto f'(a)$ is called the *derivative* of f. We will usually denote this function by f', but occasionally we will also use the notation df/dx, where again x is the variable that is being used to denote the argument to f.

Note that exercise 1.1.5 makes the continuity on S° automatic.

Caution. Some authors don't define what it means for a function to be differentiable on a non-open subset of $\mathbb{R}$ at all; some define it slightly differently than we have above. One should be cautious when talking about differentiability on a non-open subset of $\mathbb{R}$, because there is potential for misunderstanding.

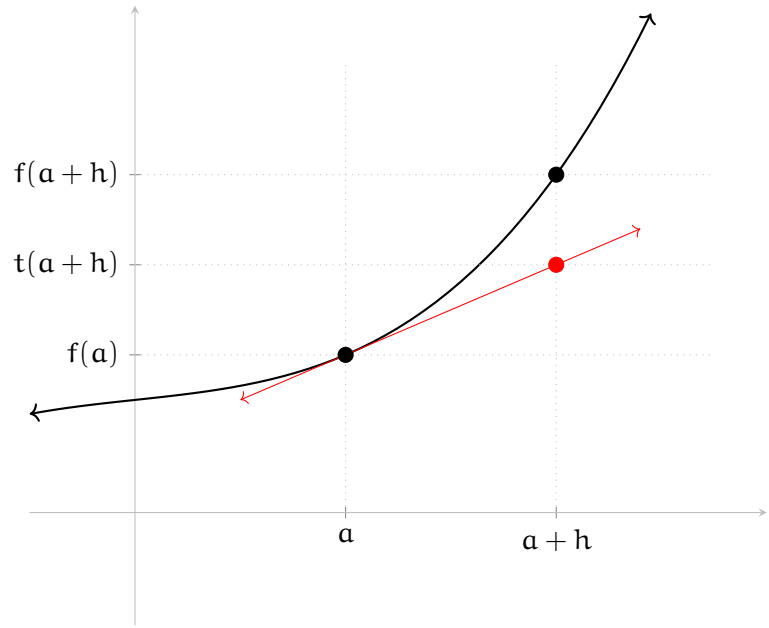

Figure 1.1.7: The graph of a function f is depicted in black, and its tangent line t at a is depicted in red. Then the vertical distance $f(a+h) - f(a)$ is approximated by the vertical distance $t(a+h) - t(a)$. Since the tangent line has slope $f'(a)$, this "rise" can be computed as $f'(a)$ times the "run" (in other words, $f'(a)h$). As h gets smaller, this approximation gets better.

1.1.B Best linear approximation

Before proceeding further, we reinterpret the definition of the derivative in a way that will be better suited to generalization to the multivariable setting. Suppose a function $f : S \to \mathbb{R}$ is differentiable at an interior point $a \in S$. The tangent line to the graph at the point $(a, f(a))$ is the graph of the function

$$t(x) = f'(a)(x - a) + f(a).$$

Since this function t is "close" to f near a, we should have that

$$f(a+h) - f(a) \approx t(a+h) - t(a) = f'(a)h$$

for small values of h. See figure 1.1.7.

Notice that the function $h \mapsto f'(a)h$ is a linear map $\mathbb{R} \to \mathbb{R}$ in the sense of linear algebra. One way to interpret the following result is that the function $h \mapsto f'(a)h$ is the *only* linear

map that is a good approximation to $h \mapsto f(a+h) - f(a)$. The precise statement of the result uses little-oh notation; if you have not seen this notation before, see definition 0.4.2.

Proposition 1.1.8. *A function $f : S \to \mathbb{R}$ is differentiable at an interior point $a \in S$ if and only if there exists a linear map $\ell : \mathbb{R} \to \mathbb{R}$ such that*

$$|f(a+h) - f(a) - \ell(h)| = o(|h|) \text{ as } h \to 0.$$

Moreover, if there exists such a linear map ℓ, then ℓ is uniquely determined: it must be given by

$$\ell(h) = f'(a)h$$

for all $h \in \mathbb{R}$.

Proof. Let ℓ be a linear map $\ell(h) = ch$, where c is some scalar. Observe that

$$\begin{aligned}\frac{|f(a+h) - f(a) - \ell(h)|}{|h|} &= \left|\frac{f(a+h) - f(a) - ch}{h}\right| \\ &= \left|\frac{f(a+h) - f(a)}{h} - c\right|.\end{aligned}$$

This leads to the following sequence of "if and only if" statements.

$$\begin{aligned}|f(a+h) - f(a) - \ell(h)| = o(|h|) &\iff \lim_{h\to 0} \frac{|f(a+h) - f(a) - \ell(h)|}{|h|} = 0 \\ &\iff \lim_{h\to 0} \left|\frac{f(a+h) - f(a)}{h} - c\right| = 0 \\ &\iff \lim_{h\to 0} \left(\frac{f(a+h) - f(a)}{h} - c\right) = 0 \\ &\iff \lim_{h\to 0} \frac{f(a+h) - f(a)}{h} = c\end{aligned}$$

The statement of the proposition follows. □

Definition 1.1.9. If $f : S \to \mathbb{R}$ is differentiable at an interior point $a \in U$, we define $df_a : \mathbb{R} \to \mathbb{R}$ to be the linear map $df_a(h) = f'(a)h$.

In other words, df_a is the only good linear approximation to the function

$$h \mapsto f(a+h) - f(a).$$

Being the *only good* linear approximation, it is also the *best* linear approximation, whence the name of this section.

The following is an important idea that we will come up frequently in proofs.

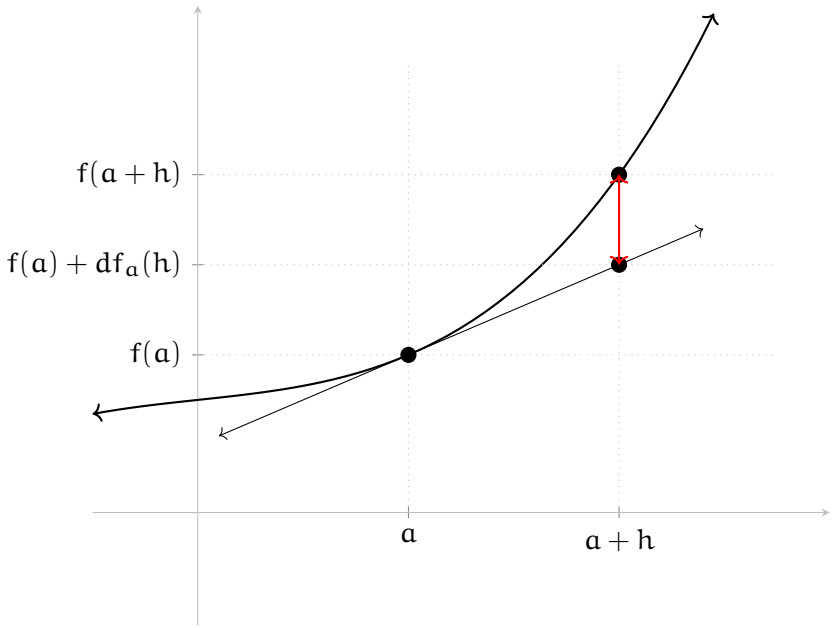

Figure 1.1.11: The thick black curve is the graph of a function f and the thinner black line is its tangent line $f(a) + df_a$ are depicted in black. On a small input h, the "error in approximation" function r outputs the vertical distance indicated in red.

Remark 1.1.10 ("Error in approximation" function)**.** Suppose $f : S \to \mathbb{R}$ is differentiable at an interior point $a \in S$. For sufficiently small values of h, we can define the "error in approximation" function

$$r(h) = f(a+h) - f(a) - df_a(h).$$

See figure 1.1.11. Rearranging the definition of r, we have

$$f(a+h) = f(a) + df_a(h) + r(h).$$

This equation just says that $f(a+h)$ is the sum of the value of the linear approximation (which is $f(a) + df_a(h)$) and the error (which is $r(h)$).

Notice that $r(0) = 0$. Moreover, since f is differentiable and therefore continuous at a by exercise 1.1.5, we can see that r is continuous at 0. Finally, r is "small." More precisely, we know that $|r(h)| = o(|h|)$ as $h \to 0$ by the discussion above, which means that

$$\lim_{h\to 0} \left| \frac{r(h)}{h} \right| = 0 \qquad \Longrightarrow \qquad \lim_{h\to 0} \frac{r(h)}{h} = 0.$$

As a first example of an application of this idea, here is a weak version of L'Hôpital's rule.

Exercise 1.1.12 (L'Hôpital's rule, weak version)**.** Suppose $f, g : S \to \mathbb{R}$ are differentiable at an interior point $a \in S$, that $f(a) = g(a) = 0$, and that $g'(a) \neq 0$. Show that

$$\lim_{x \to a} \frac{f(x)}{g(x)} = \frac{f'(a)}{g'(a)}.$$

Possible hint. If $x = a + h$, we have

$$f(x) = f(a + h) = f(a) + df_a(h) + r(h) = f'(a)h + r(h),$$

where r is the "error in approximation" function for f, as defined in remark 1.1.10.

1.2 Rules of differentiation

The goal in this section is to prove the rules for differentiation that one normally encounters in an introductory calculus course.

1.2.A Sum and scalar multiples rule

The following can be proved using either the definition of the derivative 1.1.2 or the equivalent "best linear approximation" characterization of proposition 1.1.8. My suggestion is that you try both methods for these exercises.

Exercise 1.2.1 (Constant functions rule)**.** Prove that, if $f : S \to \mathbb{R}$ is constant (ie, there exists a real number c such that $f(x) = c$ for all $x \in S$), then $f'(a) = 0$ for all interior points $a \in S$.

When you use proposition 1.1.8 to prove the following, you may find it useful to have done exercise 0.4.7.

Exercise 1.2.2 (Sum rule)**.** Prove that, if $f, g : S \to \mathbb{R}$ are both differentiable at an interior point $a \in U$, then $f + g$ is also differentiable at a and

$$(f + g)'(a) = f'(a) + g'(a).$$

Exercise 1.2.3 (Scalar multiples rule)**.** Prove that, if c is a constant and $f : S \to \mathbb{R}$ is differentiable at an interior point $a \in S$, then cf is also differentiable at a and

$$(cf)'(a) = cf'(a).$$

1.2.B Product and quotient rules

Proposition 1.2.4 (Product rule)**.** *Suppose* $f, g : U \to \mathbb{R}$ *are both differentiable at* $a \in U$*. Then* fg *is also differentiable at* a *and*

$$(fg)'(a) = f'(a)g(a) + f(a)g'(a).$$

Proof. The proof is a clever algebraic manipulation of difference quotients. The key trick is introducing a "cross term" of the form $f(a)g(a+h)$, and then cancelling it out by also introducing its negative. More precisely, for any sufficiently small real number h, we have

$$\begin{aligned}
\frac{(fg)(a+h) - (fg)(a)}{h} &= \frac{f(a+h)g(a+h) - f(a)g(a)}{h} \\
&= \frac{f(a+h)g(a+h) \color{blue}{- f(a)g(a+h) + f(a)g(a+h)} - f(a)g(a)}{h} \\
&= \frac{f(a+h)g(a+h) - f(a)g(a+h)}{h} + \frac{f(a)g(a+h) - f(a)g(a)}{h} \\
&= \frac{f(a+h) - f(a)}{h} \cdot g(a+h) + f(a) \cdot \frac{g(a+h) - g(a)}{h},
\end{aligned}$$

where the cross term and its negative is indicated in blue. Taking the limit as $h \to 0$ yields the result. □

Exercise 1.2.5. Check your understanding of the above proof of the product rule.

(a) Could you have instead introduced a cross term of the form $f(a+h)g(a)$? If so, rewrite the proof using this cross term. If not, explain why not.

(b) At what point is exercise 1.1.5 tacitly used in the above proof?

Exercise 1.2.6. Reprove the scalar multiples rule (exercise 1.2.3) using the product rule.

Exercise 1.2.7 (Quotient rule)**.** Suppose $f, g : S \to \mathbb{R}$ are both differentiable at an interior point $a \in U$, and that $g(a) \neq 0$. Prove that f/g is also differentiable at a and

$$(f/g)'(a) = \frac{f'(a)g(a) - f(a)g'(a)}{g(a)^2}.$$

Possible hint. Observe that

$$\frac{(f/g)(a+h) - (f/g)(a)}{h} = \frac{\frac{f(a+h)}{g(a+h)} - \frac{f(a)}{g(a)}}{h}.$$

Multiply both the numerator and denominator by $g(a+h)g(a)$ to clear the denominators inside the numerator, and then introduce the cross term $f(a)g(a)$. Be sure to notice at what point you use exercise 1.1.5.

Exercise 1.2.8 (Power rule for integer exponents)**.** Extend the result of exercise 1.1.4 to arbitrary integer exponents.

Possible hint. Use the quotient rule for negative integers. The exponent zero is a special but easy case; deal with it separately.

1.2.C Chain rule

Proposition 1.2.9 (Chain rule)**.** *Suppose that* S *and* T *are both subsets of* $\mathbb{R}$*, that* $f : S \to T$ *is differentiable at an interior point* a*, that* $f(a)$ *is an interior point of* T*, and that* $g : T \to \mathbb{R}$ *is differentiable at* $f(a)$*. Then the composite* $g \circ f : S \to \mathbb{R}$ *is also differentiable at* a*, and*

$$(g \circ f)'(a) = g'(f(a))f'(a).$$

Let us discuss two different (but similar) proofs of this important result.

First proof

The first proof uses the definition of the derivative 1.1.2, but there is a subtle technicality involved; so, let us discuss it informally first before diving into the formal proof.

The idea involved is again a clever rewriting of difference quotients. Again, we introduce a cross term, which in this is $f(a+h) - f(a)$. But this time we introduce this cross term "multiplicatively" rather than "additively." More precsiely, we have

$$\begin{aligned}\frac{(g \circ f)(a+h) - (g \circ f)(a)}{h} &= \frac{g(f(a+h)) - g(f(a))}{h} \\ &= \frac{g(f(a+h)) - g(f(a))}{f(a+h) - f(a)} \cdot \frac{f(a+h) - f(a)}{h}.\end{aligned}$$

The difference quotient on the right is precisely $f'(a)$. Thus, it would be sufficient to prove

$$\lim_{h \to 0} \frac{g(f(a+h)) - g(f(a))}{f(a+h) - f(a)} = g'(f(a)). \tag{1.2.10}$$

This formula looks an awful lot like the difference quotient that is used to define the derivative of g at $f(a)$. In fact, since f is continuous at a by exercise 1.1.5, we have $f(a+h) \approx f(a) + h$ for small values of h, which means that

$$\frac{g(f(a+h)) - g(f(a))}{f(a+h) - f(a)} \approx \frac{g(f(a)+h) - g(f(a))}{f(a)+h-f(a)} = \frac{(g \circ f)(a+h) - (g \circ f)(a)}{h}$$

for small values of h. Taking the limit as $h \to 0$ on the right-hand side yields precisely $g'(f(a))$, so equation (1.2.10) seems like a reasonable conjecture.

The problem is that equation (1.2.10) isn't true. Issues arise when $f(a+h) = f(a)$ for arbitrarily small values of h, which leads to an infinite sequence of divisions by 0.

Example 1.2.11. Let

$$f(x) = \begin{cases} x^2 \sin(1/x) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

and let $g(x) = x^2$. Observe that $g'(f(0)) = 0$. You will show later (in exercise 1.2.24) that f is in fact differentiable at 0, so these functions do satisfy the hypothesis of the chain rule. However, the limit as $h \to 0$ of

$$\frac{g(f(a+h)) - g(f(a))}{f(a+h) - f(a)} = \frac{(h^2 \sin(1/h))^2}{h^2 \sin(1/h)}$$

does not exist. For example, the sequence $h_k = 1/k\pi$ converges to 0 as $k \to \infty$, but the expression above is undefined along this sequence.

We will get around the kind of issue that is described by the above example by carefully studying the expression

$$\frac{g(f(a+h)) - g(f(a))}{f(a+h) - f(a)}. \tag{1.2.12}$$

Observe that, when $f(a+h) \neq f(a)$, then this expression is precisely the difference quotient that calculates the slope of the secant line passing through $(f(a), g(f(a))$ and $(f(a+h), g(f(a+h))$. When $f(a+h) = f(a)$, this secant line doesn't make sense; but what does make sense is the tangent line! So we'll remove the undefinedness of expression (1.2.12) by extending it so that it takes the value $g'(f(a))$ whenever $f(a+h) = f(a)$.

This leads us to our first proof of the chain rule.

First proof of the chain rule. Consider the function ϵ defined for small values of h by $\epsilon(h) = f(a+h) - f(a)$. See figure 1.2.13. You will verify in exercise 2.3.5 below that ϵ is continuous at 0.

Next, define a function σ by the following.

$$\sigma(k) = \begin{cases} \dfrac{g(f(a)+k) - g(f(a))}{k} & \text{if } k \neq 0 \\ g'(f(a)) & \text{if } k = 0 \end{cases}$$

In other words, σ is the function that, on some small nonzero input k, outputs the slope of the secant line connecting the two points on the graph of g where $x = f(a)$ and $x = f(a) + k$.

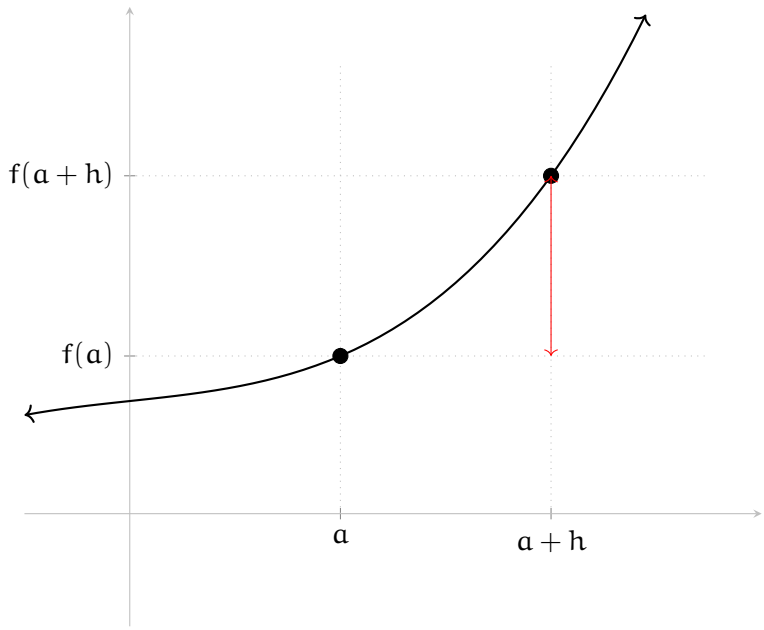

Figure 1.2.13: The graph of f is depicted in black. The value of the function t on a small nonzero input value h is the vertical distance depicted in red.

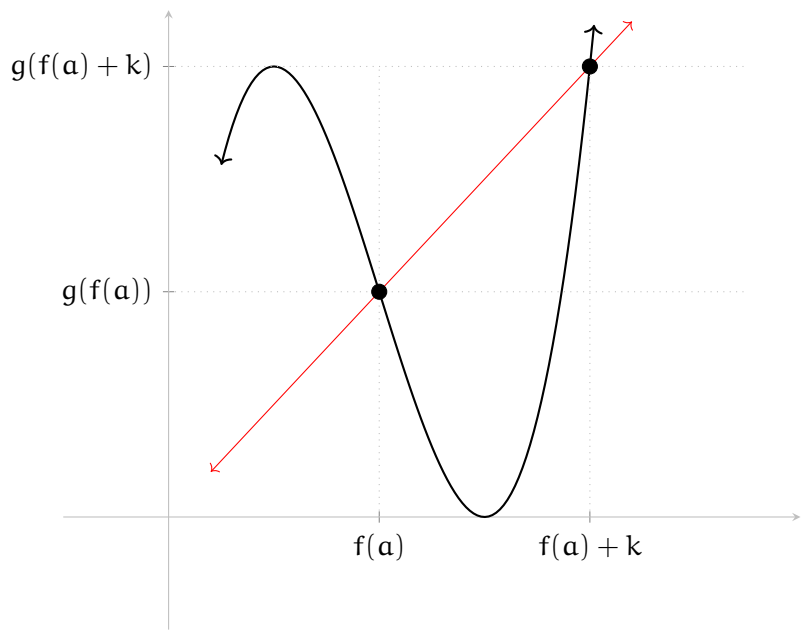

Figure 1.2.14: The graph of g is depicted in black. The output of the function σ on a small nonzero input value k is the slope of the red secant line passing through the two points $(f(a), g(f(a))$ and $(f(a)+k, g(f(a)+k))$.

See figure 1.2.14. It follows from the definition of the derivative of g at $f(a)$ that the function σ is continuous at 0.

Observe that

$$\frac{(g \circ f)(a+h) - (g \circ f)(a)}{h} = \sigma(\epsilon(h)) \cdot \frac{f(a+h) - f(a)}{h} \tag{1.2.15}$$

for all small values of h, as you will verify in exercise 2.3.5 below. Thus

$$\begin{aligned}\lim_{h \to 0} \frac{(g \circ f)(a+h) - (g \circ f)(a)}{h} &= \lim_{h \to 0} \sigma(\epsilon(h)) \cdot \frac{f(a+h) - f(a)}{h} \\ &\overset{(*)}{=} \sigma(\epsilon(0)) f'(a) \\ &= g'(f(a)) f'(a)\end{aligned}$$

where we have used the continuity of ϵ and σ and the definition of the derivative of f at a for the step labeled $(*)$ above. This proves the chain rule. □

Exercise 1.2.16. Verify the following facts used in the proof above.

(a) Prove that ϵ is continuous at 0.

(b) Prove equation (1.2.15).

Possible hint. You might split up your proof of equation (1.2.15) into two cases: when $f(a+h) \neq f(a)$, and when $f(a+h) = f(a)$.

Second proof

Our second proof of the chain rule will use the "best linear approximation" interpretation of derivatives provided by proposition 1.1.8.

Second proof of the chain rule. By proposition 1.1.8, it is sufficient to prove that

$$g(f(a+h)) - g(f(a)) - dg_{f(a)}(df_a(h)) = o(|h|) \text{ as } h \to 0.$$

Define the following "error in approximation" functions, as in remark 1.1.10.

$$\begin{aligned} r(h) &= f(a+h) - f(a) - df_a(h) \\ s(k) &= g(f(a)+k) - g(f(a)) - dg_{f(a)}(k) \end{aligned}$$

Since

$$f(a+h) = f(a) + df_a(h) + r(h)$$

by definition of r, we have

$$\begin{aligned} g(f(a+h)) &= g(f(a) + df_a(h) + r(h)) \\ &= g(f(a)) + dg_{f(a)}(df_a(h) + r(h)) + s(df_a(h) + r(h)) \\ &= g(f(a)) + dg_{f(a)}(df_a(h)) + dg_{f(a)}(r(h)) + s(df_a(h) + r(h)) \end{aligned}$$

where we use the definition of s on input $k = df_a(h) + r(h)$ for the second step, and the fact that $dg_{f(a)}$ is linear for the third. Thus

$$g(f(a+h)) - g(f(a)) - dg_{f(a)}(df_a(h)) = dg_{f(a)}(r(h)) + s(df_a(h) + r(h)).$$

We therefore want to prove that the right-hand side is $o(|h|)$.

Since

$$dg_{f(a)}(r(h)) = g'(f(a)) \cdot r(h)$$

is just a scalar multiple of r, and $r = o(|h|)$, we see that $dg_{f(a)}(r) = o(|h|)$ also. So it is sufficient to prove that

$$s(df_a(h) + r(h)) = o(|h|).$$

We've now hit the technical part, so let's hit pause.

To see why this is technical, and for essentially the same reasons as the previous proof, notice that we're trying to prove that

$$\lim_{h \to 0} \frac{|s(df_a(h) + r(h)|)}{|h|} = 0.$$

To prove this, it's tempting to introduce a multiplicative cross-term as follows.

$$\frac{|s(df_a(h) + r(h)|)}{|h|} = \frac{|s(df_a(h) + r(h)|)}{|df_a(h) + r(h)|} \cdot \frac{|df_a(h) + r(h)|}{|h|}$$

But notice that $df_a(h) + r(h) = f(a+h) - f(a)$ by definition of r, so this cross term is problematic for the same reasons as before! If $f(a+h) = f(a)$ for h arbitrary close to 0, we have an infinite sequence of divisions by zeroes that causes problems.

We'll deal with this problem in essentially the same way as before: namely, we extend the definition of

$$\frac{|s(df_a(h) + r(h)|)}{|df_a(h) + r(h)|}$$

so that it is also defined when $f(a+h) = f(a)$.

Second proof of the chain rule, continued. Define $\eta(k) = s(k)/k$ for nonzero k. If we set $\eta(0) = 0$, the fact that $s(k) = o(|k|)$ implies that η is continuous at 0. Then

$$\begin{aligned}\frac{|s(df_a(h) + r(h))|}{|h|} &= \frac{|\eta(df_a(h) + r(h))| \cdot |df_a(h) + r(h)|}{|h|} \\ &\leqslant |\eta(df_a(h) + r(h))| \cdot \left(|f'(a)| + \frac{|r(h)|}{|h|}\right).\end{aligned}$$

Observe that r and therefore $df_a + r$ is continuous at 0 (cf. remark 1.1.10). Moreover, we have just seen that η is continuous at $(df_a + r)(0) = 0$. Thus the composite $\eta \circ (df_a + r)$ is also continuous at 0. Thus

$$\lim_{h \to 0} |\eta(df_a(h) + r(h))| \cdot \left(|f'(a)| + \frac{|r(h)|}{|h|}\right) = 0,$$

where we have used the fact that $|r(h)| = o(|h|)$ to see that the parenthetical expression has a finite limit (namely, $|f'(a)|$). So, by the squeeze theorem, we can therefore conclude that $|s(df_a(h) + r(h))| = o(|h|)$. □

Notice that in the first proof, the trick to overcoming the technicality was defining the function σ at $k = 0$. In the second proof, the trick to overcoming the same technicality was defining η at $k = 0$. These two tricks are really the same trick, as shown by the following.

Exercise 1.2.17. Prove that the function η from the second proof and the function σ from the first proof are related by the equation

$$\sigma(k) = g'(f(a)) + \eta(k).$$

1.2.C.1 Applications of the chain rule

Exercise 1.2.18 (Derivatives of inverses)**.** Suppose S and T are both subsets of $\mathbb{R}$ and $f : S \to T$ is bijective, with inverse function $f^{-1} : T \to S$. Suppose further that f is differentiable at an interior point $a \in S$, that $f'(a) \neq 0$, and that $b = f(a)$ is an interior point of T. Prove that f^{-1} is differentiable at b and

$$(f^{-1})'(b) = \frac{1}{f'(a)} = \frac{1}{f'(f^{-1}(b))}.$$

Exercise 1.2.19 (Power rule for fractional exponents)**.** Extend the result of exercise 1.2.8 to arbitrary rational exponents (ie, to exponents that can be written in the form m/n where m and n are both integers).

1.2.D Some rules without proofs

In this section, we discuss differentiation of some common "transcendental functions," but without complete proofs. The proofs are omitted because they would involve long-ish detours away from the intended object of meditation: the derivative.

First off, here is the most general statement of the power rule, but you should note that exercise 1.2.19 already covers the most common and most important cases.

Proposition 1.2.20 (Power rule for real exponents)**.** *If* r *is any real number and* $f : (0, \infty) \to \mathbb{R}$ *is the function* $f(x) = x^r$*, then* f *is differentiable and* $f'(x) = rx^{r-1}$*.*

Proving this result requires having a rigorous definition of what x^r even means for irrational exponents r. This would be a long detour, so it is omitted. In fact, once we have a sufficiently rigorous definition of x^r for irrational r, it turns out that proposition 1.2.20 is actually a relatively straightforward *consequence* of exercise 1.2.19.

Next up, here is a fact about differentiating power series. The proof is omitted because it is rather technical.

Proposition 1.2.21. *Suppose the power series* $\sum_{n=0}^{\infty} a_n x^n$ *has radius of convergence* R*. Then the function* $f : (-R, R) \to \mathbb{R}$ *given by*

$$f(x) = \sum_{n=0}^{\infty} a_n x^n$$

is differentiable and

$$f'(x) = \sum_{n=0}^{\infty} n a_n x^{n-1}.$$

Moreover, the radius of convergence of the power series $\sum_{n=0}^{\infty} n a_n x^{n-1}$ *is also* R*.*

The exponential, sine and cosine functions are usually defined using power series. More precisely, we have the following.

$$\exp(x) = \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

$$\cos(x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}$$

$$\sin(x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}$$

All three of these power series have infinite radius of convergence, so the following three facts are immediate consequences of proposition 1.2.21.

(1) If $f(x) = \exp(x)$, then $f'(x) = \exp(x)$.

(2) If $f(x) = \sin(x)$, then $f'(x) = \cos(x)$.

(3) If $f(x) = \cos(x)$, then $f'(x) = -\sin(x)$.

Remark 1.2.22. Defining the exponential, sine, and cosine functions using power series is mathematically fine, but there are numerous facts about these functions that are not clear from the power series, and one needs to prove these facts separately. For example, it's useful knowing that, if $e := \exp(1)$, then $\exp(x) = e^x$. This fact is not immediately clear from the power series definition; it requires some proof, and we won't prove it here, but you should feel free to use it. It's obviously also useful to know that sines and cosines have something to do with trigonometry. Again, this is not clear from the power series definition; they require some proof, and we won't prove these facts, but you should feel free to use them.

We can now compute the derivative of the logarithm.

Exercise 1.2.23. If $f(x) = \ln(x)$, show that $f'(x) = 1/x$.

Possible hint. The natural logarithm $\ln : (0, \infty) \to \mathbb{R}$ is the inverse of the exponential function.

Here is some further practice with differentiation of transcendental functions.

Exercise 1.2.24. (a) Prove that the function

$$f(x) = \begin{cases} x \sin(1/x) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

is continuous, but that it is not differentiable at 0.

(b) Prove that the function

$$f(x) = \begin{cases} x^2 \sin(1/x) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

is differentiable, but that the derivative f' is not continuous at 0.

1.3 Derivatives, extremums, and monotonicity

1.3.A Derivatives and local extremums

We first recall the following definition.

Definition 1.3.1. If X is a metric space and $f : X \to \mathbb{R}$ is a function, a point $a \in X$ is said to be a *local maximum* of f if there exists a neighborhood U of a in X such that $f(x) \leqslant f(a)$ for all $x \in U$. We define *local minimum* by reversing the inequality. A *local extremum* is a point that is either a local maximum or a local minimum.

The following is a key property of derivatives that you probably remember from calculus.

Exercise 1.3.2 (Interior extremum theorem)**.** Suppose an interior point $a \in S$ is a local extremum of a function $f : S \to \mathbb{R}$. If f is differentiable at a, prove that $f'(a) = 0$.

Possible hint. Consider the following function (which we've seen before, during the first proof of the chain rule in section 1.2.C).

$$\sigma(h) = \begin{cases} \dfrac{f(a+h) - f(a)}{h} & \text{if } h \neq 0 \\ f'(a) & \text{if } h = 0 \end{cases}$$

Since f is differentiable at $x = a$, we know that σ is continuous at $h = 0$. If a is a local minimum, what can you say about the values of σ for nonzero values of h? What about if a is a local maximum?

But remember, the converse to the interior extremum theorem 1.3.2 is *not* true.

Exercise 1.3.3. Give an example of a function $f : \mathbb{R} \to \mathbb{R}$ where $f'(0) = 0$, but f does not have a local extremum at 0.

1.3.B Mean value theorem

The mean value theorem 1.3.5 is an extremely important foundational result; it comes up often during proofs. Rolle's theorem 1.3.4 is a special case of the mean value theorem, but the general case can be derived from this special case.

Exercise 1.3.4 (Rolle's theorem)**.** Suppose $a < b$ and let $f : [a, b] \to \mathbb{R}$ be a differentiable function. If $f(a) = f(b) = 0$, prove that there exists $c \in (a, b)$ such that $f'(c) = 0$.

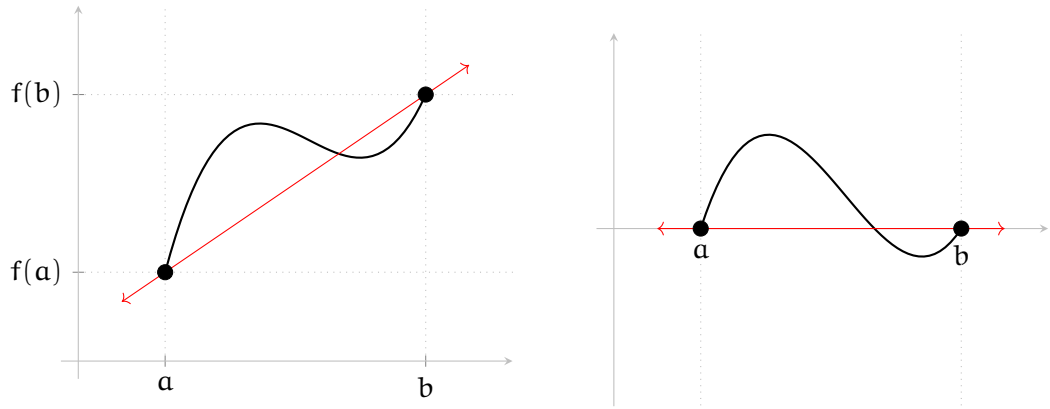

Figure 1.3.6: On the left is depicted the graph of a function f and the secant line ℓ. Subtracting ℓ from the picture on the left yields the picture on the right.

Possible hint. Any function must fall into at least one of the following three cases: (i) f is constantly zero, (ii) f takes on positive values somewhere on $[a, b]$, and (iii) f takes on negative values somewhere on $[a, b]$. Deal with the first case using exercise 1.2.1, and the latter two cases using the extreme value theorem[1] together with exercise 1.3.2.

Exercise 1.3.5 (Mean value theorem)**.** Suppose $a < b$ and let $f : [a, b] \to \mathbb{R}$ be a differentiable function. Prove that there exists $c \in (a, b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Possible hint. Let ℓ be the secant line passing through $(a, f(a))$ and $(b, f(b))$. Then consider the function $g = f - \ell$. See figure 1.3.6. Then $g(a) = g(b) = 0$, so we can apply Rolle's theorem 1.3.4 to g.

Here is the first of many, many consequences of the mean value theorem.

Exercise 1.3.7. Suppose I is an open interval and $f : I \to \mathbb{R}$ is a differentiable function such that there exists a number M such that $|f'(x)| \leqslant M$ for all $x \in I$. Show that $|f(b) - f(a)| \leqslant M|b - a|$ for all $a, b \in I$.

[1] The *extreme value theorem* says that if K is a compact metric space and $f : K \to \mathbb{R}$ is continuous, then f achieves its global maximum and its global minimum somewhere on K (cf. [PM91, theorems 3.17 and 6.30]).

1.3.C Derivatives and monotonicity

Here is a standard property of derivatives from calculus; the proof follows quickly from the mean value theorem.

Proposition 1.3.8. *Suppose* I *is an interval and that* $f : I \to \mathbb{R}$ *is differentiable. If* $f' = 0$*, then* f *is constant.*

Proof. Suppose we have $a < b$ are two distinct points in I. Since f is differentiable on I and $[a, b] \subseteq I$, we know in particular that f is also differentiable on $[a, b]$, so the mean value theorem applies: there exists a point $c \in (a, b)$ such that

$$\frac{f(b) - f(a)}{b - a} = f'(c).$$

But $f'(c) = 0$ by assumption, so $f(a) = f(b)$. □

Next, we recall the following definition.

Definition 1.3.9. Suppose $f : S \to \mathbb{R}$ is a function and $x, y \in S$.

- f is *increasing* if $x < y$ implies $f(x) \leqslant f(y)$.
- f is *decreasing* if $x < y$ implies $f(x) \geqslant f(y)$.
- f is *strictly increasing* if $x < y$ implies $f(x) < f(y)$.
- f is *strictly decreasing* if $x < y$ implies $f(x) > f(y)$.

Further, f is *monotonic* if it is either increasing or decreasing, and *strictly monotonic* if it is either strictly increasing or strictly decreasing.

The following standard property of differentiable functions again follows quickly from the mean value theorem.

Exercise 1.3.10. Suppose I is an interval and that $f : I \to \mathbb{R}$ is differentiable. Prove the following four statements.

(a) If $f' \geqslant 0$, then f is increasing.

(b) If $f' > 0$, then f is strictly increasing.

(c) If $f' \leqslant 0$, then f is decreasing.

(d) If $f' < 0$, then f is strictly decreasing.

Possible hint. For all four, imitate the proof of proposition 1.3.8.

1.3.D Darboux's theorem ⋆

Darboux's theorem is not commonly discussed in elementary calculus, nor is it strictly necessary for what follows. But it is an interesting fact about derivatives, and is sometimes useful to know. Roughly, the theorem says that derivatives are "close" to being continuous. Before getting to the precise statement of the theorem, we make the following definition.

Definition 1.3.11. Suppose I is an interval and $f : I \to \mathbb{R}$ is a function. Then f *has the intermediate value property* if, for every $a < b$ in I and every $y \in \mathbb{R}$ strictly between $f(a)$ and $f(b)$, there exists $x \in (a, b)$ such that $f(x) = y$.

Using this vocabulary, the intermediate value theorem asserts that every continuous function has the intermediate value property [PM91, theorem 3.3]. Simple examples of discontinuous functions don't have the intermediate value property (cf. exercise 1.3.12 below). But, it turns out that some discontinuous functions do have the intermediate value property (cf. exercise 1.3.13 below).

Exercise 1.3.12. Let I be an interval and $f : I \to \mathbb{R}$ a function.

(a) Recall that f has a *removable discontinuity* at a point $a \in I$ if

$$\lim_{x \to a} f(x)$$

exists, but the value of this limit does not equal $f(a)$. Show that, if f has a removable discontinuity, then f does *not* have the intermediate value property.

(b) Recall that f has a *jump discontinuity* at a point $a \in I$ if the two one-sided limits

$$\lim_{x \to a^-} f(x) \text{ and } \lim_{x \to a^+} f(x)$$

both exist, but their values are not equal to each other. Show that, if f has a jump discontinuity, then f does *not* have the intermediate value property.

Exercise 1.3.13. Show that the function $f : \mathbb{R} \to \mathbb{R}$ defined by

$$f(x) = \begin{cases} \sin(1/x) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

is discontinuous at $x = 0$ but has the intermediate value property.

Possible hint. First show that for any non-degenerate interval I containing 0, we have $f(I) = [-1, 1]$. Then explain how this implies the intermediate value property.

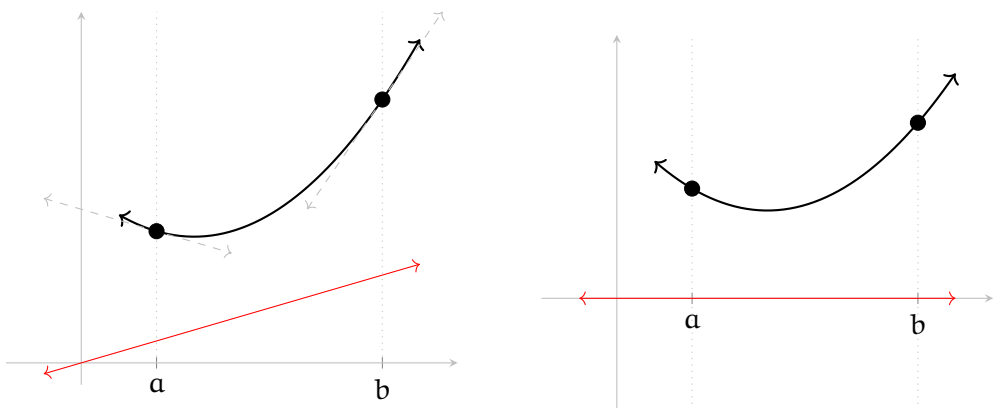

Figure 1.3.15: On the left is depicted the graph of a function f, the two tangent lines at a and b, and the graph of the line $\ell(x) = mx$ for some m in between $f'(a)$ and $f'(b)$. Subtracting ℓ from this picture yields the picture on the right.

We know that the derivative f' of a differentiable function f need not be continuous (cf. exercise 1.2.24). Darboux's theorem asserts that, even though f' might not be continuous, it must still have the intermediate value property! This means, for example, that derivatives cannot have removable or jump discontinuities, since we saw in exercise 1.3.12 that functions with these kinds of discontinuities do not have the intermediate value property.

Theorem 1.3.14 (Darboux)**.** *Suppose* I *is an open interval and* $f : I \to \mathbb{R}$ *is differentiable. Then* f' *has the intermediate value property.*

Proof. Suppose $a < b$ are two elements of I and m is a real number strictly between $f'(a)$ and $f'(b)$. We want to show that there exists $c \in (a, b)$ such that $f'(c) = m$. This statement somewhat resembles the statement of the mean value theorem 1.3.5, and indeed, our proof will also somewhat resemble the proof we gave of the mean value theorem.

Consider the differentiable function $g : I \to \mathbb{R}$ defined by $g(x) = f(x) - mx$. See figure 1.3.15. Observe that $f'(c) = m$ if and only if $g'(c) = 0$. Thus we would like to show that there exists $c \in (a, b)$ such that $g'(c) = 0$.

There are two cases two consider: either (i) $f'(a) < m < f'(b)$ or (ii) $f'(a) > m > f'(b)$. Suppose we are in case (i). By the extreme value theorem, g attains its minimum on the compact interval $[a, b]$ at some point $c \in [a, b]$. We claim that c must be an interior point of $[a, b]$, and then interior extremum theorem 1.3.2 will imply that $g'(c) = 0$.

If $c = a$ were the minimum of g on $[a, b]$, then we would have

$$\frac{g(a+h) - g(a)}{h} > 0$$

for all sufficiently small positive values of h. This would mean that $g'(a) \geqslant 0$, but we know that $g'(a) = f'(a) - m < 0$, which is a contradiction. Similarly, if $c = b$ were the minimum of g on $[a, b]$, then we would have

$$\frac{g(b+h) - g(b)}{h} < 0$$

for all sufficiently small negative values of h. This would imply that $g'(b) \leqslant 0$, but we know that $g'(b) = f'(b) - m > 0$.

In case (ii), we instead let c be the *maximum* of g on $[a, b]$, and argue similarly to show that c must be an interior point. □

Here is an example application.

Exercise 1.3.16. Suppose I is an open interval and $f : I \to \mathbb{R}$ is a differentiable function such that $f'(x) \neq 0$ for all $x \in I$. Show that f is strictly monotonic.

1.4 Levels of differentiability

1.4.A C^k functions

Definition 1.4.1. Suppose U is an open subset of $\mathbb{R}$ and $f : U \to \mathbb{R}$ is differentiable. If the derivative $f' : U \to \mathbb{R}$ is also continuous, then f is said to be *continuously differentiable*.

Definition 1.4.2. Let U be an open subset of $\mathbb{R}$ and $f : U \to \mathbb{R}$ a function. We make the following string of definitions.

- f is C^0 if it is continuous.
- f is C^1 if it is continuously differentiable.
- f is C^2 if it is differentiable and the derivative f' is C^1. In other words, this means that the derivative of the derivative $(f')'$, typically denoted f'' or $f^{(2)}$ and called the *second derivative*, exists and is continuous.
- In general, for any positive integer k, we say that f is C^k if f' exists and is C^{k-1}. This means that the k*th derivative* $f^{(k)} = (f^{(k-1)})'$ exists and is continuous.

Exercise 1.4.3. Prove that the set of all C^k functions $U \to \mathbb{R}$ is a vector space.

Exercise 1.4.4 (General Leibniz Rule)**.** Suppose that $f, g : U \to \mathbb{R}$ are C^k functions. Prove that the product fg is also C^k, and that

$$(fg)^{(k)} = \sum_{i=0}^{k} \binom{k}{i} f^{(i)} g^{(k-i)}.$$

Exercise 1.4.5. Show that the composite of two C^k functions is also C^k.

Possible hint. Use induction on k. The base case $k = 0$ is clear. For the inductive step, use the chain rule 1.2.9 and exercise 1.4.4.

Observe that we have an infinite chain of implications

$$\cdots \implies C^3 \implies C^2 \implies C^1 \implies C^0.$$

The following exercise shows that all of these implications are strict; in other words, for every k, there exists a C^k function which is not C^{k+1}.

Exercise 1.4.6. For every non-negative integer k, let

$$f_k(x) = x^k|x|.$$

Prove that f_k is C^k but not C^{k+1}.

Possible hint. First prove that $f'_{k+1} = (k+2)f_k$.

1.4.B Smooth functions

Definition 1.4.7. A function $f : U \to \mathbb{R}$ is said to be C^∞, or *infinitely differentiable*, or *smooth*, if it is in C^k for all k.

It's a little annoying that there are three different words that all mean the same thing, but that's just how it is; all three are commonly used in the mathematical literature, so it's best to get used to all of them. We'll usually use the word "smooth." It follows immediately from exercises 1.4.3 to 1.4.5 that sums, scalar multiples, products, and composites of smooth functions are also smooth.

An important principle is that we can tailor smooth functions to almost arbitrary specifications. Our first example of this principle is the following smooth bijection $(-1, 1) \to \mathbb{R}$ whose inverse is also smooth. A smooth bijection with a smooth inverse is also called a "diffeomorphism."

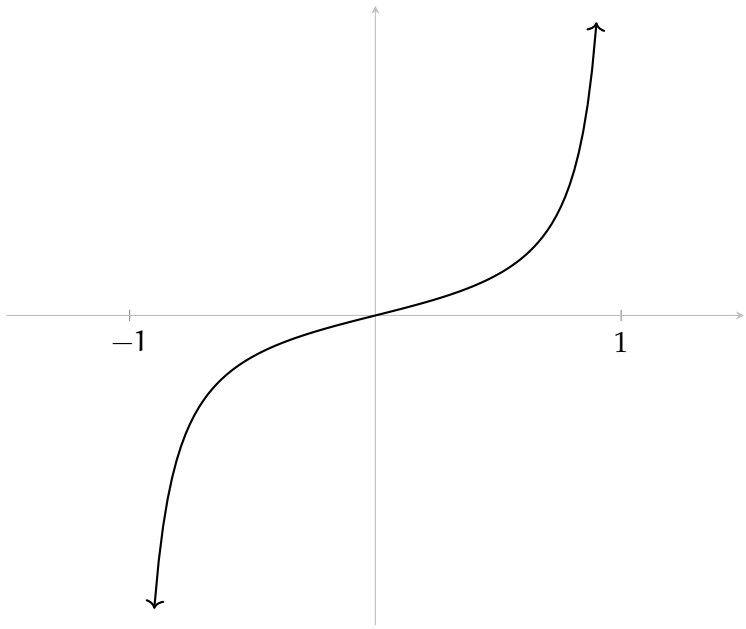

Figure 1.4.9: The graph of the function f from example 1.4.8.

Example 1.4.8. Consider the function $f : (-1, 1) \to \mathbb{R}$ defined by

$$f(x) = \frac{x}{1 - x^2}.$$

See figure 1.4.9. Since this is a quotient of two smooth functions and the denominator is nonzero on $(-1, 1)$, this function is also smooth. We can calculate that

$$f'(x) = \frac{1 + x^2}{(1 - x^2)^2}$$

and we can see from this formula that the derivative is always strictly positive. Thus f is strictly increasing (cf. exercise 1.3.10); in partcular, it is injective. Moreover, we have

$$\lim_{x \to -1^+} f(x) = -\infty \quad \text{and} \quad \lim_{x \to 1^-} f(x) = +\infty$$

so the intermediate value theorem guarantees that f is bijective. Thus the inverse function $f^{-1} : \mathbb{R} \to (-1, 1)$ exists. Finally, since f' is always nonzero, it follows from lemma 1.4.10 that f^{-1} must be smooth. It is possible to verify that

$$f^{-1}(x) = \frac{-1 + \sqrt{4x^2 + 1}}{2x}$$

by check that composing this formula with f yields the identity, but, thanks to lemma 1.4.10, we don't actually need this formula for f^{-1} at all to know that f^{-1} is smooth.

Lemma 1.4.10. *Suppose U and V are open subsets of $\mathbb{R}$, that $f : U \to V$ is a smooth bijection, and that f' is always nonzero. Then f is a diffeomorphism.*

Proof. We want to show that $f^{-1} : V \to U$ is smooth. Since f' is always nonzero, we know from exercise 1.2.18 that f^{-1} is differentiable and that

$$(f^{-1})'(a) = \frac{1}{f'(f^{-1}(a))}.$$

In other words, $(f^{-1})'$ is the composite of three functions: f^{-1}, f', and inversion (ie, the function $x \mapsto 1/x$). Inversion is clearly smooth; since f is smooth, so is f'.

We now prove by induction that f^{-1} is C^k for all integers k. For the base case, we know from above that it is C^0 (in fact, we even know more: we know that f^{-1} is even differentiable, but all we need to know for the base case is that it is continuous). Suppose inductively that f^{-1} is C^k for some non-negative integer k. Then, as we saw above, $(f^{-1})'$ is the composite of f^{-1}, f', and inversion. The latter two are smooth, so in particular are C^k. By the inductive hypothesis, we know that f^{-1} is C^k. Thus $(f^{-1})'$, being the composite of three C^k functions, is also C^k by exercise 1.4.5. This means that f^{-1} is C^{k+1}, which completes the induction. □

Exercise 1.4.11. For any $a < b$, construct a diffeomorphism $f : (a, b) \to \mathbb{R}$.

Our next example of a slightly strange smooth function is the following one, which gets "infinitely flat" at 0 but is not constantly equal to 0.

Remark 1.4.12. Here is a fact that you might recognize:

$$\lim_{u\to\infty} \frac{u^m}{e^u} = 0 \tag{1.4.13}$$

for any non-negative integer m. We'll use this fact, and also the related fact that

$$\lim_{x\to 0^+} \frac{e^{-1/x}}{x^m} = 0$$

for any non-negative integer m, which follows immediately from equation (1.4.13) by making the substitution $u = 1/x$.

You probably remember proving equation (1.4.13) using L'Hôpital's rule in your calculus class. We haven't proved L'Hôpital's rule (except for the weak version in exercise 1.1.12,

which you can check is not enough to prove equation (1.4.13)), but you can find proofs of L'Hôpital's rule in many places (eg, [PM91, theorem 4.15], [Rud76, theorem 5.13], or even Wikipedia). The proof is a clever application of the mean value theorem 1.3.5.

An alternative proof of equation (1.4.13) uses the power series definition of the exponential function. The idea is to notice that

$$\frac{e^u}{u^m} = \frac{1}{u^m} + \cdots + \frac{1}{(m-1)!u} + \sum_{k=0}^{\infty} \frac{u^k}{(m+k)!}$$

As $u \to \infty$, the first few summands all tend to 0, and the series at the end tends to ∞. Equation (1.4.13) follows from this; details are omitted, but you should be able to work them out yourself if you're interested.

Exercise 1.4.14. Consider the function $f : \mathbb{R} \to \mathbb{R}$ defined as follows.

$$f(x) = \begin{cases} e^{-1/x} & \text{if } x > 0 \\ 0 & \text{if } x \leqslant 0 \end{cases}$$

See figure 1.4.15.

(a) Use remark 1.4.12 to show, using the definition of the derivative 1.1.2, that $f'(0) = 0$.

(b) Check that

$$f'(x) = \begin{cases} \dfrac{e^{-1/x}}{x^2} & \text{if } x > 0 \\ 0 & \text{if } x \leqslant 0. \end{cases}$$

You've already computed $f'(0)$. You can compute formulas for the derivative $f'(x)$ for $x > 0$ and for $x < 0$ using the usual rules for differentiation.

(c) Use remark 1.4.12 to show that $f''(0) = 0$, and then check that

$$f''(x) = \begin{cases} p_2(x) \cdot \dfrac{e^{-1/x}}{x^4} & \text{if } x > 0 \\ 0 & \text{if } x \leqslant 0 \end{cases}$$

where p_2 is the function $p_2(x) = 1 - 2x$.

(d) Inductively, prove that

$$f^{(k)}(x) = \begin{cases} p_k(x) \cdot \dfrac{e^{-1/x}}{x^{2k}} & \text{if } x > 0 \\ 0 & \text{if } x \leqslant 0 \end{cases}$$

where p_k is a polynomial function of degree $k - 1$ such that $p_k(0) = 1$.

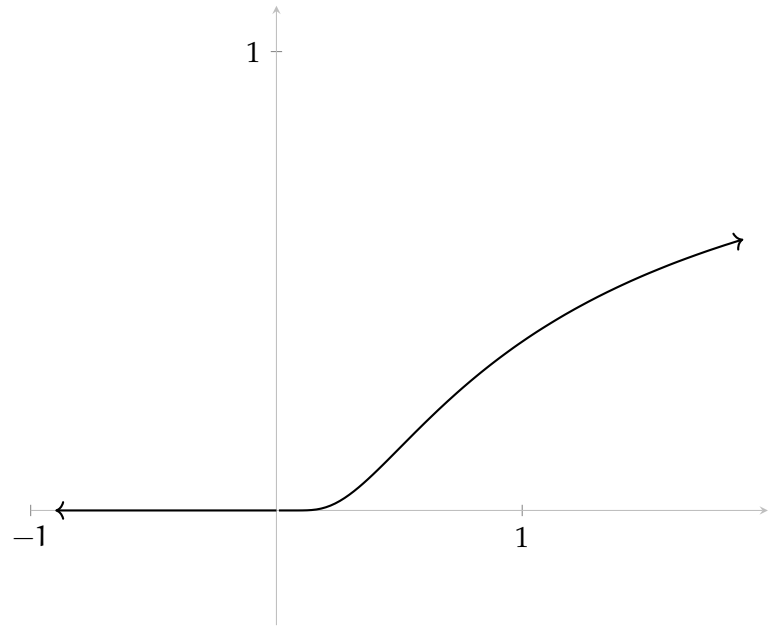

Figure 1.4.15: The graph of the function f from exercise 1.4.14.

Thus f is an example of a smooth function that is "infinitely flat" at $x = 0$, in the sense that $f^{(k)}(x) = 0$ for all k, but is not constantly equal to 0.

Next up, we have "bump functions." Recall that the "support" of a function $f : U \to \mathbb{R}$ is the closure of the set $\{x \in U : f(x) \neq 0\}$. A "bump function" is a smooth function whose support is compact.

Exercise 1.4.16. Consider the function $\psi : \mathbb{R} \to \mathbb{R}$ given by the following.

$$\psi(x) = \begin{cases} e^{-1/(1-x^2)} & \text{if } x \in (-1, 1) \\ 0 & \text{otherwise.} \end{cases}$$

See figure 1.4.17 for its graph. Check that ψ is the composite of the "infinitely flat" function from exercise 1.4.14 and the polynomial $p(x) = 1 - x^2$. Conclude that ψ is a bump function whose support is $[-1, 1]$.

Exercise 1.4.18. For any $a < b$ in $\mathbb{R}$, construct a bump function whose support is $[a, b]$.

Exercise 1.4.19. Let F be a closed subset of $\mathbb{R}$. Construct a smooth function which is zero on F and nonzero outside of F.

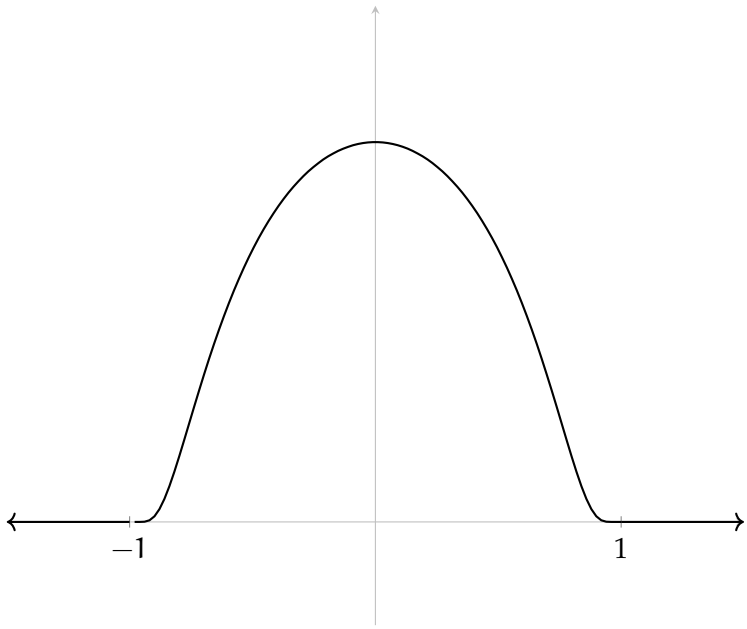

Figure 1.4.17: The graph of the function ψ from exercise 1.4.16.

Possible hint. Recall that the complement of F is a countable disjoint union of open intervals (cf. [PM91, theorem 6.17]).

In the following two examples, we will have to invoke the fundamental theorem of calculus, even though we have not proved it (or even defined integrals, for that matter). If you're unhappy with using things you haven't proved, you can find rigorous discussions of integration in many places (eg, [PM91, chapter 5], [Rud76, chapter 6], etc).

Example 1.4.20. Let ψ be the bump function from exercise 1.4.16, and then consider

$$\eta(x) = \int_{-1}^{x} \psi(t)\,dt.$$

Then

$$\eta'(x) = \frac{d}{dx}\int_{-1}^{x} \psi(t)\,dt = \psi(x)$$

by the fundamental theorem of calculus. Thus η is smooth, since $\eta' = \psi$ is smooth. Note moreover that η is increasing, since $\eta' = \psi \geqslant 0$. Finally, it is clear from the geometric interpretation of integrals as "area under the curve" that η is constantly 0 for all $x \leqslant -1$ and that it is constantly $\eta(1)$ for all $x \geqslant 1$.

Exercise 1.4.21. Let $I = [a, b]$ be a compact interval in $\mathbb{R}$ and let U be an open subset containing I. Prove that there exists a bump function $f : \mathbb{R} \to \mathbb{R}$ such that $f(x) = 1$ for all $x \in K$ and $f(x) = 0$ for all $x \notin U$.

Possible hint. The characteristic function

$$\chi_I(x) = \begin{cases} 1 & \text{if } x \in I \\ 0 & \text{if } x \notin I \end{cases}$$

is close to having all of the properties we want, but it's not smooth (it's clearly not even continuous). Since U is open, there exist $a' < a$ and $b' > b$ such that $I := [a', b'] \subseteq U$. Modify χ_I on the intervals $[a', a]$ and $[b, b']$ to obtain the desired function. Use inspiration from the function η of example 1.4.20.

2 Multivariable derivatives

In this chapter, we will study derivatives of functions $S \to \mathbb{R}^n$ where S is a subset of $\mathbb{R}^m$. As it turns out, the most important case is when $n = 1$. In other words, the "multi" in the name of this chapter (as opposed to the "single" in the name of the previous chapter) is really referring to the fact that we might have $m > 1$ (rather than to the fact that we might have $n > 1$). Moreover, as it turns out, the single variable case is quite important for the multivariable case; we'll use results from chapter 1 frequently in this chapter.

We'll begin by analyzing a particular example to build up some geometric intuition in section 2.1, and then proceed with the abstract discussion of multivariable derivatives.

2.1 Example

Recall that we started off our discussion of single variable derivatives starting with a very geometric idea of tangent lines to graphs. For large values of m (and n), graphs become hard to visualize and it is not so clear what "tangent" should mean. But there is at least one multivariable situation where, by exerting some strain on the three-dimensional visualization sectors of our brain, we can geometrically formalize what "tangent" might mean. This is the $m = 2, n = 1$ situation. The graph of such a function is a surface in $\mathbb{R}^3$, and its "tangent" at a point should be a plane.

By analyzing a specific function $f : \mathbb{R}^2 \to \mathbb{R}$, we can get some valuable insights into multivariable derivatives; the analysis will presage many of the concepts we will discuss later in the chapter. Any function would do, but let's focus on the function $f : \mathbb{R}^2 \to \mathbb{R}$ defined by

$$f(x_1, x_2) = x_1^2 + x_2^2$$

for our analysis.

Describing the graph of f

First off, let's try to get a solid understanding of the graph of f. The graph Γ is a subset of $\mathbb{R}^2 \times \mathbb{R} = \mathbb{R}^3$ defined by

$$\Gamma = \{(x_1, x_2, y) : y = f(x_1, x_2)\} = \{(x_1, x_2, y) : y = x_1^2 + x_2^2\}.$$

To visualize Γ, we will repeatedly slice Γ along various planes.

If we consider the slice obtained by setting $x_1 = 0$, we find a parabola $y = x_2^2$. Similarly, if we consider the slice obtained by setting $x_1 = 1$, we obtain the parabola $y = 1 + x_2^2$. Fixing $x_1 = 5$, we obtain the parabola $y = 25 + x^2$. In fact, we can see that no matter what value of x_1 we fix, the resulting slice is always a parabola, just translated up from the origin by differing amounts (that amount being x_1^2).

It's also worth considering the "level sets," which are the slices of Γ obtained by fixing various values of y. When $y = 0$, there is just a single point $x_1 = x_2 = 0$. Fixing $y = 1$, the slice is given by $1 = x_1^2 + x_2^2$, which is a circle of radius 1. Fixing $y = 17$, the slice is $17 = x_1^2 + x_2^2$, which is circle of radius $\sqrt{17}$. In fact, all of the level sets are circles.

Putting these slices together, we see that the graph Γ is a "big parabolic bowl."

Tangent plane at $a = (2, 1)$

Now that we understand what the graph of f looks like, let's try to figure out what the "tangent plane" T at a point $a = (2, 1)$ will look like. Of course, it's a plane inside $\mathbb{R}^3$ that passes through the point $(2, 1, f(2, 1)) = (2, 1, 5)$. To specify which plane it actually is, we start by describing some lines that lie on this plane.

Consider the $x_2 = 1$ slice. We know that the graph of Γ along this slice is the parabola $y = x_1^2 + 1$. So, slicing the tangent plane T along $x_2 = 1$ should yield the tangent line to this function of one variable, which we know how to compute from chapter 1. The slope of this tangent line is

$$\left.\frac{df(x_1, 1)}{dx_1}\right|_{x_1=2} = 4.$$

This quantity is called the *first partial derivative of* f, and will be denoted $\partial_1 f(a)$ later. Thus the tangent line is the line parametrized by

$$h_1 \mapsto (2, 1, 5) + h_1 \cdot (1, 0, 4).$$

Similarly, we can consider the $x_1 = 2$ slice. We know that the graph Γ along this slice is the

parabola $y = 4 + x_2^2$.The slope of the corresponding tangent line is

$$\partial_2 f(a) = \left.\frac{df(2, x_2)}{dx_2}\right|_{x_2=1} = 2.$$

Thus this tangent line is the line parametrized by

$$h_2 \mapsto (2, 1, 5) + h_2 \cdot (0, 1, 2).$$

These two lines must uniquely determine the entire plane T. One way of describing T that immediately falls out of the parametrizations we've found of the tangent lines is the plane parametrized by

$$\begin{bmatrix} h_1 \\ h_2 \end{bmatrix} \mapsto \begin{bmatrix} 2 \\ 1 \\ 5 \end{bmatrix} + \begin{bmatrix} h_1 \\ h_2 \\ 4h_1 + 2h_2 \end{bmatrix}.$$

The important part of this expression is the bottom entry on the far right, the $4h_1 + 2h_2$. To see why, consider the function $\mathbb{R}^2 \to \mathbb{R}$ given by $(h_1, h_2) \mapsto 4h_1 + 2h_2$. This is what we will call the *(total) derivative* of f and denote by df_a. Notice that df_a is a linear map $\mathbb{R}^2 \to \mathbb{R}$. Moreover, its graph is a plane passing through the origin that is parallel to T. In other words, if we take the graph of df_a and translate it over to the point $(2, 1, 5)$, we obtain exactly the tangent plane T. Speaking more loosely, the derivative $df_a : \mathbb{R}^2 \to \mathbb{R}$ records all of the "slopey information" about the tangent plane T.

Notice moreover that, if e_1, e_2 are the standard basis vectors of $\mathbb{R}^2$ (cf. section 0.3), then

$$\begin{aligned} df_a(e_1) &= 4 = \partial_1 f(a) \\ df_a(e_2) &= 2 = \partial_2 f(a) \end{aligned}$$

so the standard matrix representation $[df_a]$ (cf. definition 0.5.14) of the linear map df_a is

$$[df_a] = \begin{bmatrix} 4 & 2 \end{bmatrix} = \begin{bmatrix} \partial_1 f(a) & \partial_2 f(a) \end{bmatrix}. \tag{2.1.1}$$

df_a as an approximation

Consider the function $\epsilon : \mathbb{R}^2 \to \mathbb{R}$ defined by

$$v \mapsto |f(a + v) - f(a) - df_a(v)|.$$

If $v = (h_1, h_2)$, we have

$$\epsilon(v) = \epsilon\left(\begin{bmatrix} h_1 \\ h_2 \end{bmatrix}\right) \mapsto |(2 + h_1)^2 + (1 + h_2)^2 - 5 - 4h_1 - 2h_2| = |h_1^2 + h_2^2| = \left|\begin{bmatrix} h_1 \\ h_2 \end{bmatrix}\right|^2 = |v|^2.$$

Notice that ϵ gets small rapidly as $v \to 0$. More precisely, notice that

$$\lim_{v \to 0} \frac{\epsilon(v)}{|v|} = \lim_{v \to 0} |v| = 0,$$

so $\epsilon = o(|v|)$ as $v \to 0$. Since ϵ gets small rapidly, the function $v \mapsto f(a) + df_a(v)$ is a good approximation of the function $v \mapsto f(a+v)$ for small values of v. Said differently, letting $x = a + v$, the function $x \mapsto f(a) + df_a(x-a)$ is a good approximation of f near a.

2.1.A Overview

In what follows, we will turn the above example on its head and *define* the derivative using approximations. In other words, the derivative df_a will be defined to be a linear function which yields a good approximation of f near a point a. Then we will prove in theorem 2.3.21 that the partial derivatives can be used to compute the standard matrix representation df_a, as we noted in equation (2.1.1) above.

This might seem a bit "backwards," given our analysis of the example above, but there is a good reason for doing this; it turns out that there are some bizarre functions where partial derivatives make sense, but tangent planes do not (cf. exercise 2.3.19 below).

As you may be able to tell, we'll be using linear algebra heavily in this chapter. We'll also need some facts about the operator norm; I encourage you to at least skim through section 0.6.A.

2.2 Definition of the derivative

Throughout, S will denote a subset of $\mathbb{R}^m$.

Definition 2.2.1 (Differentiability at a point)**.** A function $f : S \to \mathbb{R}^n$ is *differentiable at* an interior point $a \in S$ if there exists a linear map $\ell : \mathbb{R}^m \to \mathbb{R}^n$ such that

$$|f(a+v) - f(a) - \ell(v)| = o(|v|) \text{ as } v \to 0.$$

It turns out that there can exist at most one linear map that has the above property.

Lemma 2.2.2. *Suppose $f : S \to \mathbb{R}^m$ is differentiable at an interior point $a \in S$. If ℓ and ℓ' are both linear maps $\mathbb{R}^m \to \mathbb{R}^n$ such that*

$$|f(a+v) - f(a) - \ell(v)| = o(|v|) \textit{ and } |f(a+v) - f(a) - \ell'(v)| = o(|v|) \textit{ as } v \to 0,$$

then $\ell = \ell'$.

Proof. Let $\phi = \ell' - \ell$. Then ϕ is also a linear map $\mathbb{R}^m \to \mathbb{R}^n$. Moreover, observe that

$$\begin{aligned}|\phi(v)| &= |\ell'(v) - \ell(v)| \\ &= |(f(a+v) - f(a) - \ell(v)) - (f(a+v) - f(a) - \ell'(v)| \\ &\leqslant |f(a+h) - f(a) - \ell(v)| + |f(a+v) - f(a) - \ell'(v)|.\end{aligned}$$

Since both $|f(a+v) - f(a) - \ell(v)|$ and $|f(a+v) - f(a) - \ell'(v)|$ are $o(|v|)$, exercises 0.4.7 and 0.4.8 imply that $|\phi(v)| = o(|v|)$ also. Then lemma 0.6.5 implies that $\phi = 0$. □

Exercise 2.2.3. Look at the exercises from chapter 0 that are invoked in the proof above (namely, exercises 0.4.7, 0.4.8 and 0.6.3) and do any of them that you haven't already done.

Lemma 2.2.2 guarantees that the following definition makes sense.

Definition 2.2.4 (Derivative at a point)**.** Suppose $f : S \to \mathbb{R}^n$ is differentiable at an interior point $a \in S$. Then the *total derivative*, or just *derivative*, of f at a, denoted df_a, is the unique linear map $\mathbb{R}^m \to \mathbb{R}^n$ satisfying

$$|f(a+v) - f(a) - df_a(v)| = o(|v|) \text{ as } v \to 0.$$

We've now defined the derivative, but it's fairly difficult to use this definition in practice. There are very few examples that can be computed directly from the definition. Here are some.

Exercise 2.2.5. Suppose $\ell : \mathbb{R}^m \to \mathbb{R}^n$ is a linear map. Prove that $d\ell_a = \ell$ for all $a \in \mathbb{R}^m$.

Exercise 2.2.6. Let $f : \mathbb{R}^2 \to \mathbb{R}$ be the map $f(x_1, x_2) = x_1 x_2$. Use the definition of the derivative to prove that

$$df_a(h_1, h_2) = a_2 h_1 + a_1 h_2$$

for all $a = (a_1, a_2) \in \mathbb{R}^2$.

Exercise 2.2.7. Suppose $v, w \in \mathbb{R}^n$ and $f : \mathbb{R} \to \mathbb{R}^n$ is given by $f(t) = v + tw$. Calculate df_a for any $a \in \mathbb{R}$. What is $df_a(1)$?

We will develop a bit more theory in order to compute derivatives more easily. Meanwhile, we can prove the following directly from the definition.

Exercise 2.2.8. If $f : S \to \mathbb{R}^n$ is differentiable at an interior point $a \in S$, show that f must be continuous at a.

2.3 Computing derivatives

2.3.A Sum and scalar multiples rule

Note that, by taking $w = 0$ in exercise 2.2.7, you have already shown that the derivative of a constant function is zero.

Exercise 2.3.1 (Sum rule)**.** Prove that, if $f, g : S \to \mathbb{R}^n$ are both differentiable at an interior point $a \in S$, then $f + g$ is also differentiable at a and

$$d(f+g)_a = df_a + dg_a.$$

Exercise 2.3.2 (Scalar multiples rule)**.** Prove that, if c is a constant and $f : S \to \mathbb{R}^n$ is differentiable at an interior point $a \in S$, then cf is also differentiable at a and

$$d(cf)_a = c \cdot df_a.$$

2.3.B Chain rule

The proof of the following is essentially the same as the second proof of the single variable chain rule given in section 1.2.C. Since this proof is basically a repeat, some details are omitted; you are asked to fill in the details in exercise 2.3.5 below.

Theorem 2.3.3 (Chain rule)**.** *Suppose that S and T are subsets of $\mathbb{R}^m$ and $\mathbb{R}^n$, respectively, that $f : S \to T$ is differentiable at an interior point a, that $f(a)$ is an interior point of T, and that $g : T \to \mathbb{R}^p$ is differentiable at $f(a)$. Then the composite $g \circ f : S \to \mathbb{R}^p$ is also differentiable at a, and*

$$d(g \circ f)_a = dg_{f(a)} \circ df_a.$$

Proof. Define the "error in approximation" functions

$$\begin{aligned} r(v) &= f(a+v) - f(a) - df_a(v) \\ s(w) &= g(f(a)+w) - g(f(a)) - dg_{f(a)}(w) \end{aligned}$$

and then observe that

$$g(f(a+v)) - g(f(a)) - dg_{f(a)}(df_a(v)) = dg_{f(a)}(r(v)) + s(df_a(v) + r(v)). \tag{2.3.4}$$

This means that

$$|g(f(a+v)) - g(f(a)) - dg_{f(a)}(df_a(v))| \leqslant |dg_{f(a)}(r(v))| + |s(df_a(h) + r(v))|.$$

Since $dg_{f(a)}$ is linear and $|r(v)| = o(|v|)$, exercise 2.3.5 guarantees that $|dg_{f(a)}(r(v))| = o(|h|)$ also. Thus, by exercise 0.4.7, it is sufficient to prove that $|s(df_a(v) + r(v))| = o(|v|)$. To do this, define $\eta(w) = |s(w)|/|w|$ and then notice that

$$\begin{aligned}\frac{|s(df_a(v) + r(v))|}{|v|} &= \eta(df_a(v) + r(v)) \cdot \frac{|df_a(v) + r(v)|}{|v|} \\ &\leqslant \eta(df_a(v) + r(v)) \left(\|df_a\| + \frac{|r(v)|}{|v|} \right).\end{aligned}$$

We now take the limit as $h \to 0$. Using the facts that $\eta \circ (df_a + r)$ is continuous at 0 and that $|r(v)| = o(|v|)$, we obtain the result. □

Exercise 2.3.5. (a) Prove equation (2.3.4).

(b) Suppose $r : \mathbb{R}^m \to \mathbb{R}^n$ is a function such that $|r(v)| = o(|v|)$ as $v \to 0$. If $\ell : \mathbb{R}^n \to \mathbb{R}^p$ is linear, prove that $|\ell(r(v))| = o(|v|)$ also.

Possible hint. Even though the operator norm does not appear anywhere in the statement of this exercise, you might find the concept useful. In particular, you might consider doing exercise 0.6.2 and then using it to prove (b).

(c) Prove that $\eta \circ (df_a + r)$ is continuous at 0.

2.3.C Differentiability by components

Recall that we asserted at the beginning of this chapter that the "multi" in the name of this chapter refers to the number of inputs (rather than the number of outputs). This section is where we show that understanding the $n = 1$ case is "enough" to understand general n.

Definition 2.3.6 (Component functions)**.** For any function $f : S \to \mathbb{R}^n$, we define the jth *component function* $f_j : S \to \mathbb{R}$ to be the composite $\pi_j \circ f$, where π_j is the jth projection map (cf. section 0.3).

Lemma 2.3.7. *A function $f : S \to \mathbb{R}^n$ is differentiable at an interior point $a \in S$ if and only if the component function $f_j : S \to \mathbb{R}$ is differentiable at a for all $j = 1, \dots n$. Moreover,*

$$df_a(v) = \begin{bmatrix} df_{1,a}(v) \\ \vdots \\ df_{n,a}(v) \end{bmatrix}. \tag{2.3.8}$$

Proof. If f is differentiable at a, then the composite $f_j = \pi_j \circ f$ is also differentiable at a by the chain rule 2.3.3. Moreover, since π_j is linear, we know that $d\pi_{j,f(a)} = \pi_j$ by exercise 2.2.5. Thus, by the chain rule,

$$df_{j,a}(v) = (d\pi_{j,f(a)} \circ df_a)(v) = \pi_j(df_a(v))$$

for all j, which is precisely equation (2.3.8). For the converse, we turn equation (2.3.8) on its head. Suppose f_j is differentiable at a for all j, and let $\ell : \mathbb{R}^m \to \mathbb{R}^n$ be the linear map

$$\ell(v) = \begin{bmatrix} df_{1,a}(v) \\ \vdots \\ df_{n,a}(v) \end{bmatrix}.$$

We now want to show that $|f(a+v) - f(a) - \ell(v)| = o(|v|)$ as $v \to 0$. Observe that the component functions of $v \mapsto f(a+v) - f(a) - \ell(v)$ are given by

$$v \mapsto \pi_j\left(f(a+v) - f(a) - \ell(v)\right) = f_j(a+v) - f_j(a) - df_{j,a}(v),$$

and we know that $|f_j(a+v) - f_j(a) - df_{j,a}(v)| = o(|v|)$. By exercise 2.3.9 below applied with the function $r(v) = f(a+v) - f(a) - \ell(v)$, we conclude that $|f(a+v) - f(a) - \ell(h)| = o(|v|)$. □

Exercise 2.3.9. Do exercise 0.2.10. Then use it to show that if S is a neighborhood of 0 in $\mathbb{R}^m$ and $r : S \to \mathbb{R}^n$ is a function such that $|r_j(v)| = o(|v|)$ as $v \to 0$ for all $j = 1, \ldots, n$, then $|r(v)| = o(|v|)$ also.

2.3.D Partial derivatives

We showed in the previous section that it's enough to understand the $n = 1$ case. So, in this section, we'll focus our attention on the $n = 1$ case and try to understand these derivatives better.

Definition 2.3.10 (Partial derivatives)**.** Suppose $f : S \to \mathbb{R}$ is a function and $a \in S$ is an interior point. The ith *partial derivative* of f at a, denoted $\partial_i f(a)$ is defined to be the limit

$$\lim_{h \to 0} \frac{f(a + he_i) - f(a)}{h},$$

where e_i denotes the ith standard basis vector (cf. section 0.3). If we're using x to denote the input to f and x_i to denote the ith component of the input, $\partial_i f(a)$ is sometimes also denoted by one of the following.

$$\left.\frac{\partial f}{\partial x_i}\right|_{x=a} \qquad \left.\frac{\partial}{\partial x_i} f(x)\right|_{x=a} \qquad \frac{\partial f}{\partial x_i}(a)$$

You may remember from multivariable calculus that partial derivatives behave a great deal like single variable derivatives. The reason for this is the following important remark, which reinterprets the partial derivative as a single variable derivative.

Remark 2.3.11. Consider the function $\xi_i : \mathbb{R} \to \mathbb{R}^m$ given by $\xi_i(h) = a + he_i$. Then 0 is an interior point of $T = \xi_i^{-1}(S) \subseteq \mathbb{R}$, and $f \circ \xi_i$ is a single variable function $T \to \mathbb{R}$. Moreover, we have

$$\partial_i f(a) = (f \circ \xi_i)'(0),$$

just by writing out the definition of both sides.

Example 2.3.12. Consider the function $f : \mathbb{R}^2 \to \mathbb{R}$ given by $f(x_1, x_2) = \sin(x_1 x_2^2)$. Let $a = (a_1, a_2) \in \mathbb{R}^2$ and define ξ_1 as in remark 2.3.11. Then

$$(f \circ \xi_1)(h) = f(a_1 + h, a_2) = \sin((a_1 + h)a_2^2).$$

Then

$$\begin{aligned}
\partial_1 f(a) &= (f \circ \xi_1)'(0) \\
&= \frac{d}{dh}\left(\sin((a_1 + h)a_2^2)\right)\Big|_{h=0} \\
&= a_2^2 \cos((a_1 + h)a_2^2)\big|_{h=0} \\
&= a_2^2 \cos(a_1 a_2^2).
\end{aligned}$$

Exercise 2.3.13. With notation as in example 2.3.12, compute $\partial_2 f(a)$.

We can also use remark 2.3.11 to prove some partial derivative analogs of results we proved in chapter 1.

Exercise 2.3.14 (Product rule for partial derivatives)**.** Suppose $f, g : S \to \mathbb{R}$ are functions and $a \in S$ is an interior point such that $\partial_i f(a)$ and $\partial_i g(a)$ both exist. Prove that $\partial_i(fg)(a)$ also exists and that

$$\partial_i(fg)(a) = \partial_i(f)(a)g(a) + f(a)\partial_i g(a).$$

Possible hint. Use remark 2.3.11 and the statement of the single variable product rule 1.4.4.

Exercise 2.3.15 (Interior extremum theorem)**.** Suppose $f : S \to \mathbb{R}$ is a function and an interior point $a \in S$ is a local extremum of f. Show that, if $\partial_i f(a)$ exists, then $\partial_i f(a) = 0$.

Possible hint. Use remark 2.3.11 and the single variable interior extremum theorem 1.3.2.

We have the same caveat to the interior extremum theorem that we had in the single variable case (cf. exercise 1.3.3), except that there are now even more ways for the converse to fail.

Example 2.3.16. Consider the function $f : \mathbb{R}^2 \to \mathbb{R}$ given by

$$f(x_1, x_2) = x_1^2 - x_2^2.$$

Let $\xi_i(h) = 0 + he_i$ for $i = 1, 2$. Then $f \circ \xi_1$ has a minimum at 0, so $(f \circ \xi_1)'(0) = \partial_1 f(0) = 0$. Similarly, $f \circ \xi_2$ has a maximum at 0, so $(f \circ \xi_2)'(0) = \partial_2 f(0) = 0$ also. But clearly these two facts together mean that 0 cannot be a local extremum of f.

Now for the main result of this section: partial derivatives is that they help us compute derivatives. Note that we will generalize this result in theorem 2.3.21 below.

Lemma 2.3.17. *If $f : S \to \mathbb{R}$ is differentiable at an interior point $a \in S$, then the ith partial derivative $\partial_i f(a)$ exists for all $i = 1, \dots m$, and the standard matrix representation $[df_a]$ of the linear map $df_a : \mathbb{R}^m \to \mathbb{R}$ is a $1 \times m$ matrix whose ith entry is $\partial_i f(a)$.*

$$[df_a] = \begin{bmatrix} \partial_1 f(a) & \partial_2 f(a) & \cdots & \partial_m f(a) \end{bmatrix}$$

Proof. Let r be the "error in approximation" function

$$r(v) = f(a + v) - f(a) - df_a(v),$$

so that $|r(v)| = o(|v|)$ as $h \to 0$. Then

$$r(he_i) = f(a + he_i) - f(a) - df_a(he_i) = f(a + he_i) - f(a) - hdf_a(e_i)$$

by linearity of df_a, so

$$\frac{r(he_i)}{h} = \frac{f(a + he_i) - f(a) - hdf_a(e_i)}{h} = \frac{f(a + he_i) - f(a)}{h} - df_a(e_i).$$

Then

$$\begin{aligned} \lim_{h \to 0} \left| \frac{f(a + he_i) - f(a)}{h} - df_a(e_i) \right| &= \lim_{h \to 0} \left| \frac{r(he_i)}{h} \right| \\ &= \lim_{h \to 0} \frac{|r(he_i)|}{|h|} \\ &= \lim_{h \to 0} \frac{|r(he_i)|}{|he_i|} \\ &= 0 \end{aligned}$$

where we used the fact that $|r(v)| = o(|v|)$ as $v \to 0$ for the final step. This tells us that

$$\partial_i f(a) = \lim_{h \to 0} \frac{f(a + he_i) - f(a)}{h}$$

exists and equals $df_a(e_i)$. □

Remark 2.3.18. If $f : S \to \mathbb{R}$ is differentiable at an interior point $a \in S$, then the vector

$$[df_a]^T = \begin{bmatrix} \partial_1 f(a) \\ \partial_2 f(a) \\ \vdots \\ \partial_m f(a) \end{bmatrix} \in \mathbb{R}^m$$

is often called the *gradient* of f at a, and is denoted $\nabla f(a)$.

Now for the important caveat: unfortunately, the converse to lemma 2.3.17 is *not* true.

Exercise 2.3.19. Consider the function $f : \mathbb{R}^2 \to \mathbb{R}$ defined as follows.

$$f(x_1, x_2) = \begin{cases} \dfrac{x_1 x_2}{x_1^2 + x_2^2} & \text{if } (x_1, x_2) \neq 0 \\ 0 & \text{if } (x_1, x_2) = 0 \end{cases}$$

Show that $\partial_1 f(0)$ and $\partial_2 f(0)$ both exist, but that f is not even continuous at 0, so cannot be differentiable (cf. exercise 2.2.8).

Possible hint. Approach the origin along the line $x_1 = x_2$.

This behavior is fairly pathological; theorem 2.4.4 later on will provide a converse to lemma 2.3.17 under a fairly mild additional hypothesis.

2.3.E Jacobian matrix

Definition 2.3.20 (Jacobian matrix)**.** Suppose $f : S \to \mathbb{R}^n$ is differentiable at an interior point $a \in S$. The *Jacobian matrix* of f at a, denoted $f'(a)$, is the $n \times m$ matrix whose (j, i)-entry is $\partial_i f_j(a)$. In other words,

$$f'(a) = \begin{bmatrix} \partial_1 f_1(a) & \partial_2 f_1(a) & \cdots & \partial_m f_1(a) \\ \partial_1 f_2(a) & \partial_2 f_2(a) & \cdots & \partial_m f_2(a) \\ \vdots & \vdots & \ddots & \vdots \\ \partial_1 f_n(a) & \partial_2 f_n(a) & \cdots & \partial_m f_n(a) \end{bmatrix}$$

Note that lemmas 2.3.7 and 2.3.17 tell us that all of these partial derivatives exist.

Theorem 2.3.21. *If* $f : S \to \mathbb{R}^m$ *is differentiable at an interior point* $a \in S$*, then* $[df_a] = f'(a)$.

Exercise 2.3.22. Prove theorem 2.3.21.

Possible hint. Use lemmas 2.3.7 and 2.3.17.

Note that, using theorem 2.3.21, the chain rule 2.3.3 states that

$$(g \circ f)'(a) = [d(g \circ f)_a] = [dg_{f(a)} \circ df_a] = [dg_{f(a)}][df_a] = g'(f(a))f'(a),$$

which is the form of the chain rule you may feel more comfortable with from single variable calculus.

2.3.F Rank of a differentiable map

Definition 2.3.23 (Rank of a differentiable map)**.** Suppose $f : S \to \mathbb{R}^n$ is differentiable at an interior point $a \in S$. The rank of the linear map $df_a : \mathbb{R}^m \to \mathbb{R}^n$ is also called the *rank of* f *at* a, denoted $\operatorname{rank}_a(f)$. Note that $\operatorname{rank}_a(f) \leqslant \min\{m, n\}$.

- If $\operatorname{rank}_a(f) = m$, then we say that f is *immersive* at a. Note that this is equivalent to asking that df_a be injective.
- If $\operatorname{rank}_a(f) = n$, then we say that f is *submersive at* a. Note that this is equivalent to asking that df_a be surjective.
- If $\operatorname{rank}_a(f) = m = n$, then we say that f is *étale at* a. Note that this is equivalent to asking that df_a be an isomorphism.
- If $\operatorname{rank}_a(f) = \min\{m, n\}$, then we say that a is a *regular point* of f.
- If $\operatorname{rank}_a(f) < \min\{m, n\}$, then we say that a is a *critical point* of f.

There are lots of relationships between these notions, depending on how m compares to n. You should convince yourself of the following facts.

- If $m > n$...
  - f cannot be immersive at any point.
  - a is a regular point of f if and only if f is submersive at a.
- If $m < n$...
  - f cannot be submersive at any point.

  - a is a regular point of f if and only if f is immersive at a.

- If $m = n$...
  - a is a regular point of f if and only if f is immersive at a, if and only if f is submersive at a, if and only if f is étale at a.

Caution. A map f being immersive or submersive at an interior point $a \in S$ is always defined as it is above, and critical points are always defined to be the opposite of regular points. But some mathematicians use a different definition of regularity: they define a point to be regular if $\operatorname{rank}_a(f) = n$, so that, by definition, it's *always* true (for all m, n) that f is a submersion at a if and only if a is a regular point of f. If $m \geqslant n$, there's no difference between these alternative definitions and our definitions. But, if $m < n$, there is a difference: these other mathematicians would say that all points are critical in this case, but that isn't true for us. So, if you see the words "regular point" and "critical point" in another text and it's possible that $m < n$, it's worth looking back to make sure you know what the author means!

Example 2.3.24. Let $f : \mathbb{R}^2 \to \mathbb{R}^2$ be the map

$$f(r, \theta) = (r\cos\theta, r\sin\theta).$$

See figure 2.3.25. We compute partial derivatives.

$$\frac{\partial f_1}{\partial r} = \cos\theta \quad \frac{\partial f_2}{\partial r} = \sin\theta \quad \frac{\partial f_1}{\partial \theta} = -r\sin\theta \quad \frac{\partial f_2}{\partial \theta} = r\cos\theta.$$

Thus

$$f'(r, \theta) = \begin{bmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{bmatrix}.$$

Since this is a square matrix, $df_{(r,\theta)}$ has non-maximal rank if and only if its determinant vanishes. But

$$\det f'(r, \theta) = r\cos^2\theta + r\sin^2\theta = r,$$

so the critical locus of f is the vertical line $r = 0$, and f is étale away from this line.

Exercise 2.3.26. Show that an interior point $a \in S$ is a critical point of $f : S \to \mathbb{R}$ if and only if $\partial_i f(a) = 0$ for all i.

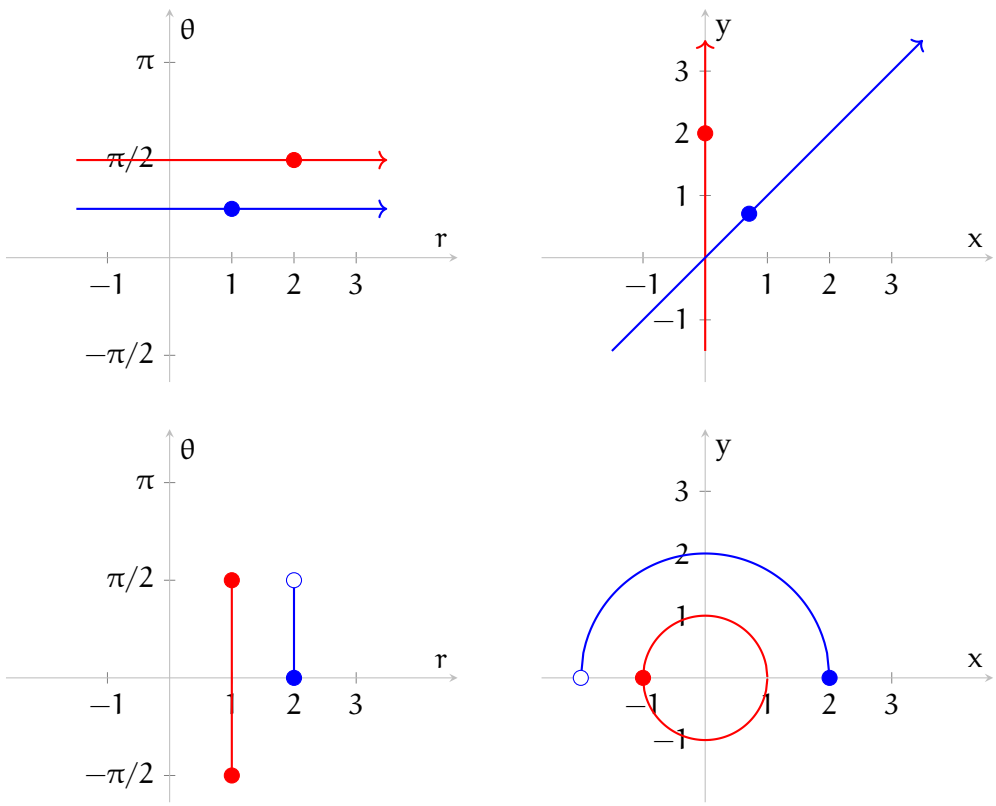

Figure 2.3.25: The function $f(r, \theta) = (r\cos\theta, r\sin\theta)$ of example 2.3.24 transforms the pictures on the right to the pictures on the left.

2.4 Levels of differentiability

Throughout, U denotes an open subset of $\mathbb{R}^m$.

Definition 2.4.1 (Differentiability)**.** If a function $f : U \to \mathbb{R}^n$ is differentiable at every $a \in U$, then f is *differentiable on* U. Then the *derivative* of f, denoted df, is the function $U \to \mathcal{L}(\mathbb{R}^m, \mathbb{R}^n)$ given by $a \mapsto df_a$. Given $i = 1, \dots, m$ and $j = 1, \dots n$, the i*th partial derivative* $\partial_i f_j$ is the function $U \to \mathbb{R}$ defined by $a \mapsto \partial_i f_j(a)$.

2.4.A Vanishing derivative

Here we prove a multivariable version of proposition 1.3.8, which stated that a differentiable function must be constant if its derivative vanishes. In fact, we will use the single variable version to prove the multivariable version.

Proposition 2.4.2. *Suppose* B *is an open ball and* $f : B \to \mathbb{R}^n$ *is a differentiable function such that* $df = 0$*. Then* f *is constant.*

Proof. We can assume, without loss of generality, that $n = 1$ (cf. exercise 2.4.3). We want to show that, for any $a, b \in U$, we have $f(a) = f(b)$. Consider the function $\gamma : [0, 1] \to B$ given by

$$\gamma(t) = tb + (1 - t)a.$$

In other words, γ is the "straight line path" starting at a and ending at b. Notice that $\gamma(t) \in B$ for all t because a and b are both inside B and B is an open ball. Also, γ is differentiable (cf. exercise 2.4.3). Thus $f \circ \gamma$ is a differentiable function $[0, 1] \to \mathbb{R}$ and

$$(f \circ \gamma)'(t) = f'(\gamma(t))\gamma'(t) = 0$$

where we used the chain rule 2.3.3 at the second step, and the fact that $df = 0$ for the final step. Thus it follows from proposition 1.3.8 that $f \circ \gamma$ is constant, which means that

$$f(a) = f(\gamma(0)) = f(\gamma(1)) = f(b). \qquad \square$$

Exercise 2.4.3. (a) Explain why proving proposition 2.4.2 for $n = 1$ implies it for general n.

(b) Show that the straight line path γ defined in the proof above is differentiable.

2.4.B Continuous differentiability

Continuity of partials

The following theorem is the promised sort-of-converse to lemma 2.3.17. Recall from exercise 2.3.19 above that $\partial_i f$ existing for all i is *not* sufficient to guarantee differentiability. But, insisting that $\partial_i f$ exist *and be continuous* turns out to be enough to guarantee differentiability of f. The proof is harder than you might expect.

Theorem 2.4.4. *Suppose $f : U \to \mathbb{R}$ is a function such that the partial derivatives $\partial_i f : U \to \mathbb{R}$ exist for all $i = 1, \dots m$. If $a \in U$ is a point such that $\partial_i f$ is continuous at a for all i, then f is differentiable at a.*

Proof. If f is to be differentiable at a, we know from theorem 2.3.21 what df_a needs to be in terms of the partial derivatives; so let's turn this on its head by writing down the linear map defined by the partial derivatives, and then proving that that linear map is in fact df_a.

More precisely, we mean the following. Let ℓ be the linear map $\mathbb{R}^m \to \mathbb{R}$ such that

$$[\ell] = \begin{bmatrix} \partial_1 f(a) & \cdots & \partial_m f(a) \end{bmatrix}.$$

In other words, $\ell : \mathbb{R}^m \to \mathbb{R}$ is the linear map given by

$$\ell(h_1, \dots, h_m) = h_1 \partial_1 f(a) + \cdots + h_m \partial_m f(a).$$

We will show that

$$|f(a + v) - f(a) - \ell(v)| = o(|v|) \text{ as } v \to 0,$$

which implies that f is differentiable at a and that $df_a = \ell$. In other words, we will show that, for any $\epsilon > 0$, there exists a $\delta > 0$ such that, for all $v \in \mathbb{R}^m$ with $|v| < \delta$, we have

$$|f(a + v) - f(a) - \ell(v)| \leqslant |v|\epsilon. \tag{2.4.5}$$

Let $\delta_0 > 0$ be small enough that the open ball $B(a, \delta_0)$ is contained in U. We will soon make it smaller to get inequality (2.4.5) to hold, but for the moment it is sufficient that $B(a, \delta_0) \subseteq U$. Suppose $v = (h_1, \dots, h_m) \in \mathbb{R}^m$ with $|v| < \delta_0$. Then consider the following

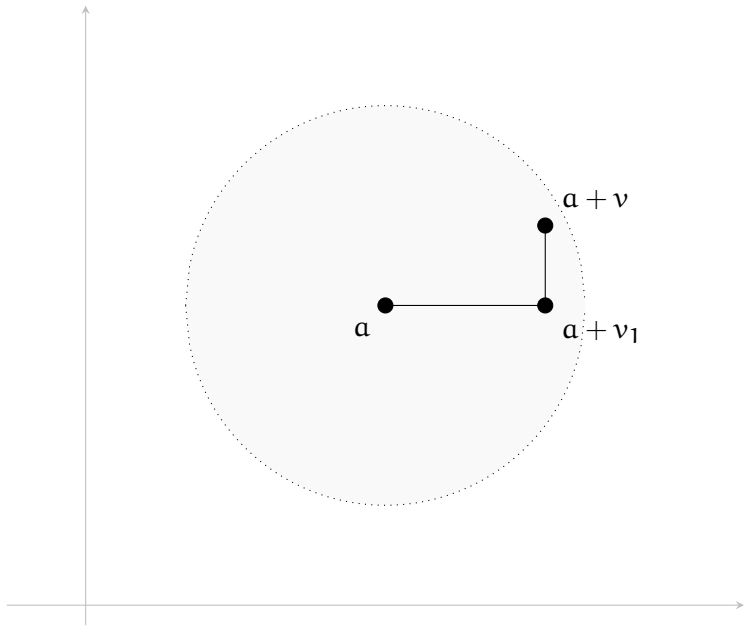

Figure 2.4.6: This is a picture of an open ball of some radius δ_0 around a point $a \in \mathbb{R}^2$. The vector $v \in \mathbb{R}^2$ has $|v| < \delta$, so that $a + v$ is a point inside this ball. Defining v_1 as in the proof of theorem 2.4.4, the vector $a + v_1$ is also inside this ball. Then, defining s_1 and s_2 as in the proof, the point $a + s_1 e_1$ is somewhere along the horizontal line joining a and $a + v_1$, and the point $a + v_1 + s_2 e_2$ is somewhere on the vertical line joining $a + v_1$ and $a + v$.

vectors in $\mathbb{R}^m$.

$$\begin{aligned} v_0 &= 0 \\ v_1 &= (h_1, 0, 0 \ldots, 0, 0) \\ v_2 &= (h_1, h_2, 0, \ldots, 0, 0) \\ &\vdots \\ v_{m-1} &= (h_1, h_2, h_3, \ldots, h_{m-1}, 0) \\ v_m &= (h_1, h_2, h_3, \ldots, h_{m-1}, h_m) = v \end{aligned}$$

See figure 2.4.6. Clearly $|v_i| \leqslant |v|$ for all i.

Notice that

$$f(a + v) - f(v) = \sum_{i=1}^{m} \left(f(a + v_i) - f(a + v_{i-1}) \right).$$

Let us fix i temporarily. Notice that $v_i = v_{i-1} + h_i e_i$, so

$$f(a + v_i) - f(a + v_{i-1}) = f(a + v_{i-1} + h_i e_i) - f(a + v_{i-1}).$$

We now apply the mean value theorem 1.3.5 to the single variable function

$$h \mapsto f(a + v_{i-1} + he_i)$$

on the closed interval between 0 and h_i. For any such h, we have $|v_{i-1} + he_i| < \delta$, so the points $a + v_{i-1} + he_i$ are all contained inside the ball $B(a, \delta_0) \subseteq U$. Thus this function is well-defined. It is also differentiable: its derivative at a value h is $\partial_i f(a + v_{i-1} + he_i)$, by definition of partial derivatives. The mean value theorem says that there exists s_i between 0 and h_i such that

$$\partial_i f(a + v_{i-1} + se_i) = \frac{f(a + v_{i-1} + h_i e_i) - f(a + v_{i-1})}{h_i} = \frac{f(a + v_i) - f(a + v_{i-1})}{h_i},$$

or, in other words,

$$f(a + v_i) - f(a + v_{i-1}) = h_i \partial_i f(a + v_{i-1} + se_i).$$

See figure 2.4.6 again. Unfixing i, we find that

$$f(a + v) - f(a) = \sum_{i=1}^{m} (f(a + v_i) - f(a + v_{i-1})) = \sum_{i=1}^{m} h_i \partial_i f(a + v_{i-1} + s_i e_i).$$

Thus

$$|f(a + v) - f(a) - \ell(v)| \leqslant \sum_{i=1}^{m} |h_i| \cdot |\partial_i(a + v_{i-1} + s_i e_i) - \partial_i f(a)|.$$

Since $\partial_i f$ is continuous at a, there exists $\delta_i > 0$ such that $|\partial_i(a + w) - \partial_i(a)| < \epsilon/m$ for all $|w| < \delta_i$. Let $\delta = \min\{\delta_0, \delta_1, \ldots, \delta_m\}$. If $|v| < \delta$, then $|v_{i-1} + s_i e_i| < |v| < \delta$ for all i, so

$$|\partial_i(a + v_{i-1} + s_i e_i) - \partial_i f(a)| < \epsilon/m,$$

which means that

$$|f(a + v) - f(a) - \ell(v)| \leqslant \sum_{i=1}^{m} \frac{|h_i|\epsilon}{m} \leqslant |v|\epsilon.$$

This proves that f is differentiable at a. □

Continuous differentiability

Definition 2.4.7 (Continuously differentiable functions)**.** We say that f is *continuously differentiable* if $\partial_i f_j : U \to \mathbb{R}$ is continuous for all $i = 1, \ldots, m$ and $j = 1, \ldots, n$. Theorem 2.4.4 guarantees that such a function is, in particular, differentiable.

Continuously differentiable functions have lots of nice properties. For example, recall the calculation we did in example 2.3.24 showing that the critical locus was a line inside $\mathbb{R}^2$. Here is a generalization of this.

Exercise 2.4.8. Suppose $f : U \to \mathbb{R}^n$ is continuously differentiable. Show that the critical locus of f is a closed subset of U.

Another important property of continuously differentiable functions is the following.

Theorem 2.4.9 (Inverse function theorem)**.** *Suppose $f : U \to \mathbb{R}^n$ is continuously differentiable, and suppose that f is étale at a point $a \in U$. Then there exists an open neighborhood V of a inside U such that the restriction $f|_V$ of f to V is injective, $f(V)$ is an open subset of $\mathbb{R}^n$, and the inverse function $(f|_V)^{-1} : f(V) \to V$ is differentiable.*

Note that, if f is étale at a, then $df_a : \mathbb{R}^m \to \mathbb{R}^n$ is an isomorphism, so we must have $m = n$. The inverse function theorem can be regarded as a multivariable version of exercise 1.2.18, but its proof is much more difficult. We defer the proof to a starred section below. Here are some applications.

Exercise 2.4.10. Suppose $f : U \to \mathbb{R}^n$ is continuously differentiably and étale at all points in U. Show that f is an open map (in other words, show that, if V is an open subset of U, then $f(V)$ is an open subset of $\mathbb{R}^n$).

Exercise 2.4.11. You might recall proving at some point in a previous class that $\mathbb{R}^2$ and $\mathbb{R}$ have the same cardinality: in other words, that there exists a bijection $f : \mathbb{R}^2 \to \mathbb{R}$. Show that any such bijection cannot be continuously differentiable.

Possible hint. Suppose for a contradition that $f : \mathbb{R}^2 \to \mathbb{R}$ is a continuously differentiable bijection. Let U be an open subset of $\mathbb{R}^2$ such that $\partial_1 f(x) \neq 0$ for all $x \in U$, and then consider the function $g : U \to \mathbb{R}^2$ given by $g(x_1, x_2) = (f(x_1, x_2), x_2)$.

The inverse function theorem constructs a local inverse to a continuously differentiable function, but only asserts that that inverse is differentiable. In fact, that inverse must also be continuously differentiable.

Lemma 2.4.12. *Suppose* $f : U \to \mathbb{R}^n$ *is continuously differentiable, injective, and that* $f(U)$ *is open in* $\mathbb{R}^n$*. If* $f^{-1} : f(U) \to U$ *is differentiable, then it is continuously differentiable.*

Proof. Since $f^{-1} \circ f = \mathrm{id}$, we have

$$\mathrm{id} = d(\mathrm{id})_x = d(f^{-1} \circ f)_x = d(f^{-1})_{f(x)} df_x$$

for all $x \in U$, where we used exercise 2.2.5 for the first step and the chain rule theorem 2.3.3 for the last. This implies, in particular, that df_x is an invertible linear map. Said differently, for any $y \in f(U)$, we have

$$d(f^{-1})_y = (df_{f^{-1}(y)})^{-1}.$$

Taking standard matrix representations, we get

$$[d(f^{-1})_y] = [(df_{f^{-1}(y)})^{-1}] = [df_{f^{-1}(y)}]^{-1}.$$

By theorem 2.3.21, the (j, i)-entry on the left hand side is $\partial_i(f^{-1})_j(y)$, so this equality says that $\partial_i(f^{-1})_j : f(U) \to \mathbb{R}$ is equal to the following composite.

$$f(U) \xrightarrow{f^{-1}} U \xrightarrow{x \mapsto [df_x]} GL_n \xrightarrow{A \mapsto A^{-1}} GL_n \xrightarrow{(j,i)-\text{entry}} \mathbb{R}$$

Here GL_n denotes the set of invertible $n \times n$ matrices as in section 0.5.B. Each of the functions being composed here is continuous (cf. exercise 2.2.8 and lemma 0.5.18), so we conclude that $\partial_i(f^{-1})_j$ is continuous. □

Proof of the inverse function theorem ⋆

There are many proofs of the inverse function, but none of them are particularly easy. The proof we give here is the one from [Spi65, chapter 2]. It begins with the following multivariable version of exercise 1.3.7 (and again, we will actually use the single variable version to prove the multivariable version).

Lemma 2.4.13. *Suppose* B *is an open ball and* $f : B \to \mathbb{R}^n$ *is a differentiable function. Suppose further that there exists a constant* M *such that* $|\partial_i f_j(x)| \leqslant M$ *for all* $i = 1, \ldots, m$ *and* $j = 1, \ldots, n$*. Then*

$$|f(b) - f(a)| \leqslant m\sqrt{n}M|b - a|$$

for all $a, b \in B$*.*

Proof. Let us first show that it is enough to prove the lemma for $n = 1$. Indeed, if it is true for $n = 1$, then it is true for all of the component functions $f_j : B \to \mathbb{R}$, so

$$|f_j(b) - f_j(a)| \leqslant mM|b - a|$$

for all $a, b \in B$, which means that

$$|f(b) - f(a)|_{\max} \leqslant mM|b - a|,$$

where $| - |_{\max}$ denotes the max norm defined in section 0.3. But then

$$|f(b) - f(a)| \leqslant \sqrt{n}|f(b) - f(a)|_{\max} \leqslant m\sqrt{n}M|b - a|,$$

where we have used exercise 0.2.10 for the first step.

So we can assume that $n = 1$. The idea of this proof is very similar to the idea we used in the proof of theorem 2.4.4. Set $v = b - a$. If $v = (h_1, \ldots, h_m)$, we define the following list of vectors.

$$\begin{aligned}
v_0 &= 0 \\
v_1 &= (h_1, 0, 0 \ldots, 0, 0) \\
v_2 &= (h_1, h_2, 0, \ldots, 0, 0) \\
&\vdots \\
v_{m-1} &= (h_1, h_2, h_3, \ldots, h_{m-1}, 0) \\
v_m &= (h_1, h_2, h_3, \ldots, h_{m-1}, h_m) = v
\end{aligned}$$

Then set $a_i = a + v_i$, so that $a_0 = a$ and $a_m = b$. Notice that

$$f(b) - f(a) = \sum_{i=1}^{m} (f(a_i) - f(a_{i-1}))$$

We're now going to restrict f to the zigzag path from a to b made by the points $a_0, \ldots, a_m$, looking at each segment in turn, just as we did in the proof of theorem 2.4.4. The function

$$h \mapsto f(a_{i-1} + he_i)$$

is a single variable function whose derivative at a point h is given by $\partial_i f(a_{i-1} + he_i)$, which we know is bounded in absolute value by M. Since this function has the value $f(a_{i-1})$ at $h = 0$ and $f(a_i)$ at $h = h_i$, exercise 1.3.7 gives us the first step of the following chain.

$$|f(a_i) - f(a_{i-1})| \leqslant M|h_i| \leqslant M|v|_{\max}.$$

But then, using exercise 0.2.10, we have

$$|f(a_i) - f(a_{i-1})| \leqslant M|v|_{\max} \leqslant M|v| = M|b-a|$$

so we have

$$|f(b) - f(a)| = \left|\sum_{i=1}^{m} f(a_i) - f(a_{i-1})\right| \leqslant \sum_{i=1}^{m} |f(a_i) - f(a_{i-1})| \leqslant mM|b-a|. \qquad \square$$

Proof of the inverse function theorem 2.4.9. Observe that

$$d(df_a^{-1} \circ f)_a = df_a^{-1} \circ df_a = \mathrm{id}$$

using the chain rule 2.3.3 and exercise 2.2.5. By the first two parts of exercise 2.4.16 with $\ell = df_a^{-1}$, we can freely replace f with $df_a^{-1} \circ f$ in order to assume that $df_a = \mathrm{id}$. In other words, we have

$$\partial_i f_j(a) = \delta_{i,j} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{otherwise.} \end{cases}$$

We will now construct an open ball B around a, and then shrink it so that all points in B have certain desirable properties. First of all, notice that

$$\lim_{v \to 0} \frac{|f(a+v) - f(a) - df_a(v)|}{|v|} = 0,$$

but if $f(a+v) = f(a)$, then

$$\frac{|f(a+v) - f(a) - df_a(v)|}{|v|} = \frac{|v|}{|v|} = 1,$$

which means that there must exist an open ball B around a such that $f(a+v) \neq f(a)$ whenever $a + v \in \bar{B} \setminus \{a\}$.

We now shrink B a bit. Consider the map $U \mapsto \mathbb{R}$ given by $x \mapsto \det(df_x)$. Since $\det(df_x)$ is a polynomial in the partial derivatives of f, which are all continuous at a since f is continuously differentiable at a, the map $x \mapsto \det(df_x)$ must also be continuous at a. But $\det(df_a) \neq 0$, so by shrinking B, we can assume that $\det(df_x) \neq 0$ for all $x \in B$. Finally, since $\partial_i f_j$ is continuous at a, we can shrink B further in order to assume that

$$|\partial_i f_j(x) - \partial_i f_j(a)| = |\partial_i f_j(x) - \delta_{i,j}| \leqslant \frac{1}{2n^{3/2}}$$

for all i, j and all $x \in B$. In other words, we have constructed an open ball B around a inside U with the following three properties.

(1) $f(x) \neq f(a)$ for all $x \in \bar{B} \setminus \{a\}$.

(2) df_x is invertible for all $x \in B$.

(3) $|\partial_i f_j(x) - \delta_{i,j}| \leqslant 1/2n^{3/2}$ for all i, j and $x \in B$.

The set V we want to construct will be an open subset of B. The remainder of the proof is a series of claims.

Claim A. $|x_1 - x_2| \leqslant 2|f(x_1) - f(x_2)|$ for all $x_1, x_2 \in B$.

Proof of claim A. Consider the function $g(x) = f(x) - x$. Then $\partial_i g_j(x) = \partial_i f_j(x) - \delta_{i,j}$, so

$$|\partial_i g_j(x)| = |\partial_i f_j(x) - \delta_{i,j}| \leqslant \frac{1}{2n^{3/2}}$$

using property (3). By lemma 2.4.13, we see that

$$|g(x_1) - g(x_2)| \leqslant \frac{1}{2}|x_1 - x_2|.$$

But

$$|g(x_1) - g(x_2)| = |(f(x_1) - x_1) - (f(x_2) - x_2)| \geqslant |f(x_1) - f(x_2)| - |x_1 - x_2|$$

by the reverse triangle inequality. Combining these two inequalities proves claim A.

Claim B. There exists a positive real number d such that $|f(x) - f(a)| \geqslant d$ for all $x \in \partial B$.

Proof of claim B. The function $\partial B \to \mathbb{R}$ given by

$$x \mapsto |f(x) - f(a)|$$

is continuous. Let d denote the infimum of the range of this function. Clearly $d \geqslant 0$, and to prove claim B, we only need to prove that $d > 0$. Suppose for a contradiction that $d = 0$. Since ∂B is compact, there must exist $x \in \partial B$ such that $|f(x) - f(a)| = 0$ (cf. [PM91, theorem 6.30]). This contradicts property (1).

Claim C. Let d be the positive real number defined in claim B, and let

$$W = \{y \in \mathbb{R}^n : |y - f(a)| < d/2\}.$$

Then for every $y \in W$, there exists a unique $x \in B$ such that $f(x) = y$.

Proof of claim C. Fix $y \in W$ and consider the function $\bar{B} \to \mathbb{R}$ given by

$$g(x) = |y - f(x)|^2 = \sum_{j=1}^{n} (\pi_j(y) - f_j(x))^2.$$

This function is continuous and $\bar{B}$ is compact, so the extreme value theorem guarantees that g achieves its minimum at some $x \in \bar{B}$. Notice that, if $x \in \partial B$, then

$$g(x) = |y - f(x)|^2 \geqslant d^2$$

whereas

$$g(a) = |y - f(a)|^2 < d^2/4,$$

so $g(a) < g(x)$, which contradicts the fact that x is the minimum of g. Thus $x \in \bar{B} \setminus \partial B = B$. By the interior extremum theorem 2.3.15, we know that

$$0 = \partial_i g(x) = \sum_{j=1}^{n} 2(\pi_j(y) - f_j(x))\partial_i f_j(x) = \pi_i \left(2f'(x)^T(y - f(x))\right)$$

for all i. Here, $f'(x)^T$ denotes the transpose of the Jacobian matrix. This means that $f'(x)^T(y - f(x)) = 0$. Now recall that df_x is an invertible linear map using property (2). Thus $f'(x)$ is an invertible matrix, so $f'(x)^T$ is also an invertible matrix, so we must have $y - f(x) = 0$. In other words, $y = f(x)$. This proves the existence assertion of the claim.

For the uniqueness assertion, if x' is any other point in B such that $f(x') = y$, then we would have

$$0 = 2|y - y| = 2|f(x') - f(x)| \geqslant |x' - x| \geqslant 0$$

using claim A, so $x' = x$. This completes the proof.

Claim D. With W as in claim C, define

$$V = f^{-1}(W) = \{x \in B : f(x) \in W\}.$$

Then V is open and $f|_V : V \to W$ is a homeomorphism. In particular, $f(V) = W$ and W is open.

Proof of claim D. Recall from claim C that W is the open ball of radius $d/2$ around $f(a)$. Since f is continuous, $f^{-1}(W)$ must also be open. Claim C tells us that $f|_V$ is injective, and claim A says that

$$|f^{-1}(y_1) - f^{-1}(y_2)| \leqslant 2|y_1 - y_2| \tag{2.4.14}$$

for all $y_1, y_2 \in W$. This implies that $(f|_V)^{-1}$ is continuous.

Claim E. $(f|_V)^{-1}$ is differentiable.

Proof of claim E. For $x \in V$, let $y = f(x) \in f(V)$. We will show that $(f|_V)^{-1}$ is differentiable at y with derivative df_x^{-1}. This means that, if we set

$$s(w) = f^{-1}(y + w) - f^{-1}(y) - df_x^{-1}(w),$$

we want to show that $|s(w)| = o(|w|)$ as $w \to 0$. In fact, since df_x is an invertible linear map, it is sufficient to show that $|df_x(s(w))| = o(|w|)$ (cf. exercise 2.3.5 part (b)).

Define $v(w) = f^{-1}(y + w) - f^{-1}(w)$. Since $f|_V$ is injective, we see that $v(w) = 0$ if and only if $w = 0$. Also, equation (2.4.14) tells us that

$$|v(w)| \leqslant 2|w|. \tag{2.4.15}$$

By definition of $s(w)$ and $v(w)$, we have

$$s(w) = v(w) - df_x^{-1}(w)$$

so

$$\begin{aligned} df_x(s(w)) &= df_x(v(w)) - w \\ &= df_x(v(w)) - y + w + y \\ &= df_x(v(w)) - f(x + v(w)) + f(x). \end{aligned}$$

In other words, if we consider the "error in approximation" function

$$r(v) = f(x + v) - f(x) - df_x(v),$$

then

$$df_x(s(w)) = -r(v(w))$$

for all w. This means that, for all $w \neq 0$, we have

$$\frac{|df_x(s(w))|}{|w|} = \frac{|r(v(w))|}{|w|} = \frac{|r(v(w))|}{|v(w)|}\frac{|v(w)|}{|w|} \leqslant 2 \cdot \frac{|r(v(w))|}{|v(w)|}.$$

Note that we have used the fact that $v(w)$ is nonzero for nonzero w, in order to avoid a division by zero. Equation (2.4.15) tells us that

$$\lim_{w \to 0} v(w) = 0$$

so

$$\lim_{w \to 0} \frac{|r(v(w))|}{|v(w)|} = 0$$

since $|r(v)| = o(|v|)$ as $v \to 0$. By the squeeze theorem, we conclude that

$$\lim_{w \to 0} \frac{|df_x(s(w))|}{|w|} = 0,$$

which is precisely the statement that $|df_x(s(w))| = o(|w|)$ as $w \to 0$. □

Exercise 2.4.16. (a) Suppose $\ell : \mathbb{R}^n \to \mathbb{R}^n$ is an invertible linear map. Show that $f : U \to \mathbb{R}^n$ is étale at a point $a \in U$ if and only if $\ell \circ f$ is étale at a.

(b) Suppose $\ell : \mathbb{R}^n \to \mathbb{R}^n$ is an invertible linear map. Show that the conclusion of the inverse function theorem is true for f if and only if it is true for $\ell \circ f$.

(c) Show that, if K is a compact subset of a metric space X and $a \in X \setminus K$, then

$$\inf\{d(a, x) : x \in K\} > 0.$$

Continuous differentiability via the operator norm ⋆

For this section, we will need to make substantial use of the operator norm. This might make it a bit challenging for those who have not used the operator norm before, but the results contained in this section are also not terribly important. The only reason it is included is because proposition 2.4.19 is aesthetically a rather satisfying statement.

Lemma 2.4.17. *Suppose* $f : U \to \mathbb{R}^n$ *is differentiable. Then* $df : U \to \mathcal{L}(\mathbb{R}^m, \mathbb{R}^n)$ *is continuous if and only if* $df_j : \mathcal{L}(\mathbb{R}^m, \mathbb{R})$ *is continuous for all* $j = 1, \ldots, n$.

Proof. We will use lemma 2.3.7, which tells us that f is differentiable if and only if f_j is differentiable for all $j = 1, \ldots, n$, and moreover that $df_{j,a} = \pi_j \circ df_a$ for all $a \in U$. In other words, the following diagram commutes.

$$\begin{tikzcd} U \arrow[r, "df"] \arrow[dr, bend right, "df_j"'] & \mathcal{L}(\mathbb{R}^m, \mathbb{R}^n) \arrow[d, "\pi_j \circ -"] \\ & \mathcal{L}(\mathbb{R}^m, \mathbb{R}) \end{tikzcd}$$

You will verify in exercise 2.4.18 below that the map "postcompose with π_j" map $\pi_j \circ - : \mathcal{L}(\mathbb{R}^m, \mathbb{R}^n) \to \mathcal{L}(\mathbb{R}^m, \mathbb{R})$ is continuous. Thus, if df is continuous, then df_j is also continuous at a.

Conversely, suppose that df_j is continuous at a for all j. Let us show that df is continuous at a. Suppose $\epsilon > 0$. Since df_j is continuous, there exists $\delta_j > 0$ such that

$$\|df_{j,x} - df_{j,a}\| < \frac{\epsilon}{\sqrt{n}}$$

for all $x \in U$ such that $|x - a| < \delta_j$. Let $\delta := \min\{\delta_1, \ldots, \delta_n\}$. Then $\|df_x - df_a\| < \epsilon$ for all $x \in U$ such that $|x - a| < \delta$, as you will verify in exercise 2.4.18 below. □

Exercise 2.4.18. (a) Check that the "postcompose with π_j" map $\mathcal{L}(\mathbb{R}^m, \mathbb{R}^n) \to \mathcal{L}(\mathbb{R}^m, \mathbb{R})$ (ie, the one given by $\phi \mapsto \pi_j \circ \phi$) is continuous. Alternatively, prove the more general statement given in exercise 0.6.9 and then derive this statement.

(b) With notation as in the latter part of the proof above, verify that $\|df_a - df_b\| < \epsilon$.

Proposition 2.4.19. *Suppose $f : U \to \mathbb{R}^n$ is differentiable. Then f is continuously differentiable if and only if $df : U \to \mathcal{L}(\mathbb{R}^m, \mathbb{R}^n)$ is continuous.*

Proof. Thanks to lemma 2.4.17, we can assume without loss of generality that $n = 1$. If f is differentiable, then lemma 2.3.17 tells us that $\partial_i f = df(e_i)$. In other words, the following diagram commutes.

$$\begin{tikzcd} U \arrow[r, "df"] \arrow[dr, bend right, "\partial_i f"'] & \mathcal{L}(\mathbb{R}^m, \mathbb{R}) \arrow[d, "\text{“evaluate at } e_i\text{”}"] \\ & \mathbb{R} \end{tikzcd}$$

The "evaluate at e_i" map is continuous by exercise 0.6.7. So, if f is continuously differentiable (ie, if df is continuous), then $\partial_i f$ must also be continuous, as it is the composite of two continuous functions.

Conversely, suppose that all of the partial derivatives $\partial_i f$ exist and are continuous. Then theorem 2.4.4 guarantees that f is differentiable. The final step is to show that $df : U \to \mathcal{L}(\mathbb{R}^m, \mathbb{R})$ is continuous; This is left for you to check; see exercise 2.4.20 below. □

Exercise 2.4.20. (a) Elaborate on the first sentence of the above proof (in other words, explain why proving theorem 2.4.4 for $n = 1$ is sufficient to prove it for all n).

(b) If you haven't already done it, do exercise 0.6.7.

(c) Complete the proof above by showing that $df : U \to \mathcal{L}(\mathbb{R}^m, \mathbb{R})$ is continuous.

Remark 2.4.21. This reformulation of continuous differentiability is often convenient in proofs. For example, in the proof of lemma 2.4.12, we calculated that

$$d(f^{-1})_y = (df_{f^{-1}(y)})^{-1}.$$

This says that the $d(f^{-1}) = \text{inversion} \circ df \circ f^{-1}$.

$$\begin{tikzcd} f(U) \arrow[r, "d(f^{-1})"] \arrow[d, "f^{-1}"'] & GL(\mathbb{R}^n) \\ U \arrow[r, "df"] & GL(\mathbb{R}^n) \arrow[u, "\text{inversion}"'] \end{tikzcd}$$

We know that f^{-1} is continuous by exercise 2.2.8, that df is continuous by proposition 2.4.19, and inversion is continuous by exercise 0.6.12. Thus $d(f^{-1})$ is continuous, so proposition 2.4.19 says that f^{-1} is continuously differentiable.

2.4.C C^k functions

Definition 2.4.22 (C^k functions)**.** We say that a function $f : U \to \mathbb{R}^n$ is C^0 if it is continuous. Then, inductively, for any positive integer k, we say that a function $f : U \to \mathbb{R}^n$ is C^k if the partial derivatives $\partial_i f_j$ all exist and are C^{k-1} functions $U \to \mathbb{R}$.

By definition, $f : U \to \mathbb{R}^n$ is C^1 if and only if it is continuously differentiable. It is clear from the definitions that a function $f : U \to \mathbb{R}^n$ is C^k if and only if its component functions $f_j : U \to \mathbb{R}$ are C^k.

Exercise 2.4.23. Show that the set of all C^k functions $U \to \mathbb{R}^n$ form a vector space.

Exercise 2.4.24. Suppose U is an open subset $\mathbb{R}^m$, that V is an open subset of $\mathbb{R}^n$, and that $f : U \to V$ and $g : V \to \mathbb{R}^p$ are both C^k. Show that $g \circ f$ is also C^k.

Exercise 2.4.25. Suppose $f, g : U \to \mathbb{R}$ are both C^k. Show that fg is also C^k.

Exercise 2.4.26. Suppose $f : U \to \mathbb{R}$ is C^k and $f(x) \neq 0$ for all $x \in U$. Show that $1/f$ is also C^k.

Possible hint. Note that $1/f$ is f composed with the single variable inversion function.

Exercise 2.4.27. Suppose $f : U \to \mathbb{R}^n$ is C^k for some $k \geqslant 1$, injective, and $f(U)$ is open. If $f^{-1} : f(U) \to U$ is differentiable, show that f^{-1} is also C^k.

Possible hint. Induct on k. The base case $k = 1$ is lemma 2.4.12. For the inductive step, if you look carefully at the proof of lemma 2.4.12 (and of lemma 0.5.18), you'll see that the partials of f^{-1} equal f^{-1} composed with a rational function applied to the partials of f.

2.4.D Equality of mixed partials

Definition 2.4.28 (Mixed partials)**.** Suppose $f : U \to \mathbb{R}$ is C^2. The partial derivative $\partial_i f : U \to \mathbb{R}$ for any $i = 1, \dots, m$ is a continuously differentiable function, so its jth partial deriative $\partial_j(\partial_i f)$ is a continuous function $U \to \mathbb{R}$ for any $j = 1, \dots m$. We denote the function $\partial_j(\partial_i f)$ more succinctly as $\partial_{j,i} f$.

Theorem 2.4.29 (Equality of mixed partials)**.** *Suppose $f : U \to \mathbb{R}$ is C^2. Then, for all $i, j = 1, \dots, m$, we have*

$$\partial_{j,i} f = \partial_{i,j} f.$$

Remark 2.4.30. There are many versions of this theorem. For instance, there is the version due to Peano [Rud76, theorem 9.41]. It assumes that $\partial_i f$, $\partial_j f$, and $\partial_{j,i} f$ all exist, and that $\partial_{j,i} f$ is continuous; it then proves that $\partial_{i,j} f$ must also exist and be equal $\partial_{j,i} f$. The version stated above will be enough for our needs.

Remark 2.4.31. If $f : U \to \mathbb{R}$ is C^2, the *Hessian* of f at a, denoted $f''(a)$, is the matrix of second partial derivatives.

$$f''(a) = \begin{bmatrix} \partial_{1,1} f(a) & \partial_{1,2} f(a) & \cdots & \partial_{1,m} f(a) \\ \vdots & \vdots & \ddots & \vdots \\ \partial_{m,1} f(a) & \partial_{m,2} f(a) & \cdots & \partial_{m,m} f(a) \end{bmatrix}$$

Theorem 2.4.29 is equivalent to the assertion that $f''(a)$ is a symmetric matrix.

2.4.E Smooth functions

Definition 2.4.32 (Smoothness)**.** We say that a function $f : U \to \mathbb{R}^n$ is C^∞, or *infinitely differentiable*, or *smooth*, if it is C^k for all k.

In section 1.4.B, we formulated the principle that smooth single variable functions can be tailored to almost arbitrary specifications. The same principle remains true of smooth multivariable functions, and we can often bootstrap up from the single variable case.

Exercise 2.4.33. Let $\psi : \mathbb{R} \to \mathbb{R}$ denote the bump function from exercise 1.4.16. Then define $\Psi : \mathbb{R}^m \to \mathbb{R}$ by

$$\Psi(x_1, \dots, x_m) = \psi(x_1)\psi(x_2) \cdots \psi(x_m).$$

Show that Ψ is a bump function (ie, a smooth function with compact support).

3 Manifolds

At this point, we will need to use the language of topological spaces. If you have seen metric spaces but not topological spaces, you should look over section 0.7.A at this point.

3.1 Definition of a manifold

Throughout, let X be a topological space.

Definition 3.1.1 (Chart)**.** A *chart* on X is a pair (U, x) consisting of an open subset $U \subseteq X$ and a function $x : U \to \mathbb{R}^n$ which is a homeomorphism onto an open subset of $\mathbb{R}^n$ for some n. We call the functions $x_i := \pi_i \circ x : U \to \mathbb{R}$ *coordinate functions* on U with respect to the chart (U, x), and for any $p \in U$, the list of numbers $x_1(p), \ldots, x_n(p)$ are the *coordinates* of p with respect to the chart (U, x).

Exercise 3.1.2. Let $X = \mathbb{R}^2$ and let U denote $\mathbb{R}^2$ minus the non-negative real axis. Let $\text{polar} : U \to \mathbb{R}^2$ be the function which sends a point p to its polar representation $(r(p), \theta(p))$, where $r(p) = |p|$ and $\theta(p)$ is the angle in radians strictly between 0 and 2π formed between P and the non-negative real axis. Show that this function is a homeomorphism onto its image.

Definition 3.1.3. Suppose (U, x) and (V, y) are charts on X. The *transition map* from (U, x) to (V, y) is the function $y \circ x^{-1}$.

$$x(U \cap V) \xrightarrow{x^{-1}} U \cap V \xrightarrow{y} y(U \cap V)$$

Definition 3.1.4 (Compatibility of charts)**.** The two charts (U, x) and (V, y) on X are *smoothly compatible*, or just *compatible*, if both of the transition maps between (U, x) and (U, y) are smooth.

Remark 3.1.5. One could also make a definition of C^k*-compatibility* for any $k = 0, 1, \ldots, \infty$, but we won't ever use this.

Exercise 3.1.6. Let $\mathrm{id} : \mathbb{R}^2 \to \mathbb{R}^2$ denote the identity map and $\mathrm{polar} : U \to \mathbb{R}^2$ the polar representation map from exercise 3.1.2. Show that $(\mathbb{R}^2, \mathrm{id})$ and (U, polar) are compatible charts.

Exercise 3.1.7. Show that the relation "is compatible with" is *not* an equivalence relation on the set of all charts on X.

Possible hint. The function $\mathbb{R} \to \mathbb{R}$ given by $x \mapsto x^3$ is a smooth homeomorphism, but its inverse is not differentiable.

Exercise 3.1.8. Suppose (U, x) and (V, y) are compatible charts, where $x : U \to \mathbb{R}^m$ and $y : V \to \mathbb{R}^n$, and that $U \cap V$ is nonempty. Show that $m = n$.

Possible hint. Pick a point $p \in U \cap V$, and consider the derivatives of $x \circ y^{-1}$ at $y(p)$ and of $y \circ x^{-1}$ at $x(p)$.

Definition 3.1.9 (Atlas)**.** An *atlas* on X is a collection $\mathcal{A}$ of charts such that covers X. In other words,

$$\bigcup_{(U,x)\in\mathcal{A}} U = X.$$

An atlas is *maximal* if there is no strictly larger atlas containing it.

Exercise 3.1.10 (Maximal atlases)**.** Prove that every atlas $\mathcal{A}$ is contained in a unique maximal atlas.

Possible hint. Let $\mathcal{A}'$ be the set of all charts that are compatible with all of the charts in $\mathcal{A}$. Prove that $\mathcal{A}'$ is an atlas (caution: keep exercise 3.1.7 in mind). Then prove that it is maximal, and that it is the only maximal atlas containing $\mathcal{A}$.

This leads us to the following definition.

Definition 3.1.11 (Manifold)**.** A *smooth manifold*, or just a *manifold*, is a pair $(X, \mathcal{A}_X)$ where X is a topological space and $\mathcal{A}_X$ is a maximal atlas. Often, write simply "X" in place of the pair $(X, \mathcal{A}_X)$, and sometimes $\mathcal{A}_X$ is called the "manifold structure on X."

Of course, exercise 3.1.10 tells us that any not-necessarily-maximal atlas on a metric space X defines a manifold structure on X in a canonical way. When constructing examples, we will usually give an explicit non-maximal atlas and then use exercise 3.1.10 to extend it to a maximal atlas. But we require maximality in the definition in part because, when dealing with abstract manifolds in proofs, we will often find it convenient to assume that the atlas that the manifold comes equipped with is already maximal.

We can also make the following definition.

Definition 3.1.12 (Dimension)**.** Suppose X is a manifold and $p \in X$ is a point. By exercise 3.1.8, there is a unique integer n such that, for every chart $(U, x) \in \mathcal{A}_X$ with $p \in U$, the codomain of x is $\mathbb{R}^n$. This integer is called the *dimension of* X *at* p, and is denoted $\dim_p(X)$. We say that X is n-*dimensional* if $\dim_p(X) = n$ for all $p \in X$.

3.2 Examples

3.2.A First examples

Example 3.2.1. The single chart $(\mathbb{R}^n, \mathrm{id})$, where $\mathrm{id} : \mathbb{R}^n \to \mathbb{R}^n$ is the identity map, defines an atlas on $\mathbb{R}^n$. We can extend this to a maximal atlas by exercise 3.1.10, called the *euclidean atlas* on $\mathbb{R}^n$. Unless explicitly specified otherwise, we always regard $\mathbb{R}^n$ as a manifold by equipping it with the euclidean atlas. By exercise 3.1.6, the polar representation chart (U, polar) is an element of the euclidean atlas.

Exercise 3.2.2. The single chart $(\mathbb{R}, f)$ where $f : \mathbb{R} \to \mathbb{R}$ is the function $f(x) = x^3$ is an atlas on $\mathbb{R}$. Show that the corresponding maximal atlas is *not* the euclidean atlas.

Exercise 3.2.3 (The circle S^1)**.** Let S^1 denote the set of points in $\mathbb{R}^2$ along the unit circle. In other words,

$$S^1 = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = 1\}.$$

(a) There are two ways you might think to regard S^1 as a topological space. First, you might restrict the euclidean metric on $\mathbb{R}^2$ to S^1, so that S^1 becomes a metric space, and then you can forget this restricted metric and just remember the resulting topology as in example 0.7.2. Alternatively, you can first forget the metric on $\mathbb{R}^2$ and regard it as a topological space as in example 0.7.2, and then regard S^1 as a subspace of $\mathbb{R}^2$ (cf. definition 0.7.13). Show that you end up with the same topology on S^1 in both cases.

(b) Let $U = S^1 \setminus \{(0, 1)\}$. For $p \in S^1$, let $x(p)$ denote the x-coordinate of the point where the line $y = -1$ intersects the line passing through $(0, 1)$ and p. See figure 3.2.4. If $p = (a, b)$, find a formula for $x(p)$ in terms of a and b.

(c) Show that $x : U \to \mathbb{R}$ is a homeomorphism. Thus (U, x) is a chart on S^1.

Possible hint. Use geometry to write down a formula for the inverse function to x.

(d) Let $U' = S^1 \setminus \{(0, -1)\}$. For $p \in V$, let $x'(p)$ denote the x-coordinate of the point where the line $y = 1$ intersects the line passing through $(0, -1)$ and p. Show that $x' : U' \to \mathbb{R}$ is also a homeomorphism.

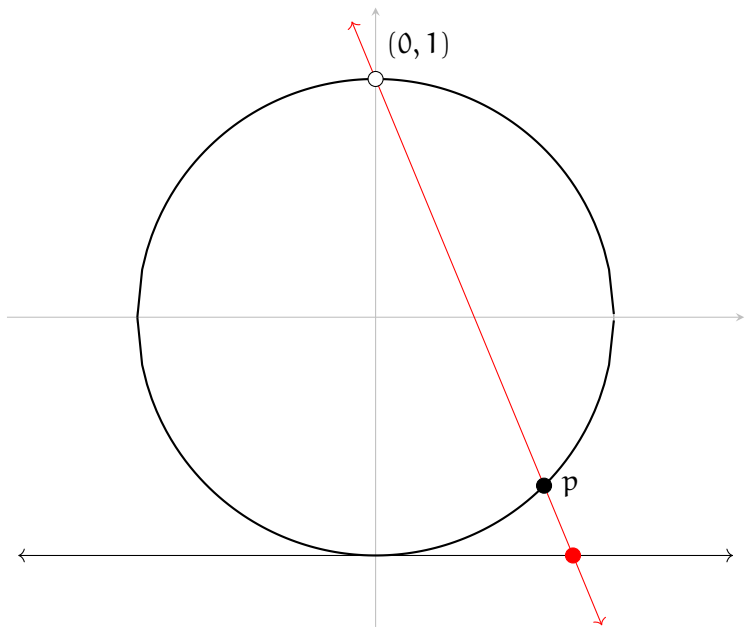

Figure 3.2.4: Given a point $p \in U = S^1 \setminus \{(0,1)\}$, we draw a line connecting p to $(0,1)$, indicated in red. The point where that line intersects the line $y = -1$ (in black) is sometimes called the *projection of* p *from* $(0,1)$ *onto* $\mathbb{R} \times \{-1\}$. The x-coordinate of this point is $x(p)$.

(e) Show that (U, x) and (U', x') are compatible charts.

Thus $\mathcal{A} = \{(U, x), (U', x')\}$ is an atlas on S^1. Extending $\mathcal{A}$ to a maximal atlas, we have defined a manifold structure on S^1. It is evidently a 1-dimensional manifold.

Exercise 3.2.5 (The n-sphere S^n)**.** Let S^n denote the set of points in $\mathbb{R}^{n+1}$ of norm 1. Generalize exercise 3.2.3 and show that "projection from the north pole" and "projection from the south pole" define an atlas on S^n.

Example 3.2.6 (Matrices)**.** There is an isomorphism of vector spaces $x : \mathcal{M}_{n\times m} \to \mathbb{R}^{mn}$ where, if $A \in \mathcal{M}_{n\times m}$ has entries $a_{i,j}$, then

$$x(A) = \begin{bmatrix} a_{1,1} \\ a_{1,2} \\ \vdots \\ a_{1,m} \\ a_{2,1} \\ \vdots \\ a_{n,m} \end{bmatrix}.$$

We regard $\mathcal{M}_{n\times m}$ as a topological space using the matrix norms defined in section 0.5.B, then x is a homeomorphism by definition. In other words, the single chart $(\mathcal{M}_{n\times m}, x)$ is an atlas on $\mathcal{M}_{n\times m}$. Extending it to a maximal atlas using exercise 3.1.10, we see that we can regard $\mathcal{M}_{n,m}$ as a manifold. It is mn-dimensional.

Exercise 3.2.7 (Open submanifolds)**.** Let X be a manifold and $U \subseteq X$ an open subset. Show that

$$\mathcal{A}_U = \{(V, x) \in \mathcal{A}_X : V \subseteq U\}$$

is a maximal atlas on U. We call U, equipped with this maximal atlas, an *open submanifold* of X.

Exercise 3.2.8 (Product manifold)**.** Let X and Y be manifolds. Let $\mathcal{A}$ be the set of charts of the form $(U \times V, x \times y)$, where $(U, x : U \to \mathbb{R}^m) \in \mathcal{A}_X$, $(V, y : V \to \mathbb{R}^n) \in \mathcal{A}_Y$, and $x \times y$ denotes the natural map $U \times V \to \mathbb{R}^m \times \mathbb{R}^n = \mathbb{R}^{m+n}$. Show that $\mathcal{A}$ is an atlas on $X \times Y$. We call $X \times Y$, equipped with the corresponding maximal atlas, the *product manifold*.

3.2.B Möbius strip

Let B denote the box $[0,1] \times (0,1)$ inside $\mathbb{R}^2$, and define an equivalence relation $\sim$ on B by declaring $(0,b) \sim (1, 1-b)$ for all $b \in (0,1)$. We will denote the equivalence class of a point $(a,b) \in B$ as $[a,b]$.

Exercise 3.2.9. Check that $\sim$ is an equivalence relation.

The *Möbius strip* is the set of equivalence classes $M := B/\sim$. In other words, we take the box $[0,1] \times (0,1)$ and "glue" its left and right edges, with a twist. If you've ever made a Möbius strip out of paper, you should be able to see that we've formalized exactly that same process. We regard M as a topological space using the quotient topology, as in example 0.7.3.

To give X the structure of a manifold, we will now produce an atlas on X. It will consist of two charts. The first chart is easy to describe. Let

$$U = \{[a,b] \in M : a \neq 0, 1\}$$

and let $x : U \to \mathbb{R}^2$ be the function $x([a,b]) = (a,b)$.

Exercise 3.2.10. Show that $x : U \to \mathbb{R}^2$ is well-defined and that it is a homeomorphism onto its image.

Intuitively, what we've done here is cut along the same edge where we glued the Mobius strip together. What we're left with is U, which is just a flat rectangle and therefore embeds into $\mathbb{R}^2$.

The second chart is a bit harder to describe mathematically, but the intuitive idea is the same. Instead of cutting the Möbius strip along the line where we glued, we'll cut along the line where $a = 1/2$. This should still leave us with a flat rectangle, which we can embed into $\mathbb{R}^2$. Formally, let

$$V = \{[a,b] \in M : a \neq 1/2\}$$

and let $y : V \to \mathbb{R}^2$ be the function defined by

$$y([a,b]) = \begin{cases} (a,b) & \text{if } a < 1/2 \\ (a-1, 1-b) & \text{if } a > 1/2 \end{cases}$$

Exercise 3.2.11. Show that $y : V \to \mathbb{R}^2$ is well-defined and that it is a homeomorphism onto its image.

Exercise 3.2.12. Show that (U, x) and (V, y) are compatible.

Thus $\mathcal{A} = \{(U, x), (V, y)\}$ is an atlas on the Möbius strip M. Letting $\mathcal{A}_M$ denote the corresponding maximal atlas, we obtain the structure of a manifold on M.

3.2.C More examples

Exercise 3.2.13 (Torus)**.** Let B denote the box $[0, 1] \times [0, 1]$ inside $\mathbb{R}^2$ and define the equivalence relation $\sim$ by $(a, 0) \sim (a, 1)$ and $(0, b) \sim (1, b)$ for all $a, b \in [0, 1]$. The *torus* T is defined to be the quotient $B/\sim$. Describe an atlas on T.

Exercise 3.2.14 (Projective line)**.** Let S^1 denote the circle as in exercise 3.2.3. Define an equivalence relation $\sim$ on S^1 which identifies antipodal points; in other words, every point on S^1 is declared to be equivalent to the point that is diametrically opposite it. The *projective line* $\mathbb{P}^1$ is defined to be the quotient $S^1/\sim$. Describe an atlas on $\mathbb{P}^1$.

Exercise 3.2.15 (Projective plane)**.** Let S^2 denote the 2-sphere as in exercise 3.2.5. Define an equivalence relation $\sim$ on S^2 which identifies antipodal points; in other words, every point on S^2 is declared to be equivalent to the point that is diametrically opposite it. The *projective plane* $\mathbb{P}^2$ is defined to be the quotient $S^2/\sim$. Describe an atlas on $\mathbb{P}^2$.

3.3 Smooth functions

We will now define what it means for a function between two manifolds to be smooth. Recall that in chapter 2 we saw that the case when the codomain is $\mathbb{R}$ was the most important. The same will be true here.

3.3.A Smooth real-valued functions

Definition 3.3.1 (Smooth real-valued function)**.** Let X be a manifold. A function $f : X \to \mathbb{R}$ is said to be *smooth* if, for any chart $(U, x) \in \mathcal{A}_X$, the function $f \circ x^{-1}$ is smooth.

$$x(U) \xrightarrow{x^{-1}} U \xrightarrow{f} \mathbb{R}$$

There are two disconcerting things about this definition. First, there is the issue that checking smoothness of a particular function seems next to impossible: the maximal atlas $\mathcal{A}_X$ will have *many, many* charts, and the definition makes it seem like we need to check smoothness on each of these charts! Thankfully, we don't need to check on all possible charts. As soon as we check smoothness on a collection of charts that cover X, we're done.

Lemma 3.3.2. *Let X be a manifold and $f : X \to \mathbb{R}$ a function. Suppose that there exists an atlas $\mathcal{A} \subseteq \mathcal{A}_X$ such that $f \circ x^{-1}$ is smooth for any $(U, x) \in \mathcal{A}$. Then f is smooth.*

Proof. Suppose $(V, y) \in \mathcal{A}_X$. For any $(U, x) \in \mathcal{A}$, we have maps as follows.

$$\begin{tikzcd} y(U \cap V) \arrow[r, "y^{-1}"] \arrow[d, "x \circ y^{-1}"'] & U \cap V \arrow[r, "f"] & \mathbb{R} \\ x(U \cap V) \arrow[ur, bend right, "x^{-1}"'] & & \end{tikzcd}$$

Since (V, y) and (U, x) are both elements of the maximal atlas $\mathcal{A}_X$, we know that the transition function $x \circ y^{-1}$ is smooth. Since $(U, x) \in \mathcal{A}$, we know by assumption that $f \circ x^{-1}$ is smooth. Thus

$$f \circ y^{-1} = (f \circ x^{-1}) \circ (x \circ y^{-1})$$

is a smooth function $y(U \cap V) \to \mathbb{R}$.

Now note that, since $\mathcal{A}$ is an atlas, its charts cover all of X, so

$$\bigcup_{(U,x) \in \mathcal{A}} y(U \cap V) = y(V).$$

Since $f \circ y^{-1}$ is smooth when restricted to each the open subset $y(U \cap V)$ for all $(U, x) \in \mathcal{A}$, we conclude that $f \circ y^{-1}$ must be smooth on all of $y(V)$. $\square$

The second disconcerting thing is the following. Suppose $U \subseteq \mathbb{R}^m$ is an open subset and $f : U \to \mathbb{R}$ is a function. On the one hand, we defined what it means for f to be smooth back in definition 2.4.32. On the other hand, we can regard U as an open submanifold of $\mathbb{R}^n$ (cf. exercise 3.2.7) and then definition 3.3.1 gives us another definition of what it means to be smooth. A priori, it could be that these definitions conflict. Thankfully, they do not.

Exercise 3.3.3. Suppose $U \subseteq \mathbb{R}^m$ is an open subset and $f : U \to \mathbb{R}$ is a function. Show that f is smooth in the sense of definition 2.4.32 if and only if it is smooth in the sense of definition 3.3.1.

Possible hint. You might find lemma 3.3.2 useful.

Definition 3.3.4. If X is a manifold, we write $\mathcal{O}(X)$ for the set of smooth functions $X \to \mathbb{R}$.

Exercise 3.3.5. Show that $\mathcal{O}(X)$ is a vector space.

Exercise 3.3.6. If $f, g \in \mathcal{O}(X)$, show that $fg \in \mathcal{O}(X)$ also.

Proposition 3.3.7. *Suppose K is a compact subset of X and U is an open neighborhood of K. Then there exists a smooth function $f : X \to [0, 1]$ such that $f(x) = 1$ for all $x \in K$ and $f(x) = 0$ for all $x \notin U$.*

3.3.B Partial derivatives of real-valued functions

Definition 3.3.8 (Partial derivatives)**.** Let X be a manifold and $(U, x) \in \mathcal{A}_X$ a chart. Recall that we then have coordinate functions $x_1, \ldots, x_n : U \to \mathbb{R}$, where $x_i = \pi_i \circ x$. Given a smooth function $f : X \to \mathbb{R}$ and a point $p \in U$, we define the *partial derivative of* f *at* p *with respect to* x_i by

$$\left.\frac{df}{dx_i}\right|_p := \partial_i(f \circ x^{-1})(x(p)).$$

In other words, $df/dx_i = \partial_i(f \circ x^{-1}) \circ x$ as functions $U \to \mathbb{R}$.

3.3.C Smooth functions between manifolds

Definition 3.3.9 (Smoothness)**.** Suppose X and Y are manifolds and $f : X \to Y$ is a function. Then f is *smooth* if, for every chart $(V, y) \in \mathcal{A}_Y$ and every chart $(U, x) \in \mathcal{A}_X$ such that $f(U) \subseteq V$, the function $y \circ f \circ x^{-1} : x(U) \to y(V)$ is smooth.

$$\begin{tikzcd}
X \arrow[r, "f"] & Y \\
U \arrow[u, hook] \arrow[r, "f"] \arrow[d, "x"'] & V \arrow[u, hook] \arrow[d, "y"] \\
x(U) \arrow[r, "y \circ f \circ x^{-1}"] \arrow[d, hook] & y(V) \arrow[d, hook] \\
\mathbb{R}^m & \mathbb{R}^n
\end{tikzcd}$$

Once again, we don't need to check all possible charts. The precise statement is a bit complicated though; there's a long string of quantifiers.

Exercise 3.3.10. Suppose X and Y are manifolds and $f : X \to Y$ is a function. Suppose further that there exists an atlas $\mathcal{A} \subseteq \mathcal{A}_Y$ such that, for every $(V, y) \in \mathcal{A}$, there exists an atlas $\mathcal{A}' \subseteq \mathcal{A}_{f^{-1}(V)}$, such that for every $(U, x) \in \mathcal{A}'$, the function $y \circ f \circ x^{-1} : x(U) \to y(V)$ is smooth. Show that f is smooth.

We should also note that we haven't redefined the word "smooth" in any conflicting ways. More precisely, we have the following two exercises.

Exercise 3.3.11. If $U \subseteq \mathbb{R}^m$ is an open subset and $f : U \to \mathbb{R}^n$ is a function, show that f is smooth in the sense of definition 2.4.32 if and only if it is smooth in the sense of definition 3.3.9.

Exercise 3.3.12. Show that a function $f : X \to \mathbb{R}$ is smooth in the sense of definition 3.3.1 if and only if it is smooth in the sense of definition 3.3.9.

Finally, we have the following result which tells us that the case when the codomain is $\mathbb{R}$ is the most important case.

Proposition 3.3.13. *Suppose $f : X \to Y$ is a function between two manifolds. Then the following are equivalent.*

(a) f is smooth.

(b) For every smooth function $g : Y \to \mathbb{R}$, the composite $g \circ f$ is a smooth function $X \to \mathbb{R}$.

3.3.D Diffeomorphisms

Definition 3.3.14 (Diffeomorphism)**.** Suppose X and Y are manifolds. A function $f : X \to Y$ is a *diffeomorphism* if it is smooth, bijective, and the inverse function $f^{-1} : Y \to X$ is also smooth. If there exists a diffeomorphism $f : X \to Y$, then X and Y are *diffeomorphic*.

Exercise 3.3.15. Regard $\mathbb{R}$ as a manifold in two different ways: once with the euclidean atlas, and next with the maximal atlas containing $x \mapsto x^3$. Show that these two manifolds are diffeomorphic.

Exercise 3.3.16. If X is a manifold and $(U, x) \in \mathcal{A}_X$, prove that $x : U \to x(U)$ is a diffeomorphism.

Exercise 3.3.17. Suppose $f : X \to Y$ is a diffeomorphism. Show that, for any open subset $U \subseteq X$, the image $f(U)$ is open in Y.

Definition 3.3.18 (Local diffeomorphism)**.** Suppose X and Y are manifolds. A function $f : X \to Y$ is a *local diffeomorphism* at a point $p \in X$ if there exists an open neighborhood U of p such that $f|_U$ is a diffeomorphism $U \to f(U)$.

4 Tangent Spaces

4.1 Tangent space

Throughout this section, we let X denote a manifold and $p \in X$ a point. We also let $n := \dim_p(X)$.

4.1.A Curves based at a point

Definition 4.1.1. A *curve in* X *based at* p is a smooth function $\gamma : (-\epsilon, \epsilon) \to X$ for some $\epsilon > 0$ such that $\gamma(0) = p$.

Suppose that $(U, x) \in \mathcal{A}_X$ is a chart containing p and that $\gamma : (-\epsilon, \epsilon) \to \mathbb{R}$ is a curve based at p. Since U is an open subset of X and γ is continuous, the preimage $\gamma^{-1}(U)$ is an open subset of $(-\epsilon, \epsilon)$ containing 0, so there exists $0 < \epsilon' \leqslant \epsilon$ such that the image of the smaller interval $(-\epsilon', \epsilon)$ is entirely contained inside U. We let $x \circ \gamma$ denote the composite function

$$(-\epsilon', \epsilon') \longrightarrow U \longrightarrow \mathbb{R}^n.$$

We can then take its derivative at 0 as in chapter 2. This derivative $d(x \circ \gamma)_0$ is a linear map $\mathbb{R} \to \mathbb{R}^n$, so its matrix representation

$$(x \circ \gamma)'(0)$$

is a column vector in $\mathbb{R}^n$.

Definition 4.1.2. Two curves γ_1, γ_2 in X based at p are *equivalent* if $d(x \circ \gamma_1)_0 = d(x \circ \gamma_2)_0$ for every chart $(U, x) \in \mathcal{A}_X$ containing p.

As usual, the first point of order is to check that we don't need to check all possible charts.

Exercise 4.1.3. Show that, if γ_1, γ_2 are two curves in X based at p and there exists a chart $(U, x) \in \mathcal{A}_X$ containing p and

$$d(x \circ \gamma_1)_0 = d(x \circ \gamma_2)_0,$$

then γ_1 is equivalent to γ_2.

Definition 4.1.4. We define the *tangent space of* X *at* p, denoted T_pX, to be the set of equivalence classes of curves in X based at p. Elements of T_pX are called *tangent vectors* at p. If γ is a curve based at p, we will write v_γ for the corresponding tangent vector in T_pX.

4.1.B Vector space structure

We will now put a vector space structure on the tangent space T_pX. There are a lot of details involved in the process, so we begin with a high-level overview. If we choose a chart $(U, x) \in \mathcal{A}_X$ containing p, there is a bijection $\sigma : T_pX \to \mathbb{R}^n$ (cf. lemma 4.1.5). This means that we can define vector space operations on T_pX by "pulling back along σ." More precisely, this mean the following: given $v \in T_pX$ and $\lambda \in \mathbb{R}$, we define

$$\lambda v := \sigma^{-1}(\lambda\sigma(v)),$$

and given $v_1, v_2 \in T_pX$, we define

$$v_1 + v_2 := \sigma^{-1}(\sigma(v_1) + \sigma(v_2)).$$

We can then show that these operations define the structure of a vector space on T_pV by showing that all of the vector space axioms are satisfied (cf. lemma 4.1.7). The final step is to show that these operations do not depend on the choice of chart (U, x) (cf. exercise 4.1.11). Thus there is a canonical, coordinate-free, vector space structure on T_pX.

Lemma 4.1.5. *Suppose* $(U, x) \in \mathcal{A}_X$ *is a chart containing* p. *There is a well-defined bijection* $\sigma : T_pX \to \mathbb{R}^n$ *given by* $v_\gamma \mapsto [d(x \circ \gamma)_0] = d(x \circ \gamma)_0(1)$.

Proof. The fact that σ is well-defined follows immediately from definition 4.1.2. For a vector $w \in \mathbb{R}^n$, define $\gamma_w : (-\epsilon, \epsilon) \to X$ by

$$\gamma_w(t) = x^{-1}(x(p) + tw)$$

where ϵ is chosen to be small enough that $x(p) + tw \in x(U)$ for all $t \in (-\epsilon, \epsilon)$ (cf. exercise 4.1.6). Then $\gamma_w(0) = x^{-1}(x(p)) = p$, so γ_w is a curve in X based at p. Notice that

$$(x \circ \gamma_w)(t) = x(x^{-1}(x(p) + tw)) = x(p) + tw$$

so $[d(x \circ \gamma_w)_0] = w$ (cf. exercise 4.1.6). It follows from this that $w \mapsto v_{\gamma_w}$ is inverse to σ, proving that σ is bijective (cf. exercise 4.1.6). □

Exercise 4.1.6. (a) Explain why there must exist $\epsilon > 0$ such that $x(p) + tw \in x(U)$ for all $t \in (-\epsilon, \epsilon)$.

(b) Explain why $[d(x \circ \gamma)_0] = w$.

(c) Explain how it follows from (b) that $w \mapsto v_{\gamma_w}$ is inverse to σ.

Lemma 4.1.7. *Suppose $(U, x) \in \mathcal{A}_X$ is a chart containing p and $\sigma : T_pX \to \mathbb{R}^n$ is the bijection of lemma 4.1.5. Then T_pX, equipped with the addition and scalar multiplication operations obtained by pulling back along σ, is a vector space. The zero element is the equivalence class of the constant curve at p.*

Proof. This follows immediately from the fact that $\mathbb{R}^n$ is a vector space and that σ is bijective, but writing out all of the details is fairly tedious. To explain the idea, let's show that vector addition is commutative. For $v_1, v_2 \in T_pX$, we have

$$\begin{aligned} v_1 + v_2 &= \sigma^{-1}(\sigma(v_1) + \sigma(v_2)) \\ &= \sigma^{-1}(\sigma(v_2) + \sigma(v_1)) \\ &= v_2 + v_1 \end{aligned}$$

where we used the definition of addition in T_pX on the first and last steps, and the fact that addition in $\mathbb{R}^n$ is commutative for the middle step.

Let γ denote the constant curve $\gamma(t) = p$ for all t. Then $x \circ p$ is also a constant function which always takes the value $x(p)$, so its derivative is 0. Thus, for any $v \in T_pX$, we have

$$\begin{aligned} v + v_\gamma &= \sigma^{-1}(\sigma(v) + \sigma(v_\gamma)) \\ &= \sigma^{-1}(\sigma(v) + 0) \\ &= \sigma^{-1}(\sigma(v)) \\ &= v, \end{aligned}$$

proving that v_γ is the zero element of T_pX.

The rest of the proof is left to you (cf. exercise 4.1.8) □

Exercise 4.1.8. Check all of the other axioms that need to be checked to ensure that T_pX is a vector space.[1]

Exercise 4.1.9. Suppose $(U, x) \in \mathcal{A}_X$ is a chart containing p and $\sigma : T_pX \to \mathbb{R}^n$ is the bijection of lemma 4.1.5. If T_pX is equipped with the vector space operations obtained by pulling back along σ, then $\sigma : T_pX \to \mathbb{R}^n$ is a linear map. Therefore, it is an isomorphism of vector spaces, and $\dim T_pX = n$.

[1] During a class I took with him, George Bergman once said something along the following lines: "If you're sure you can write the details of a proof, you probably don't need to. But if you're not sure you can formalize the details, you need to do it." I encourage you to apply this principle for exercise 4.1.8.

Lemma 4.1.10. *Suppose* $(U, x), (U', x') \in \mathcal{A}_X$ *are both charts containing* p *and that* $\sigma, \sigma' : T_pX \to \mathbb{R}^n$ *are the bijections from lemma 4.1.5 corresponding to* $(U, x), (U', x')$*, respectively. Then* $\sigma' = d(x' \circ x)_{x(p)} \circ \sigma$*, where* $x' \circ x$ *denotes the transition map (cf. definition 3.1.3).*

$$\begin{tikzcd} T_pX \arrow[r, "\sigma"] \arrow[dr, bend right, "\sigma'"'] & \mathbb{R}^n \arrow[d, "d(x' \circ x^{-1})_{x(p)}"] \\ & \mathbb{R}^n \end{tikzcd}$$

Proof. Let γ be a curve in X based at p. Then

$$x' \circ \gamma = x' \circ (x^{-1} \circ x) \circ \gamma = (x' \circ x) \circ (x \circ \gamma),$$

so the chain rule 2.3.3 says that

$$d(x' \circ \gamma)_0 = d(x' \circ x)_{(x \circ \gamma)(0)} \circ d(x \circ \gamma)_0 = d(x' \circ x)_{x(p)} \circ d(x \circ \gamma)_0.$$

Evaluating at 1, we find that

$$\sigma'(v_\gamma) = d(x' \circ \gamma)_0(1) = d(x' \circ x)_{x(p)}(d(x \circ \gamma)_0(1)) = d(x' \circ x)_{x(p)}(\sigma(v_\gamma)).$$

Since this is true for all γ, we conclude that $\sigma' = d(x' \circ x)_{x(p)} \circ \sigma$. □

Exercise 4.1.11. Suppose $(U, x), (U', x') \in \mathcal{A}_X$ are both charts containing p and that $\sigma, \sigma' : T_pX \to \mathbb{R}^n$ are the bijections from lemma 4.1.5 corresponding to $(U, x), (U', x')$, respectively. Show that the vector space operations on T_pX defined by pulling back along σ are the same as those defined by pulling back along σ'.

Possible hint. Use lemma 4.1.10. The key is that $d(x' \circ x^{-1})_{x(p)}$ is an invertible linear map.

This completes our construction of a vector space structure on T_pX. Notice that T_pX is an example of a finite dimensional vector space that does not come equipped with any canonical basis! More precisely, this means the following. If we choose a chart $(U, x) \in \mathcal{A}_X$ containing p, the bijection $\sigma : T_pX \to \mathbb{R}^n$ of lemma 4.1.5 is an isomorphism of vector spaces (cf. exercise 4.1.9), so $\sigma^{-1}(e_1), \ldots, \sigma^{-1}(e_n)$ is a basis of T_pX. But then, if we choose a different chart $(U', x') \in \mathcal{A}_X$ containing p and σ' is the corresponding isomorphism $T_pX \to \mathbb{R}^n$, then $\sigma'^{-1}(e_1), \ldots, \sigma'^{-1}(e_n)$ will in general be a different basis of T_pX, even though the vector space structures defined by pulling back along σ and σ' are the same! So, for a general manifold X and point p, since there's no "canonical" choice of a chart containing p, there is also no "canonical" choice of a basis on T_pX. I strongly encourage you to work through the following exercise to make sense of this.

Exercise 4.1.12. Consider the unit circle S^1, regarded manifold using the charts (U, x) and (U', x') defined in exercise 3.2.3. Any point $p \in S^1$ can be written as $(\cos\theta, \sin\theta)$ for some $\theta \in \mathbb{R}$.

(a) Observe that $\gamma : \mathbb{R} \to S^1$ given by

$$\gamma(t) = (\cos(\theta + t), \sin(\theta + t))$$

is a curve in S^1 based at p. Explain why v_γ is a basis for T_pS^1.

(b) Suppose $p \in U$ and let $\sigma : T_pS^1 \to \mathbb{R}$ be the isomorphism of lemma 4.1.5 corresponding to the chart (U, x). Find $\lambda \in \mathbb{R}$ such that $\sigma(\lambda v_\gamma) = 1$.

(c) Suppose $p \in U'$ and let $\sigma' : T_pS^1 \to \mathbb{R}$ be the isomorphism of lemma 4.1.5 corresponding to the chart (U', x'). Find $\lambda' \in \mathbb{R}$ such that $\sigma'(\lambda' v_\gamma) = 1$.

So, if $p \in U \cap U'$, we see that there are three possible bases we might choose on T_pS^1 (v_γ, or λv_γ, or $\lambda' v_\gamma$), none of these is any more "canonical" than another. This is what is meant when we say that T_pS^1 has no canonical choice of basis.

Remark 4.1.13. Suppose U is an open subset of X containing p regarded as an open submanifold (cf. exercise 3.2.7). A curve γ in U based at p is also automatically a curve in X, which means there is a natural "inclusion" map $T_pU \to T_pX$ given by $v_\gamma \mapsto v_\gamma$. But we can "shrink" curves in X to curves in U without changing their equivalence class, so $T_pU \to T_pX$ is actually bijective. Moreover, if we choose a chart in $\mathcal{A}_U$ containing p, we can use this chart to define the vector space structure on both T_pU and T_pX, so $T_pU \to T_pX$ is actually an isomorphism of vector spaces. Since this isomorphism is tautological, we sometimes write $T_pU = T_pX$.

Remark 4.1.14. There is one situation where we do have a canonical choice of a chart: $\mathbb{R}^n$. Namely, the identity map $\mathrm{id} : \mathbb{R}^n \to \mathbb{R}^n$ defines a chart. If γ is a curve in $\mathbb{R}^n$ based at a point p, then $d(\mathrm{id} \circ \gamma)_0 = d\gamma_0$, so the isomorphism $\sigma : T_p\mathbb{R}^n \to \mathbb{R}^n$ is given by $\sigma(v_\gamma) = [d\gamma_0]$. We will call this the "canonical isomorphism" $T_p\mathbb{R}^n \to \mathbb{R}^n$, which we sometimes also write as $T_p\mathbb{R}^n = \mathbb{R}^n$. But this is fairly abusive and it's worth remembering that $T_p\mathbb{R}^n$ is not literally equal to $\mathbb{R}^n$. The former is equivalence classes of curves, the latter is column vectors, and the relationship between the two is given by taking the derivative at 0.

4.1.C Derivations ⋆

There is another perspective on tangent vectors which is often useful, though it is decidedly less geometric at first glance. It begins with the following definition.

Definition 4.1.15. Let X be a manifold and $p \in X$ a point. A *derivation* on X based at p is a linear function $\partial : \mathcal{O}(X) \to \mathbb{R}$ such that

$$\partial(fg) = \partial(f)g(p) + f(p)\partial(g).$$

We let $\mathrm{Der}_p(X)$ denote the set of all derivations on X based at p.

The following is not very hard to prove.

Exercise 4.1.16. Show that $\mathrm{Der}_p(X)$ is a vector space.

It turns out that T_pX and $\mathrm{Der}_p(X)$ are isomorphic, as we will soon prove.

Remark 4.1.17. Sometimes, people define the tangent space to be $\mathrm{Der}_p(X)$. It's not very geometric, but it at least makes the vector space operations on the tangent space fairly obvious (cf. exercise 4.1.16); compare this with our geometric definition of the tangent space in terms of curves, where we had to do a lot of work to regard the tangent space as a vector space (cf. section 4.1.B). That said, even if one defines the tangent space to be $\mathrm{Der}_p(X)$, it's not clear how to produce examples of derivations, or to prove that $\mathrm{Der}_p(X)$ is finite dimensional. One way or another, one needs to show that derivations are somehow linked to curves. This is what we do next.

Definition 4.1.18. Suppose γ is a curve in X based at p. If $f : X \to \mathbb{R}$ is a smooth function, then $f \circ \gamma$ is a single variable function $(-\epsilon, \epsilon) \to \mathbb{R}$. We define $\partial_\gamma : \mathcal{O}(X) \to \mathbb{R}$ by

$$\partial_\gamma(f) = (f \circ \gamma)'(0).$$

Lemma 4.1.19. *If γ is a curve in X based at p, the function ∂_γ is a derivation.*

Proof. Suppose $f, g \in \mathcal{O}(X)$ and $\lambda \in \mathbb{R}$. To see that ∂_γ is linear, we use the sum and scalar multiple rules from exercises 1.2.2 and 1.2.3.

$$\begin{aligned}
\partial_\gamma(f + \lambda g) &= ((f + \lambda g) \circ \gamma)'(0) \\
&= ((f \circ \gamma) + \lambda(g \circ \gamma))'(0) \\
&= (f \circ \gamma)'(0) + \lambda(g \circ \gamma)'(0) \\
&= \partial_\gamma(f) + \lambda\partial_\gamma(g)
\end{aligned}$$

To see that it is a derivation, we use the single variable product rule 1.2.4.

$$\begin{aligned}\partial_\gamma(fg) &= (fg \circ \gamma)'(0) \\ &= ((f \circ \gamma)(g \circ \gamma))'(0) \\ &= (f \circ \gamma)'(0)(g \circ \gamma)(0) + (f \circ \gamma)(0)(g \circ \gamma)'(0) \\ &= \partial_\gamma(f)g(p) + f(p)\partial_\gamma(g)\end{aligned}$$

This completes the proof. □

Lemma 4.1.20. *If γ_1 and γ_2 are equivalent curves in X based at p, then $\partial_{\gamma_1} = \partial_{\gamma_2}$.*

Proof. Choose a chart $(U, x) \in \mathcal{A}_X$ containing p and let γ be a curve in X based at p. Then

$$\begin{aligned}\partial_\gamma(f) &= (f \circ \gamma)'(0) \\ &= d(f \circ \gamma)_0(1) \\ &= d(f \circ x^{-1} \circ x \circ \gamma)_0(1) \\ &= d(f \circ x^{-1})_{x(p)}\left(d(x \circ \gamma)_0(1)\right)\end{aligned}$$

where we used the chain rule 2.3.3 at the last step. Since $\partial_\gamma(f)$ only depends on $d(x \circ \gamma)_0$, we see that equivalent curves must give rise to the same derivation. □

Theorem 4.1.21. *The function $T_pX \to \mathrm{Der}_p(X)$ given by $v_\gamma \mapsto \partial_\gamma$ is an isomorphism.*

Definition 4.1.22. If $v \in T_pX$, we let ∂_v denote the corresponding element in $\mathrm{Der}_p(X)$.

4.2 Pushforward

4.2.A Defining the pushforward

Let $f : X \to Y$ be a smooth map of manifolds and $p \in X$ a point. Given a curve γ in X based at p, observe that $f \circ \gamma$ is a curve in Y based at $f(p)$.

Lemma 4.2.1. *Let $f : X \to Y$ be a smooth map of manifolds and $p \in X$ a point. If γ_1 and γ_2 are equivalent curves in X based at p, then $f \circ \gamma_1$ and $f \circ \gamma_2$ are equivalent curves in Y based at $f(p)$.*

This allows us to make the following definition. We will shortly see that this is the "ultimate" generalization of derivatives.

Definition 4.2.2. Let $f : X \to Y$ be a smooth map of manifolds and $p \in X$ a point. If γ is a curve in X based at p, we define its *pushforward* $f_*(v_\gamma) \in T_{f(p)}Y$ to be the equivalence class of the curve $f \circ \gamma$. This defines the *pushforward map* $f_* : T_pX \to T_{f(p)}Y$.

Proposition 4.2.3. *Let $f : X \to Y$ be a smooth map of manifolds and $p \in X$ a point. The pushforward map $f_* : T_pX \to T_{f(p)}Y$ is linear.*

The following result states that the pushforward generalizes the derivative at a point.

Proposition 4.2.4. *Let U is an open subset of $\mathbb{R}^m$, $p \in U$ is a point, and $f : U \to \mathbb{R}^n$ is a smooth map. Then, using the identifications of remarks 4.1.13 and 4.1.14, we have $f_* = df_p$.*

$$\begin{tikzcd}
T_pU \arrow[r, equal] \arrow[d, "f_*"'] & T_p\mathbb{R}^m \arrow[r, equal] & \mathbb{R}^m \arrow[d, "df_p"] \\
T_{f(p)}\mathbb{R}^n \arrow[rr, equal] & & \mathbb{R}^n
\end{tikzcd}$$

Proof. Let σ denote the isomorphisms $T_p\mathbb{R}^m \to \mathbb{R}^m$ and $T_p\mathbb{R}^n \to \mathbb{R}^n$ of remark 4.1.14. Suppose γ is a curve in U based at p. Then

$$(df_p \circ \sigma)(v_\gamma) = df_p(d\gamma_0(1)) = (df_p \circ d\gamma_0)(1) = d(f \circ \gamma)_0(1),$$

where we used the chain rule 2.3.3 for the final step. On the other hand, we also have

$$(\sigma \circ f_*)(v_\gamma) = \sigma(v_{f\circ\gamma}) = d(f \circ \gamma)_0. \qquad \square$$

In light of proposition 4.2.4, the following is a generalization of the chain rule.

Proposition 4.2.5. *Suppose $f : X \to Y$ and $g : Y \to Z$ are smooth maps of manifolds, and $p \in X$ is a point. Then $g_* \circ f_* = (g \circ f)_*$.*

$$\begin{tikzcd}
T_pX \arrow[r, "f_*"] \arrow[dr, bend right, "(g\circ f)_*"'] & T_{f(p)}Y \arrow[d, "g_*"] \\
& T_{g(f(p))}Z
\end{tikzcd}$$

Proof. Suppose γ is a curve in X based at p. Then

$$g_*(f_*(v_\gamma)) = g_*(v_{f\circ\gamma}) = v_{g\circ f\circ\gamma} = (g \circ f)_*(v_\gamma). \qquad \square$$

This proof looks absurdly easy, and that's because it is. You might be wondering why the proof of this generalization of the chain rule looks so easy, while the proof of the chain rule in chapter 2 was significantly more work. It's because the pushforward is so abstractly defined that it's basically useless and uncomputable until we know that it actually generalizes the derivative from chapter 2 and therefore can be computed using techniques from chapter 2. This was the content of proposition 4.2.4, and notice that we used the chain rule 2.3.3 to prove proposition 4.2.4.

4.2.B Rank of a smooth map

In light of proposition 4.2.4, the following generalizes definition 2.3.23.

Definition 4.2.6. Suppose $f : X \to Y$ is a smooth map between manifolds and $p \in X$ is a point. The rank of the linear map $f_* : T_pX \to T_{f(p)}Y$ is also called the *rank of* f *at* p, denoted $\operatorname{rank}_p(f)$. Note that $\operatorname{rank}_p(f) \leqslant \min\{\dim_p(X), \dim_{f(p)}(Y)\}$.

- If $\operatorname{rank}_p(f) = \dim_p(X)$, then we say that f is *immsersive at* p. This is equivalent to requiring that f_* is an injective map $T_pX \to T_pY$.
- If $\operatorname{rank}_p(f) = \dim_{f(p)}(Y)$, then we say that f is *submersive at* p. This is equivalent to requiring that f_* is a surjective map $T_pX \to T_pY$.
- If $\operatorname{rank}_p(f) = \dim_p(X) = \dim_{f(p)}(Y)$, then we say that f is *étale at* p. This is equivalent to requiring that f_* is an isomorphism $T_pX \to T_pY$.
- If $\operatorname{rank}_p(f) = \min\{\dim_p(X), \dim_{f(p)}(Y)\}$, then we say that p is a *regular point* of f.
- If $\operatorname{rank}_p(f) < \min\{\dim_p(X), \dim_{f(p)}(Y)\}$, then we say that p is a *critical point* of f.

Definition 4.2.7. Suppose $f : X \to Y$ is a smooth map between manifolds.

- If f is immersive at every $p \in X$, then f is an *immersion*.
- If f is submersive at every $p \in X$, then f is a *submersion*.
- if f is étale at every $p \in X$, then f is *étale*.

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